

Aldosterone and Mineralocorticoid Receptors: Renal and Extrarenal Roles

David Pearce | Vivek Bhalla | John W. Funder

CHAPTER OUTLINE

GENERAL INTRODUCTION TO ALDOSTERONE AND MINERALOCORTICOID RECEPTORS, 336
 ALDOSTERONE SYNTHESIS, 337
 MECHANISMS OF MINERALOCORTICOID RECEPTOR FUNCTION AND GENE REGULATION, 338
 REGULATION OF SODIUM ABSORPTION AND POTASSIUM SECRETION, 342
 NONRENAL ALDOSTERONE-RESPONSIVE TIGHT EPITHELIA, 348
 ROLE OF SERUM- AND GLUCOCORTICOID-REGULATED KINASE IN MEDIATING ALDOSTERONE EFFECTS, 349

11 β -HYDROXYSTEROID DEHYDROGENASE TYPE 2, 352
 NONGENOMIC EFFECTS OF ALDOSTERONE, 353
 DISEASE STATES, 353
 NONEPITHELIAL ACTIONS OF ALDOSTERONE, 354
 RECENT ADVANCES IN NONRENAL MINERALOCORTICOID RECEPTOR-MEDIATED DISEASE PATHOLOGY, 355

KEY POINTS

- Inappropriate aldosterone hypersecretion relative to sodium status is much more common in hypertensives (5%–15%) than is generally appreciated and is even found in a significant percentage of normotensives.
- Most unilateral aldosterone-producing adenomas harbor disease-causing gene mutations, the most common of which is in the K⁺ channel gene, *KCNJ5*. Germline mutations in *KCNJ5* cause familial bilateral adrenal hyperplasia. Most recently, the chloride channel gene, *CLCN2*, has been implicated in early-onset primary aldosteronism, with clinical findings most consistent with bilateral hyperplasia.
- Recent evidence supports the idea that the actions of aldosterone on the distal convoluted tubule are indirect and mediated by changes in K⁺. Aldosterone stimulates ENaC in connecting tubule and cortical collecting ducts, and the ensuing hypokalemia acts directly in DCT cells to stimulate the Na-Cl cotransporter, NCC.
- SGK1 gene transcription is regulated by aldosterone, and it in turn stimulates ENaC, thereby enhancing Na⁺ reabsorption and K⁺ secretion. SGK1 also responds to and integrates a variety of other hormonal and nonhormonal signals, including insulin and IL-17, which stimulate SGK1 activity and contribute to salt-sensitive hypertension.
- Studies in mice demonstrate a role for T cells in the pathogenesis of hypertension through a mechanism involving IL-17A and SGK1. Mice lacking T cell SGK1 are protected from angiotensin II induced hypertension and renal inflammation.
- The classic thinking that Na⁺ is dissolved, fully exchangeable, and osmotically active in both intracellular and extracellular compartments has been proving inadequate. Recent studies have identified a nonexchangeable pool of Na⁺ in extracellular compartments, most notably in subcutaneous tissues in an osmotically inactive form, perhaps sequestered within negatively charged glycosaminoglycans.

In mammals, the control of extracellular fluid volume and blood pressure is intimately intertwined with the regulation of epithelial ion transport. Aldosterone, which is essential for survival, is the central hormone regulating the relevant epithelial transport processes, particularly of ions such as Na^+ , K^+ , and Cl^- . All circulating aldosterone is generated in the adrenal glomerulosa, where its synthesis and secretion are under the control of angiotensin II and potassium, and its major epithelial actions occur in the distal nephron and colon. The former extends from the late distal convoluted tubule (DCT) through the connecting segment and the entire cortical and medullary collecting ducts. These segments, rich in mineralocorticoid receptor (MR), are often referred to as the “aldosterone-sensitive distal nephron” (ASDN).¹ Most, if not all, effects of aldosterone are mediated by MR, a hormone-regulated transcription factor related closely to the glucocorticoid receptor and more distantly to other members of the nuclear receptor superfamily. The physiologic effects of aldosterone on epithelia entail direct gene-regulatory actions of MR. Thus, a sound foundation for understanding aldosterone’s physiologic effects on the extracellular fluid, blood pressure, and electrolyte concentrations can be understood through familiarity with the MR-dependent effects on the transcription of various genes, which, in turn, alter epithelial ion transport. Aldosterone actions in certain disease states involve both genomic and nongenomic effects in epithelial and nonepithelial tissues. Furthermore, physiologic and pathophysiologic effects of cortisol are also mediated in part by MR, which binds cortisol with high affinity. This chapter addresses the cellular and molecular mechanisms underlying aldosterone—and, to some extent, cortisol—action, focusing primarily on effects on ion transport in epithelia but also highlighting key aspects of nonepithelial actions, which are of substantial importance to its pathophysiologic effects.

GENERAL INTRODUCTION TO ALDOSTERONE AND MINERALOCORTICOID RECEPTORS

Steroid hormones are derived from cholesterol and produced in systemically relevant amounts in a relatively narrow range

of tissues (e.g., adrenal glands, gonads, placenta, skin). In mammalian physiology, six classes of steroid hormones are commonly recognized—mineralocorticoid, glucocorticoid, androgen, estrogen, progestin, and the secosteroid vitamin D_3 . This classification was based on observed effects of these hormones and has proven robust, despite current appreciation of a much more diverse physiology of steroid hormones over and above their classic roles. In further support of this original classification is the characterization of six intracellular receptors—MR; glucocorticoid, GR; androgen, AR; estrogen, ER; progestin, PR; and vitamin D_3 , VDR. As further addressed later, it is now appreciated that a one-to-one relationship between receptor and hormone does not hold, and this is particularly the case for MR.

Aldosterone was isolated and characterized in 1953. Crucial for its isolation was the application of radioisotopic techniques to measure $[\text{Na}^+]$ and $[\text{K}^+]$ flux across epithelia in the laboratory of Sylvia Simpson, a biologist, and Jim Tait, a physicist.^{2,3} Because of this, the active principle was initially called “electrocortin”; the name was soon changed to aldosterone when its unique aldehyde (rather than methyl) group at carbon 18 was discovered in collaborative studies between investigators in London and Basel.⁴ Aldosterone is commonly depicted so as to highlight this aldehyde group (Fig. 12.1, right). In vivo, the very reactive aldehyde group cyclizes with the β -hydroxyl group at carbon 11 to form the 11,18-hemiacetal and, in addition, may exist in an 11,18-hemiketal form. This cyclization of the 11 β -hydroxyl group protects aldosterone from dehydrogenation by the enzyme 11 β -hydroxysteroid dehydrogenase in epithelial tissue and by some neuronal and smooth muscle cells, which enables it to activate epithelial MR and thus regulate ion transport at very low (subnanomolar) circulating levels.^{5,6}

There is broad evidence that aldosterone is not the only cognate ligand for MR, its essential effects via MR on epithelial ion transport notwithstanding. MR is found in high abundance in the hippocampus and cardiomyocytes and, in these nonepithelial tissues—which lack 11 β -hydroxysteroid dehydrogenase type 2 (11 β -HSD2; see later, “11 β -Hydroxysteroid Dehydrogenase Type 2”)—they are essentially constitutively occupied by glucocorticoids (cortisol in humans, corticosterone in rodents). This is due to the comparable affinity and markedly higher plasma free levels (≥ 100 -fold) of endogenous

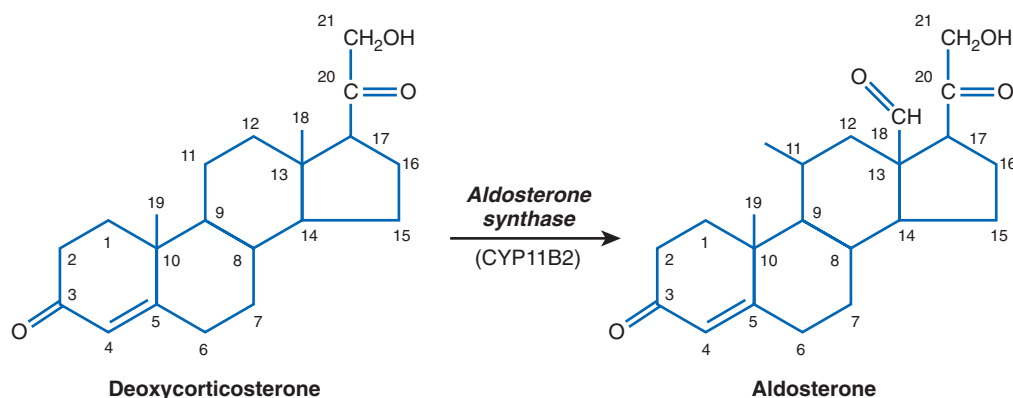


Fig. 12.1 Final step in aldosterone synthesis. Note that the aldehyde form of aldosterone is shown. Most aldosterone (>99%) exists as the hemiacetal form, which is cyclized and does not allow access of 11 β -hydroxysteroid dehydrogenase type 2 (11 β -HSD2) to the 11-hydroxyl. See text for details.

glucocorticoids compared with those of aldosterone. In terms of evolution, MR appeared well before aldosterone synthase (e.g., in fish).⁷ It was commonly assumed that MR and GR share a common immediate evolutionary precursor,⁸ although this has been challenged on sequence grounds,⁹ which implicate MR as the first of the MR-GR-AR-PR subfamily to branch off an ancestral receptor. A final reason not to equate MR and aldosterone action derives from a comparison of the MR knockout and aldosterone synthase knockout ($AS^{-/-}$) phenotypes. MR knockout mice (which lack all functional MR) cannot survive sodium restriction; $AS^{-/-}$ mice (which have no detectable aldosterone) survive even stringent sodium restriction but die when their fluid intake is restricted to that of wild-type animals.¹⁰ The survival of $AS^{-/-}$ mice on a low- Na^+ intake may reflect, in part, Na^+ retention via renal tubular intercalated cells, in which MRs (not 11β -HSD2-protected) are activated by glucocorticoids in the context of high ambient angiotensin concentrations.^{11,12} Their inability to survive fluid restriction suggests an as yet poorly defined dependence on aldosterone for vasopressin action.¹³ Potassium homeostasis is also surprisingly intact in $AS^{-/-}$ mice, although they do not tolerate extremes of K^+ loading.^{13a}

ALDOSTERONE SYNTHESIS

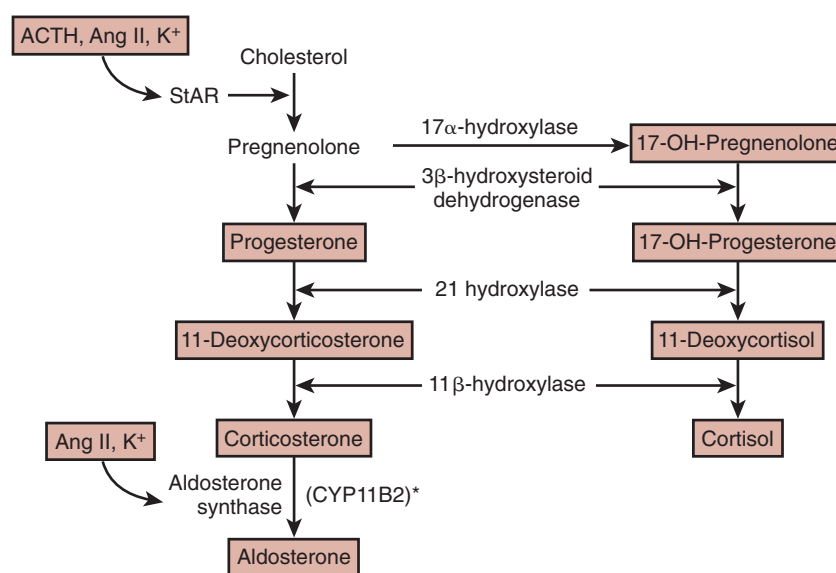
Aldosterone is synthesized in the adrenal cortex, which has three functional zones. The outermost layer of cells represents the zona glomerulosa, which is the unique site of aldosterone biosynthesis in normal physiology (see later; aldosterone is produced in excessive amounts in patients with glucocorticoid-remediable aldosteronism). Cortisol is synthesized in the middle zone, the zona fasciculata, and the innermost zona reticularis secretes adrenal androgens

in many species, including humans, but not in rats or mice. Normally, the glomerulosa secretes aldosterone at the rate of 50 to 200 μ g/day to give plasma levels of 4 to 21 μ g/dL; in contrast, secretion of cortisol is at levels 200- to 500-fold higher. Underlying the separate synthesis of cortisol and aldosterone is expression of the enzyme 17α -hydroxylase uniquely in the zona fasciculata and that of aldosterone synthase uniquely in the glomerulosa.

In most species, aldosterone synthase, or cytochrome P450 (CYP) enzyme $11B2$, is responsible for the conversion of deoxycorticosterone to aldosterone in a three-step process of sequential 11β -hydroxylation, 18 -hydroxylation, and 18 -methyl oxidation, to produce the characteristic C 18 -aldehyde from which aldosterone derives its name (see Fig. 12.1, left). Although CYP $11B2$ is distinct from CYP $11B1$ (11β -hydroxylase) in most species,^{14,15} in some species (e.g., bovine), only a single CYP $11B$ is expressed. How this enzyme is responsible for the three-step process of aldosterone synthesis in the glomerulosa but not the fasciculata has yet to be determined.

Fig. 12.2 also illustrates key steps in the biosynthesis of cortisol to illustrate the overlap and similarities with those of aldosterone. The genes encoding CYP $11B1$ and CYP $11B2$ lie close to one another on human chromosome band 8q24.3, so that an unequal crossing over at meiosis has been shown to be responsible for the syndrome of glucocorticoid-remediable aldosteronism (now known as “familial hyperaldosteronism type I”), in which the 5′ end of the CYP $11B1$ is fused to the 3′ end of CYP $11B2$. The chimeric gene product¹⁶ is expressed in the fasciculata and responds to adrenocorticotrophic hormone (ACTH) with aldosterone synthesis, producing a syndrome of juvenile-onset hyperaldosteronism and hypertension.

Normal glomerulosa secretion of aldosterone is primarily regulated by angiotensin II in response to posture and acute



*Aldosterone synthase = CYP11B2

Fig. 12.2 Overview of aldosterone synthetic pathway showing key regulatory nodes. Note that adrenocorticotrophic hormone (ACTH), angiotensin II (Ang II), and K^+ regulate steroidogenic acute regulatory protein (StAR), which stimulates cholesterol uptake by mitochondria and thus substrate availability for synthesis of all of the steroid hormones. Aldosterone synthase (gene name, CYP $11B2$), which is selectively expressed in the adrenal glomerulosa, mediates the final step in aldosterone synthesis. It is also regulated by Ang II and K^+ . Aldosterone synthesis is shown on the left. Cortisol synthesis is also shown (right) to emphasize the interconnections and similarities between these pathways.

lowering of circulating volume, to plasma $[K^+]$ in response to elevated potassium levels, particularly in settings of Na^+ deficiency,¹⁷ and to ACTH to the extent of entrainment of the circadian fluctuation in plasma aldosterone levels with those of cortisol. Aldosterone secretion is lowered by high levels of atrial natriuretic peptide and by the administration of heparin, somatostatin, and dopamine. As yet, incompletely characterized molecules of adipocyte origin have been shown to stimulate aldosterone secretion in vitro, and roles in the metabolic syndrome have been proposed on this basis.¹⁸

Angiotensin and plasma $[K^+]$ stimulate aldosterone secretion primarily by increasing the expression and activity of key steroidogenic enzymes, as well as the steroidogenic acute regulatory protein (StAR).¹⁹ StAR is required for cholesterol transport into mitochondria and hence for its availability for steroid synthesis.²⁰ Regulated steroidogenic enzymes include side chain cleavage enzyme 3β -hydroxysteroid dehydrogenase and, most notably, aldosterone synthase. Common to the mechanism of stimulation by angiotensin II and $[K^+]$ is elevation of intracellular $[Ca^{2+}]$. Angiotensin II activates the G-protein-coupled angiotensin type I receptor (AT1R) in the glomerulosa cell membrane, which in turn activates phospholipase C, which catalyzes the hydrolysis of phosphatidylinositol 4,5-bisphosphate (PIP2) to inositol trisphosphate (IP3) and diacylglycerol (DAG). DAG stimulates protein kinase C, whereas IP3 stimulates Ca^{2+} release from intracellular stores, both of which affect the aldosterone biosynthetic pathway. AT1R also separately stimulates Ca^{2+} influx, which is important for sustained stimulation of aldosterone secretion.²¹ Elevated $[K^+]$ increases intracellular $[Ca^{2+}]$ by depolarizing the cell membrane and activating voltage-sensitive Ca^{2+} channels.^{22,23} Patients taking angiotensin-converting enzyme inhibitors or angiotensin receptor blockers usually show a degree of suppression of aldosterone secretion, reflected in a modest (0.2–0.3 mEq/L) elevation in plasma $[K^+]$ levels. This is often sufficient to establish a new steady state, with plasma aldosterone levels rising into the normal range, a process best termed “breakthrough” rather than “escape,” given the time-honored usage of the latter to refer to the escape from progression of the salt and water effect of mineralocorticoid excess in the medium and long term.²⁴

Studies have shed new light on the regulation of aldosterone production by the adrenal glomerulosa in health and disease. Choi and associates have found recurrent somatic mutations in the K^+ channel Kir3.4 (encoded by the gene *KCNJ5*), which were present in more than one-third of spontaneous human aldosterone-producing adenomas studied.²⁵ These mutations increased Na^+ conductance through Kir3.4 and resulted in increased Ca^{2+} entry and enhanced aldosterone production and glomerulosa cell proliferation. Interestingly, an inherited mutation in *KCNJ5* is associated with hypertension associated with marked bilateral adrenal hyperplasia (now known as “familial hyperaldosteronism type III” [FH-III]).²⁶ These findings suggest that *KCNJ5* may provide tonic inhibition of aldosterone production and glomerulosa cell proliferation. In glomerulosa cells harboring the mutant channel, both proliferation rate and aldosterone synthesis are increased. These initial studies were continued and extended by a much wider survey by Boulkroun and colleagues²⁷; subsequently, less common but similarly somatic mutations in the adrenal cortex (*ATP1A1*, *ATP2A3*, *CACNA1D*, and *CTNNB1*) have

been associated with hyperaldosteronism caused by adrenal adenomas.^{28,29}

MECHANISMS OF MINERALOCORTICOID RECEPTOR FUNCTION AND GENE REGULATION

Mammals cannot survive without MR, except with substantial NaCl supplementation. This member of the nuclear receptor superfamily appears to have both genomic and nongenomic actions; however, the latter do not appear to play a significant role in the control of epithelial ion transport. This section thus focuses exclusively on the function of MR as a hormone-regulated transcription factor.

MINERALOCORTICOID RECEPTOR FUNCTION AS A HORMONE-REGULATED TRANSCRIPTION FACTOR: GENERAL FEATURES AND SUBCELLULAR LOCALIZATION

In the presence of agonists, MR binds to specific genomic sites and alters the transcription rate of a subset of genes. Fig. 12.3 shows the fundamental paradigm of MR function. All nuclear receptors shuttle in and out of the nucleus; however, in the absence of hormone, some, such as the estrogen and vitamin D receptors, are predominantly nuclear, whereas others, like GR, are almost exclusively cytoplasmic. In the absence of hormone, MR is distributed relatively evenly between nuclear and cytoplasmic compartments but, in the presence of hormone, it is highly concentrated in the nucleus (Fig. 12.4).^{30,31} It is also notable that in addition to this marked

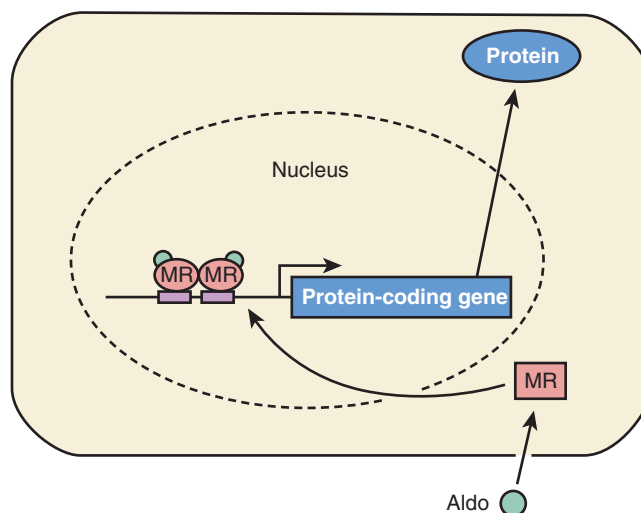


Fig. 12.3 General mechanism of aldosterone action through the mineralocorticoid receptor (MR). This simple schematic shows the general features of MR regulation of a “simple” hormone response element (HRE), common to a large subset (but not all) of aldosterone-stimulated genes. Note that in the absence of hormone, MR is found in both nucleus and cytoplasm (see Fig. 12.4). Aldosterone (Aldo) triggers nuclear translocation of cytoplasmic MR, binding as a dimer to HREs, and stimulation of transcription initiation complex formation (arrow, upstream of the so-called protein-coding gene, defines the site of transcription initiation). See text for further details.

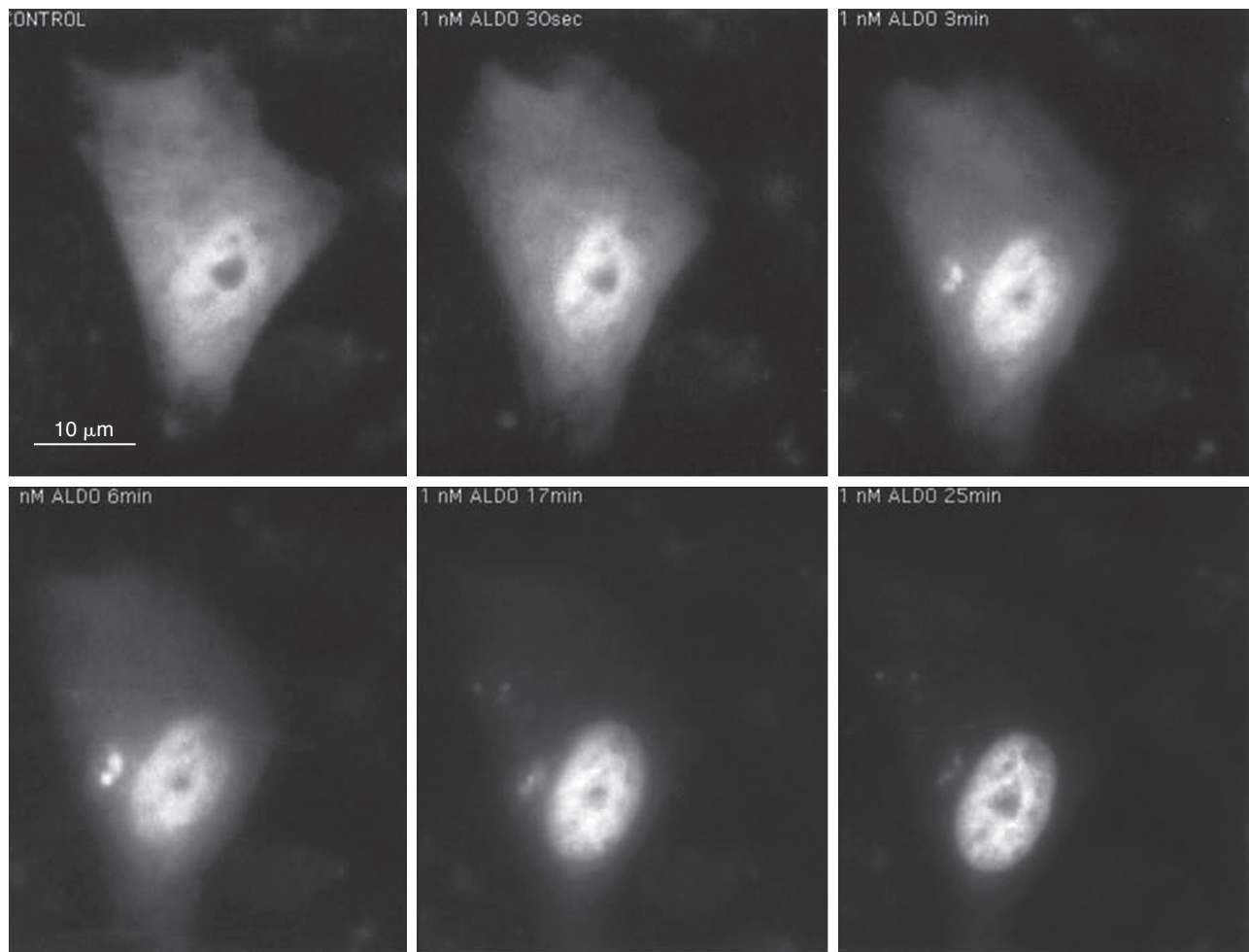


Fig. 12.4 Time-dependent nuclear translocation of the mineralocorticoid receptor (MR) in the presence of aldosterone. Cultured cells expressing green fluorescent protein–MR fusion protein (GFP–MR) were grown in a steroid-free medium and treated with 1 nmol/L aldosterone. Translocation of GFP–MR was followed in real time, and images were captured at indicated times. It is notable that the nuclear accumulation of GFP–MR started within 30 seconds, was half-maximal at 7.5 minutes, and was complete by 10 minutes. Control: MR distribution before addition of aldosterone. (From Fejes-Toth G, Pearce D, Naray-Fejes-Toth A. Subcellular localization of mineralocorticoid receptors in living cells: effects of receptor agonists and antagonists. *Proc Natl Acad Sci U. S. A.* 1998; 95[6]:2973–2978.)

change in MR cellular distribution, its subnuclear organization and protein-protein interactions are also changed.³¹ Like its close cousin the GR, the unliganded MR (in the absence of hormone) is complexed with a set of chaperone proteins, which include the heat shock proteins hsp90, hsp70, and hsp56 and immunophilins.^{32,33} This chaperone complex is essential for several aspects of MR function, notably high-affinity hormone binding and trafficking to the nucleus.³³ It was thought for many years that after binding hormone, the hsp90-containing chaperone complex is jettisoned. However, it has become clear that this complex remains associated with the receptor and plays an important role in nuclear trafficking.³² Several members of the immunophilin family, including FKBP52, FKBP51, and CyP40, are present in the chaperone complex and provide a bridge between hsp90 and the cytoplasmic motor protein dynein, which moves the receptor-hsp90 complex retrogradely along microtubules to the nuclear envelope.³³ Here, the receptor is handed off to the nuclear pore protein, importin- α , and translocated into the nucleus, where it functions as a transcription factor, stimulating the transcription of certain genes and repressing

the transcriptional activity of others. In the regulation of ion transport, stimulation of key target genes is paramount. Transcriptional repression may be essential for effects in nonepithelial cells, including neurons, cardiomyocytes, smooth muscle cells, and macrophages.³⁴

DOMAIN STRUCTURE OF MINERALOCORTICOID RECEPTORS

The MRs of all vertebrates are highly conserved. There are only minor differences between MRs in rodents and MRs in humans.³⁴ In general, the steroid and nuclear receptors have been divided into three major domains (Fig. 12.5):

1. An N-terminal transcriptional regulatory domain
2. A central DNA-binding domain (DBD)
3. A C-terminal ligand/hormone-binding domain (LBD)

Each of these broadly defined domains has more than one function, and not all the functions can be neatly assigned to separate distinct domains; however, much of the action

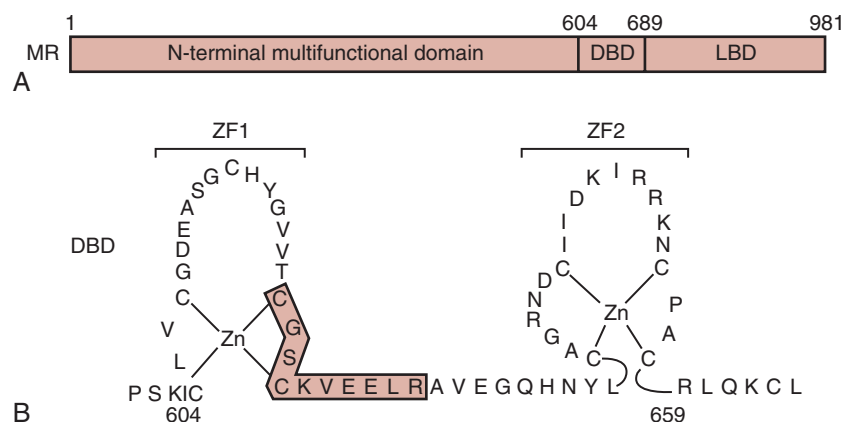


Fig. 12.5 Domain structure of the mineralocorticoid receptor (MR). Three major domains have been defined, which are common to all steroid and nuclear receptors. Further refinements have led some to use a six-letter system, which is shown; however, this adds little to the understanding of receptor structure or function, and we prefer the three-global domain system. These large receptor sections should not be confused with the many small functional domains that have been identified, as discussed in the text. The size and amino acid designations used here are for rat MR (981 amino acids total); they apply with minor variations to human MR (984 amino acids total). (A) Strip diagram of MR. The N-terminus is to the *left*, C-terminus to the *right*. (B) Schematic of MR DNA-binding domain (DBD), showing the two zinc fingers and the positions of the coordinating Zn ions. *Boxed region* is the alpha helix, which intercalates into the major groove of the DNA and provides the major protein–DNA interaction contacts. The dimerization interface comprises amino acids within the second zinc finger, which form van der Waals and salt bridge interactions. *LBD*, Ligand/hormone-binding domain.

of MRs can be understood from this point of view. In the following sections, the domains are described roughly in the historical order in which they were characterized, which also parallels the clarity of functional and structural knowledge about them.

DNA-BINDING DOMAIN

The essential quality of MR function as a transcription factor is its ability to bind specifically to DNA. This protein–DNA interaction is mediated by the receptor's compact modular DBD (amino acids 603–688 of human MR; Fig. 12.5 shows a strip diagram and two-dimensional structure), which forms a variety of contacts with a specific 15-nucleotide DNA sequence termed a “hormone response element” (HRE). Receptor binding to the HRE in the vicinity of regulated genes promotes the recruitment of coactivators and components of the general transcription machinery, such as the TATA-binding protein, which binds to the thymidine- and adenosine-rich DNA sequence found upstream of many genes and is required for correct transcription initiation. These types of HREs have been identified near or in many of the key MR-regulated genes, such as serum- and glucocorticoid-regulated kinase 1 (*SGK1*), glucocorticoid-induced leucine zipper (*GILZ*), and amiloride-sensitive sodium channel subunit α (α -*ENaC*). Although, in many cases, differential binding to HREs is a key determinant of the specificity of many transcription factors, it should be noted that some steroid receptors (notably MR and GR) have only minor (<10%) differences within this domain and have identical DNA-binding properties.³⁵ Specificity in these cases is determined through other mechanisms.^{34,36}

The canonical MR HRE is a 15-nucleotide sequence that forms a partial palindrome (inverted repeat), which binds a receptor homodimer. A dimer interface embedded within the DBD is essential for MR to form these requisite homodimers, as well as to form heterodimers with GR.^{37,38} Mutations that disrupt this interface have complex effects on receptor

activity in animals³⁹ and cultured cells,⁴⁰ and similar mutations in other receptors (AR in particular) result in disease states.⁴¹ Also, in at least one kindred of the autosomal dominant form of pseudohypoaldosteronism type I, an MR DBD mutation appears to be causative, although the mechanistic basis has not been elucidated.⁴²

In addition to supporting DNA binding and dimerization, the DBD also harbors a nuclear localization signal,^{43,44} as well as surfaces that contact distant parts of the receptor and that mediate interactions with other proteins, as has been shown for GR and, in some cases, MR.^{45–47}

LIGAND/HORMONE-BINDING DOMAIN

The MR LBD comprises amino acids 689 to 981 (see Fig. 12.5A). Like the DBD, the LBD has multiple functions; in addition to binding with high affinity to various MR agonists and antagonists, it also harbors interaction surfaces for coactivators, dimerization, and N–C interactions.^{48–50} MR is distinct from GR in that it binds with equally high affinity to cortisol, corticosterone (the physiologic glucocorticoid in rats and mice), and aldosterone. Indeed, as discussed later, MR appears to function as a high-affinity glucocorticoid receptor in some tissues, including the brain and heart.

High-resolution representations of the crystal structures of wild-type and mutant MR have identified the structural features of the LBD and specific amino acid contacts involved in binding to the mineralocorticoid desoxycorticosterone (Fig. 12.6).^{51,52} Key features include the following:

1. The LBD of MR, like that of other nuclear receptors, is arranged into 11 α -helices and four small β -strands.
2. The C-terminal alpha helix (α -helix), H12, contains the activation function AF2.
3. Ligand is deeply embedded into a pocket comprising α -helices H3, H4, H5, H7, and H10 and two beta strands (β -strands); numerous contacts are made between amino acids of the pocket and hormone.

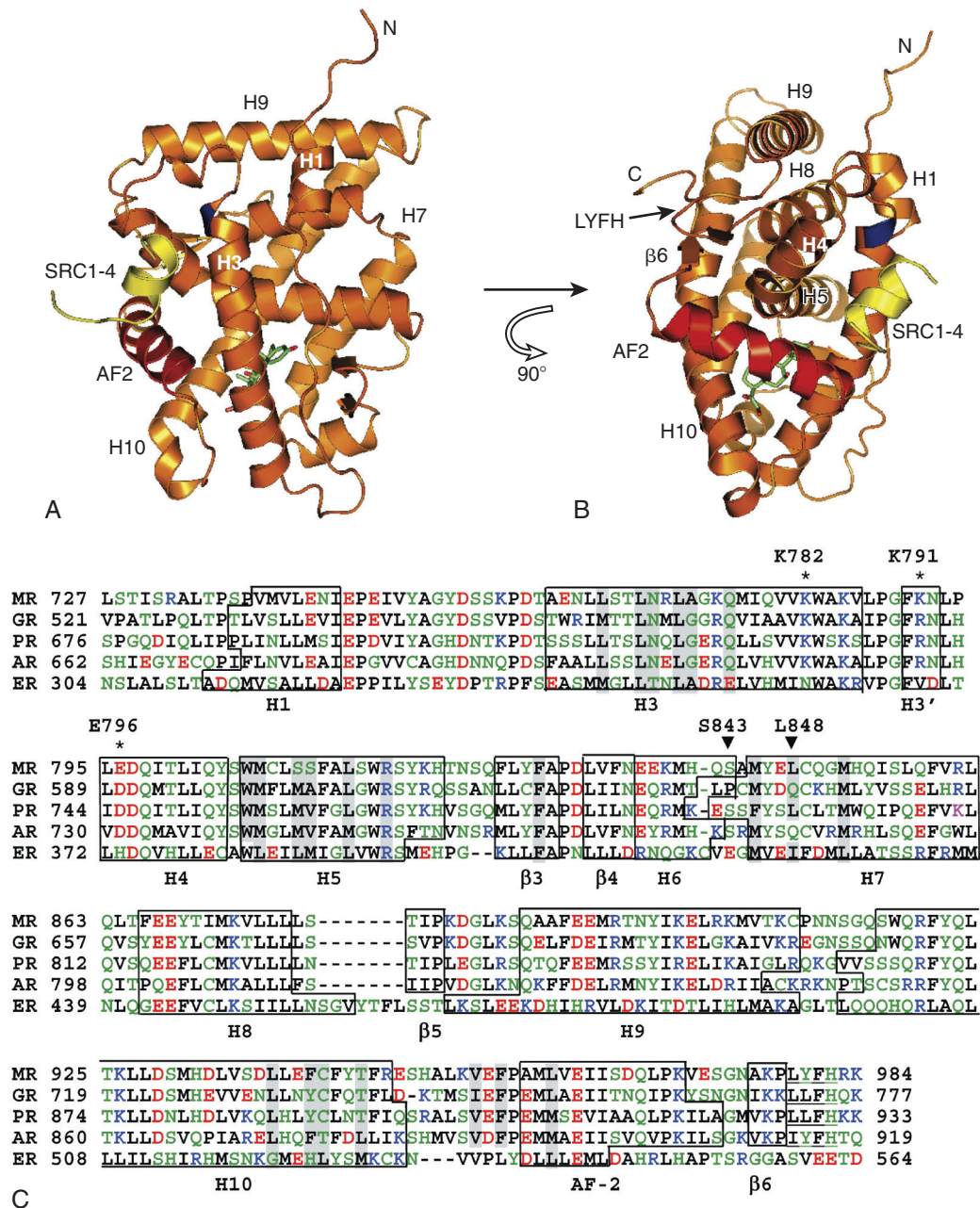


Fig. 12.6 Mineralocorticoid receptor (MR) ligand/hormone-binding domain (LBD) crystal structure. Structure of the MR LBD bound to corticosterone and coactivator peptides (SRC1-4). (A and B) Two views of the complex (rotated by 90 degrees about the vertical axis) are shown in this ribbon representation. MR is colored in *gold* and SRC1-4 peptide in *yellow*. Corticosterone, which binds MR with an affinity comparable with that of aldosterone, is shown in a *ball-and-stick* representation. Note that hormone is located in a deep pocket formed by helices 3, 5, and 7 (H3, H5, and H7), which explains the slow off-rate and high affinity. (C) Sequence alignment of the human MR LBD with other steroid hormone receptors (GR, glucocorticoid; AR, androgen; PR, progesterone; and ER, estrogen). Residues that form the steroid-binding pockets are shaded in *gray*. Key structural features for the binding of SRC peptides are noted with *stars*, and the residues that determine MR-GR hormone specificity are labeled by *arrowheads*. See Li and others³⁹⁴ for further details. (From Li Y, Suino K, Daugherty J, et al. Structural and biochemical mechanisms for the specificity of hormone binding and coactivator assembly by mineralocorticoid receptor. *Mol Cell*. 2005;19:367–380.)

This accounts for the slow off-rate and high affinity of aldosterone, corticosterone, and cortisol for MR.⁵¹ The crystal structure of the mutant (S810L) MR, in which progesterone acts as an agonist rather than an antagonist,⁵³ reveals that H12 is stabilized with AF2 in the active conformation.⁵² The crystal structure of wild-type MR LBD also provides insight into the mechanisms underlying some forms of pseudohy-

poaldosteronism type I. Notably, MR/S810L has an LBD mutation in helix 5, which is predicted to disrupt interaction with the steroid ring structure,⁵⁴ whereas Q776R and L979P have been demonstrated to have markedly reduced aldosterone binding.⁴² Structural analysis reveals that Q776 is located in helix H3 at the extremity of the hydrophobic ligand-binding pocket and anchors the steroid C3-ketone group.

MR binds cortisol and corticosterone with an affinity similar to that of aldosterone. 11β -HSD2 is an essential determinant of aldosterone specificity, through its effect in metabolizing glucocorticoids to their receptor-inactive keto-congeners in collecting duct principal cells, as discussed later. In tissues that do not coexpress 11β -HSD2, the physiologic ligand for MR is cortisol (corticosterone in rats and mice).^{55–57} The extent to which such “unprotected” MR can be pathophysiologically activated in aldosterone excess states is discussed later (see “Disease States: Primary Aldosteronism” and “Nonepithelial Actions of Aldosterone”).

N-TERMINAL DOMAIN

As its noncommittal name implies, the N-terminal region of MR has diverse functions, which appear to revolve primarily around protein–protein interactions and the recruitment of coactivators and corepressors. It has two potent transcriptional regulatory motifs, usually termed *AF1a* and *AF1b*.^{48,58} This domain bears some functional and sequence similarity to the homologous region of GR and is capable of stimulating gene transcription when fused to an unrelated DBD.⁴⁸ Overall, however, MR and GR differ markedly in the N-terminal domain, and this region of the receptor is a central determinant of specificity.⁵⁹ Other studies have supported the idea that this domain has functional sequences that limit receptor activity through the recruitment of corepressors, in addition to transcriptional activation functions.^{47,60} Its role in coactivator and corepressor recruitment is addressed further in the following section.

MINERALOCORTICOID RECEPTOR REGULATION OF TRANSCRIPTION INITIATION: COACTIVATORS AND COREPRESSORS

The major mechanism of MR action is its effects on transcription initiation; however, there may also be effects on transcript elongation.^{61,62} Much has been learned about the generation of an initiation complex and the particular roles that steroid receptors play in this process. Several review articles and book chapters have provided in-depth examinations of the biochemistry of the general transcription machinery, transcription initiation, promoter escape, and processive elongation.^{63–65} Most of the coactivators identified so far interact with the C-terminal AF2 domain and include the prototypical GRIP1/TIF2 and SRC,^{66,67} which sequentially recruit a series of different components of the transcriptional machinery and result in the formation of a preinitiation complex (PIC). This PIC includes all the key components of the transcription machinery, including RNA polymerase II. A detailed picture of MR-dependent PIC formation has not been determined. However, the general features are likely similar to those for ER⁶⁸ and involve the sequential recruitment by the receptor of the following: (1) chromatin-remodeling SWI/SNF and CARM1/PRTM1 proteins, which promote chromatin remodeling and initiation of complex formation; (2) histone acetylase CBP/P300 (cyclic adenosine monophosphate [cAMP]–responsive element-binding protein), which promotes an active chromatin conformation⁶⁴; and (3) direct or indirect recruitment of the TATA-binding protein and other components of the general transcription machinery.⁶⁸

The aforementioned mechanisms are generic and are used by many transcription factors, including all steroid receptors,

through interactions with the C-terminal AF2 domain. The N-terminal region of MR, which harbors the AF1 domain, diverges from the other steroid receptors, and other studies have identified coregulators that interact selectively with this receptor domain. ELL (11-19 lysine-rich leukemia factor) is a coactivator for MR that specifically interacts with AF1b and assists in PIC formation.⁶¹ It was originally identified as an elongation factor, and it may also affect transcript elongation. Other specific coregulators include the synergy inhibitory protein PIAS1,⁶⁰ Ubc9,⁶⁹ and p68 RNA helicase.⁷⁰ In many cases, interactions of these regulators with MR require receptor posttranslational modifications—for example, by phosphorylation, acetylation, or sumoylation.⁷¹

REGULATION OF SODIUM ABSORPTION AND POTASSIUM SECRETION

GENERAL MODEL OF ALDOSTERONE ACTION

Aldosterone enters the cell passively, binds to MR, triggers changes in gene transcription (as addressed later; see “Mechanisms of Mineralocorticoid Receptor Function and Gene Regulation”), and potentially has nongenomic effects. Aldosterone effects in the ASDN have been divided into three major phases: latent, early, and late.¹ This designation goes back to the early observations by Ganong and Mulrow that after aldosterone infusion into experimental animals, no effect was observed for at least 15 to 20 minutes.⁷² A similar delay was observed in isolated epithelia.⁷³ The early phase, which is now known to involve primarily MR-dependent regulation of signaling mediators such as SGK1, culminates in increased apical localization—and, possibly, increased probability of the open state—of ENaC. In the late phase, aldosterone stimulates transcription of a variety of effector genes, including those that encode components of the ion transport machinery, notably the epithelial sodium channel (ENaC) and Na^+ - K^+ -adenosine triphosphatase (Na^+ - K^+ -ATPase) subunits. The major direct effect is to increase Na^+ reabsorption, which is accompanied variably by Cl^- reabsorption and/or K^+ secretion and, ultimately, water reabsorption. Aldosterone’s actions in the principal cells of the connecting segment and collecting duct (Fig. 12.7) are of primary significance; however, this also has been shown to influence fluid and electrolyte transport in other tubule segments, as well as in other organs. These actions of aldosterone can be surmised from the clinical features of individuals with aldosterone-secreting tumors; they have volume expansion with high blood pressure and are commonly (>50% of patients) hypokalemic.^{74,75} In general, the effects of aldosterone on Na^+ absorption and K^+ secretion work together. However, there are ways whereby these actions can be separated, as discussed later.

The two basic cell processes that aldosterone regulates— Na^+ absorption and K^+ secretion—are depicted in Fig. 12.7. Most aspects of this mechanism are relevant to the various aldosterone target tissues.

Na^+ - K^+ -ATPase, located on the basolateral membrane (blood side), establishes the essential electrochemical gradients that drive ion transport (see Chapters 5 and 6). Importantly, it operates well below its V_{max} , and is seldom, if ever, the rate-limiting step in transepithelial Na^+ transport.⁷⁶ Rather, apical

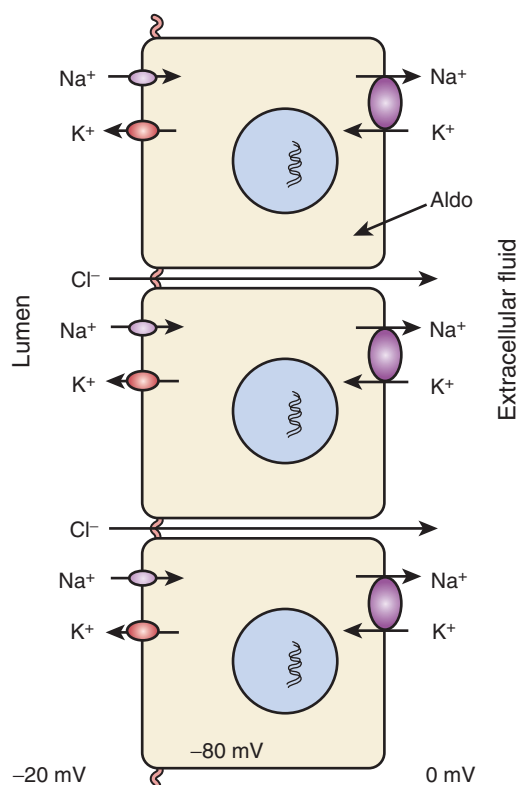


Fig. 12.7 Schematic of principal cells in the aldosterone-sensitive distal nephron (ASDN). The ASDN includes the distal third of the distal convoluted tubule (DCT), connecting tubule, and collecting duct. The $\text{Na}^+\text{-K}^+$ -adenosine triphosphatase ($\text{Na}^+\text{-K}^+\text{-ATPase}$) establishes the gradients for passive apical entry of sodium through the epithelial sodium channel (ENaC). Transport of sodium through the ENaC creates a negative lumen potential that drives potassium secretion into the lumen. Potassium is also recycled at the basolateral surface, which facilitates potassium exchange across the $\text{Na}^+\text{-K}^+\text{-ATPase}$. Chloride (Cl^-) moves via paracellular and transcellular pathways. There is evidence to support aldosterone (*Aldo*) actions in other segments, particularly the sodium-chloride cotransporter-expressing portions of the DCT (DCT1 and DCT2).

Na^+ entry into the cell via the epithelial Na^+ channel, ENaC, is the rate-limiting step for Na^+ reabsorption by the ASDN and the key locus of regulation.

The discovery of the molecular composition of ENaC in 1993^{77,78} opened the door to understanding how aldosterone functions to regulate this critically important ion channel. Most Na^+ transporters are encoded by a single gene product. In contrast, ENaC is composed of three similar but distinct subunits, each encoded by a unique gene. All three subunits come together (probably as a heterotrimer) to form an ion channel with unique biophysical characteristics, the most striking of which is the relatively long time it stays open or closed.⁷⁹ The complete loss of any one of these subunits in mice is incompatible with life,^{80–82} and mutations in channel subunits cause profound disease manifestations in humans.⁸³ The apical entry of Na^+ into the cell via ENaC is the rate-limiting step in both Na^+ absorption and K^+ secretion.⁸⁴ Na^+ enters the cell down a steep electrochemical gradient; intracellular $[\text{Na}^+]$ is approximately 10 mmol/L, and the

membrane voltage is high (inside negative). Intracellular Na^+ is pumped out across the basolateral membrane by $\text{Na}^+\text{-K}^+\text{-ATPase}$, as addressed in detail in Chapter 6. Most epithelial cells have a greater density of K^+ channels on the basolateral membrane and thus recycle K^+ back into the blood. The distal nephron is unique in that it has an unusually high density of K^+ channels on the apical membrane (primarily Kir 1.1 [renal outer medullary potassium (ROMK)] and BK channels) relative to the apical membrane of other epithelia.^{85,86} This distribution of K^+ channels permits a large amount of K^+ that enters the cell via $\text{Na}^+\text{-K}^+\text{-ATPase}$ to exit the cell into the lumen and be excreted into the urine. The vast majority of K^+ that appears in the urine is secreted by the distal nephron.

Much attention has been focused on the early phase of aldosterone action because it appears to be more tractable to dissection, and most changes in Na^+ current occur during this phase. This separation is probably somewhat artificial, however, and there is considerable overlap in events that define the early and late phases. Moreover, many efforts to manipulate mediators of the early phase (through overexpression and knockdown experiments) have been evaluated after prolonged alteration. Nevertheless, there is some heuristic value in considering the early and late phases of aldosterone action separately.

In cultured collecting duct cells deprived of corticosteroids and then exposed to high concentrations of aldosterone, an increase in ENaC-mediated Na^+ transport can be observed in well under 1 hour, which is consistent with animal studies.^{72,87} Na^+ transport continues to increase for 2 to 3 hours, then plateaus for a few hours, and then gradually increases over the next several hours. After 12 hours of exposure to saturating aldosterone concentrations, the increase in ENaC activity is near maximal. The molecular basis for this increase in ENaC activity has been intensively investigated, and several key events are now apparent.

For aldosterone to increase ENaC activity, a change in gene transcription must occur. One of the earliest response genes is *SGK1*.^{88–90} This serine–threonine kinase, which mediates a substantial portion of the early effects of aldosterone,^{91,92} is addressed in greater detail later in this chapter, together with its major target, Nedd4-2. The genetic disease Liddle syndrome provided key first clues to ENaC, as addressed further later^{93,94} and in Chapter 44.

ALDOSTERONE AND EPITHELIAL SODIUM CHANNEL TRAFFICKING

The major action of aldosterone is to increase the number of functional ENaC units on the apical membrane. This process can involve an increase in the number of channel complexes on the surface, activation of existing complexes, or both. There is evidence to support both, although the bulk of evidence favors the idea that a change in the number of ENaC units predominates.^{95–97} The redistribution of ENaC to the apical membrane can be detected in less than 2 hours after aldosterone exposure.⁸⁷

It is less well established whether the number of channels is increased through increased insertion, decreased removal, or both. Aldosterone probably contributes to both processes. Rapid insertion of ENaC is best understood with regard to the actions of cAMP.⁹⁸ The extent to which the molecules

involved in cAMP-mediated insertion are also involved in aldosterone action is uncertain, but some common mechanisms are probably used. Trafficking to the apical membrane appears to involve hsp70,⁹⁹ SNARE (soluble NEM-sensitive factor attachment protein receptor) proteins,¹⁰⁰ and the aldosterone-induced protein melanophilin.¹⁰¹ The mitogen-activated protein kinase pathway may also be involved, because interruption of ERK phosphorylation by GILZ¹⁰² increases ENaC surface expression.

Considerably more is known about how ENaC complexes are retrieved from the apical membrane. This understanding is the direct result of dissecting the molecular consequences of Liddle syndrome, in which mutations in the C terminus of ENaC lead to increased residence time in the apical membrane.^{93,94} The missing or mutated domains in the β - or γ -subunit of ENaC in this syndrome normally bind to Nedd4-2, a ubiquitin ligase, which ultimately is responsible for initiating endocytosis and degradation.^{103,104} The interaction of Sgk1 and Nedd4-2 in the actions of aldosterone is discussed later. ENaC is internalized via clathrin-coated vesicles, processed into early endosomes, and then further processed into recycling endosomes and late endosomes.^{105,106} Degradation is via lysosomes or proteasomes.^{107,108} The processing of ENaC by vesicular trafficking and its regulation by aldosterone has been reviewed by Butterworth et al.¹⁰⁹

Phosphatidylinositol-3-kinase (PI3K)-dependent signaling is essential for epithelial Na^+ transport. It controls SGK1 activity (see later) and also appears to have independent effects on ENaC open probability through direct actions of 3-phosphorylated phosphoinositides, particularly phosphatidylinositol (3,4,5)-trisphosphate.^{110,111} Ras-dependent signaling may also regulate ENaC and the pump in complex ways that depend on downstream signaling through Raf, MEK, and ERK, as well as through PI3K.^{112–117}

The late phase of ENaC activation by aldosterone is less well understood than the early phase. A simple evaluation of the late phase is that aldosterone increases the transcription and protein abundance of the ENaC α -subunit. This idea comes from the fact that aldosterone increases the mRNA and protein abundance of α -ENaC in the kidney^{96,118} after a lag of several hours.¹¹⁹ Although less well studied, aldosterone appears to produce an increase in β - and γ -subunit expression in the colon.^{119,120} Dietary Na^+ restriction, a physiologically relevant maneuver that increases aldosterone secretion, clearly increases ENaC surface expression in the renal distal nephron.⁹⁷ However, there appear to be some important differences between chronic aldosterone administration to a Na^+ -replete animal and chronic dietary Na^+ restriction.^{118,121} Furthermore, it should be noted that increased α -ENaC expression, in and of itself, does not increase ENaC activity in models of collecting duct and lung epithelia, although limiting its expression does restrict aldosterone stimulation.¹²² It appears that increased expression of α -ENaC may be important for the consolidation of the increase, but it is not sufficient to reproduce the steroid-mediated increase in ENaC activity.

BASOLATERAL MEMBRANE EFFECTS OF ALDOSTERONE

Over the years, research on aldosterone action has focused with varying degrees of intensity on apical effects,^{123,124}

basolateral effects,^{125–127} and effects on metabolism.⁷³ There is general agreement now that the early effects of aldosterone are on apical events, primarily on ENaC, and that basolateral and metabolic effects occur later. In addition, although it is somewhat less settled whether the basolateral effects are direct or result indirectly from the enhanced entry of Na^+ into cells, the bulk of evidence favors the latter view. Notably, increased Na^+ entry has been found to control more than 80% of increased Na^+ - K^+ -ATPase activity and basolateral membrane density in the rat^{128,129} and rabbit¹²⁹ cortical collecting tubules. Furthermore, striking increases in basolateral membrane folding and surface area occur in aldosterone-treated animals,¹³⁰ an effect that is markedly attenuated in animals fed a low- Na^+ diet. This result strongly suggests that apical Na^+ entry is required for basolateral changes to occur. However, there is good evidence for direct transcriptional stimulation of Na^+ - K^+ -ATPase subunit expression,^{131,132} as well as reports supporting some direct effects of aldosterone in increasing basolateral pump activity^{125,133} or at least in constituting the pool of latent pumps, which are then recruited to the basolateral membrane in response to a rise in intracellular $[\text{Na}^+]$.¹³⁴

ACTIVATION OF THE EPITHELIAL SODIUM CHANNEL BY PROTEOLYTIC CLEAVAGE

There is now clear evidence that when ENaC is delivered to the apical membrane, it can be activated by proteolytic cleavage. The first hint of this process was the demonstration that rats fed a low- Na diet or given aldosterone over the long term showed the appearance of a proteolytic fragment of the γ -ENaC subunit.⁹⁶ Subsequently, investigators have shown that both the α - and the γ but not the β -ENaC subunits can be cleaved. Furthermore, cleavage at each site apparently initiates a degree of activation of the channel complex. ENaC complexes are activated by cleavage because specific regions of the large extracellular domain (26 residues in the α -ENaC and 46 residues in the γ -ENaC) are excised. These regions contain inhibitory sequences that when introduced exogenously, can inhibit ENaC function. Removing these regions by proteolytic cleavage releases this inhibition.¹³⁵

Several proteases can cleave either the α - or γ -ENaC subunits. Among them are furin, prostatic, CAP2, kallikrein, elastase, matriptase, plasmin, and trypsin. It is not clear whether activation of ENaC by proteolytic cleavage can be regulated by aldosterone, but the idea certainly has attractive features. If aldosterone could regulate expression of one or more rate-limiting proteases, it would be able to regulate both the number of complexes in the apical membrane and the ability of the channel complex to be active. It appears that aldosterone may regulate the expression of prostatic.¹³⁶ Aldosterone may also regulate the expression of the protease nexin-1 (an inhibitor of prostatic) and other proteases.¹³⁷

The discovery of ENaC activation by cleavage helps explain how aldosterone might increase ENaC activity by increasing both surface expression and the activity of a single ENaC complex. By phosphorylating Nedd4-2 via SGK1 and reducing its ability to bind to the PY domains of the ENaC subunits, aldosterone increases ENaC residence time on the apical membrane. This additional time permits proteolytic activation by one or more endogenous proteases.¹³⁸

POTASSIUM SECRETION AND ALDOSTERONE

One of the major effects of aldosterone is to increase K^+ secretion (and thus excretion). This phenomenon has been demonstrated in countless patients with aldosterone-secreting tumors and in hundreds of studies in animals given excess amounts of aldosterone. The general mechanism whereby aldosterone increases K^+ secretion is depicted in Fig. 12.7. The key feature of this process—with respect to the direct effects of aldosterone *per se*—involves the stimulation of Na^+ absorption via ENaC. The dependence of K^+ secretion on Na^+ absorption is the basis of the action of the so-called K-sparing diuretics, amiloride and triamterene, both of which inhibit ENaC. These drugs have no direct effect on apical K^+ channels.

Increasing ENaC activity produces two major secondary effects that in turn enhance K^+ secretion. First, the enhanced Na^+ conductance of the apical membrane produces depolarization and hence a more favorable electrical driving force for K^+ efflux into the lumen. The second effect relates to the activity of the Na^+ - K^+ pump on the basolateral membrane. The more Na^+ that enters across the apical membrane, the more that must be extruded by the pump. Because the pump operates well below its V_{max} under baseline conditions, a slight increase in intracellular Na^+ concentration markedly stimulates pump activity and more K^+ enters the cell. In isolated, perfused cortical collecting ducts, the amount of secreted K^+ is highly related to the amount of absorbed Na^+ when the stimulus for Na^+ absorption is mineralocorticoid hormone.¹³⁹

Two types of K^+ channels are found in the apical membrane of the ASDN: small conductance (SK, 30–40 picosiemens [pS]) channels encoded by the *ROMK* gene, and large conductance (BK, 100–200 pS) channels found in many other cell types, including the apical membrane of the colon. Most of the K^+ channels on the apical membrane of the principal cells appear to be SK, at least as far as can be assessed by patch-clamp analysis. The activity of either channel is not directly increased by aldosterone.^{140,141}

A feature of K^+ secretion is that although apical K^+ channels are abundant in the proximal portion of the ASDN (connecting tubule and cortical collecting duct), they are strikingly less abundant in the medullary collecting duct.^{142–144} Because apical K channels are not regulated by aldosterone, their absence in the medullary collecting duct might uncouple aldosterone-regulated Na^+ reabsorption from K^+ secretion in this segment.

SEPARATION OF SODIUM ABSORPTION AND POTASSIUM SECRETION BY THE ALDOSTERONE-SENSITIVE DISTAL NEPHRON

The preceding sections establish a picture that parsimoniously accounts for the effect of aldosterone to stimulate Na^+ reabsorption and K^+ secretion at the same rate. The simple stimulation of electrogenic Na^+ reabsorption (via ENaC) is sufficient to stimulate K^+ secretion, which fits well for organisms faced with a combined low- Na^+ , high- K^+ diet, which was maintained most of the time through millions of years of vertebrate evolution. However, organisms do not ingest a fixed amount of Na^+ and K^+ , so an inexorable linkage between Na^+ absorption and K^+ secretion by the ASDN cannot possibly

occur all the time. Investigators have proposed several possibilities to explain how these processes can be separated.

ROLE OF DISTAL TUBULE FLUID DELIVERY

A traditional view for differential Na^+ and K^+ handling stems from the differing role that aldosterone plays in potassium secretion depending on the tubular flow (and thus sodium flow) rate. Studies from adrenalectomized dogs have demonstrated that the primary regulators of potassium excretion are the serum potassium concentration and tubular flow rate. A higher serum potassium concentration yields a higher filtered load of potassium. Hyperkalemia also stimulates natriuresis from upstream segments of the nephron.^{145,146} This latter effect can increase tubular flow rate, which, in turn, diminishes potassium concentration in the lumen and activates flow-stimulated BK channel-mediated potassium secretion in the collecting duct. In the setting of sufficient distal delivery of sodium, potassium loading will not yield a higher steady-state concentration of aldosterone because the two mechanisms previously mentioned are sufficient to normalize the serum potassium level.¹⁴⁷ However, under conditions of sodium depletion, proximal Na^+ reabsorption is increased, which further diminishes distal delivery of sodium and hence tubular flow rate.¹⁴⁸ This diminishes flow-mediated potassium secretion, so that aldosterone secretion is necessary to normalize potassium balance.

INDEPENDENT REGULATION OF SODIUM AND POTASSIUM TRANSPORTERS

Other possible mechanisms have been suggested, which involve separate regulation of sodium and potassium transport (e.g., ENaC and ROMK) by specific stimuli, depending on the state of Na^+ and K^+ intake. With a constant Na^+ intake, one could envision that a high-potassium diet could enhance the activity of ROMK, whereas a low-potassium diet would reduce its activity. Such an effect would cause more or less of the K^+ entering the cell via the Na^+ - K^+ pump to be recycled across the basolateral membrane. This mechanism, although probably very complex in its execution, is appealing in its simplicity.

ROLE OF WNKs

Advances in understanding genetic forms of hypertension have uncovered a key functional role for a family of kinases that have potent effects on pathways regulating Na^+ and K^+ transport in the distal nephron. These kinases, called “WNKs” (*with no lysine*; K is the one-letter code for lysine), were initially thought to lack a conserved lysine residue in the catalytic domain, which binds adenosine triphosphate (ATP), and is essential for catalysis. Interestingly, a lysine performing the same function is indeed present, but is shifted from its canonical location in subdomain II to subdomain I to permit entry of chloride into the catalytic domain.¹⁴⁹ This modification contributes to the chloride regulation of WNK activity.

The spectrum of regulatory roles of WNKs is complex and remains controversial. It is clear that the Na^+ , Cl^- cotransporter (NCC) is regulated by WNK4, and it is likely that ROMK and ENaC are regulated by WNK1.^{150,151} Recent evidence has also supported the idea that the chloride-binding properties of WNKs provide the basis for their functional regulation by extracellular K^+ concentration.¹⁵² In the DCT, in particular,

low plasma (and hence extracellular) K^+ concentration depolarizes the basolateral membrane via Kir4.1, which leads to a reduction in intracellular chloride, thereby activating WNK4 and SPAK (STE20/SPS1-related proline-alanine-rich kinase) and hence NCC phosphorylation.^{152,153} A high plasma K^+ concentration has the opposite effect and inhibits NCC phosphorylation; however, the mechanism appears to be chloride-independent. Thus, through its effects to raise the plasma K^+ concentration, a high- K^+ diet has effects both locally in the kidneys and through aldosterone to stimulate K^+ excretion. Aldosterone acts directly in the ASDN to stimulate ENaC, which increases the driving force for K^+ secretion through ROMK and BK channels.¹⁵⁴ In conjunction, an elevated K^+ concentration acts directly in DCT cells to inhibit NCC phosphorylation and enhance delivery of Na^+ to the ENaC-expressing segments. According to this view, aldosterone does not have a direct effect on NCC-mediated electroneutral Na^+ transport. Further details of the mechanisms of K transport and the role of WNKs in the distal nephron are presented in Chapter 6.

ROLE OF CHLORIDE TRANSPORT REGULATION

There is also evidence supporting the independent regulation of Cl^- transport in the collecting duct. Cl^- can be absorbed by the paracellular pathway (i.e., between cells) driven by the lumen-negative voltage across the epithelium. This pathway can be influenced by aldosterone.¹⁵⁵ Cl^- can also be absorbed through the cells by specific transporters. One example of a Cl^- transporter in the collecting duct is pendrin, an anion exchanger present on the apical membrane of intercalated cells. Mice that lack this transporter do not tolerate NaCl restriction as well as normal mice.¹⁵⁶ Its activity is dependent on Cl^- delivery to the distal nephron, and it is coregulated by angiotensin II and aldosterone.^{157,158,159}

As discussed in the next section, the ability of the MR in intercalated cells (but not in principal cells) to respond to aldosterone is regulated by the phosphorylation of its ligand-binding domain. It should also be noted that modulation of Na^+ absorption in the medullary collecting duct may also play a role in the balance of Na^+ reabsorption and K^+ secretion; this segment has little capacity to secrete K^+ , and endogenous paracrine factors such as prostaglandins E_2 and transforming growth factor- β , which have potent inhibitory effects on Na^+ transport, are increased in response to a high-NaCl diet.^{160,161}

DIFFERENTIAL REGULATION OF INTERCALATED CELL MINERALOCORTICOID RECEPTOR

It has become increasingly clear that intercalated cell Cl^- transport contributes to collecting duct NaCl absorption, and this plays an important role in allowing distinct responses to aldosterone in states of volume depletion versus hyperkalemia.^{162,163} A central feature of this proposed regulation is differential phosphorylation of the MR LBD in intercalated cells, as shown schematically in Fig. 12.8. When phosphorylated at S843 in the LBD, MR cannot bind aldosterone (or cortisol) and thus cannot be activated. This phosphorylation, which is stimulated by hyperkalemia, occurs selectively in intercalated cells but not in principal cells. Angiotensin II, on the other hand, induces S843 dephosphorylation in intercalated cells, markedly increasing ligand binding and, therefore, activation. Intercalated cells are known predominantly to mediate H^+ transport; however, other studies have

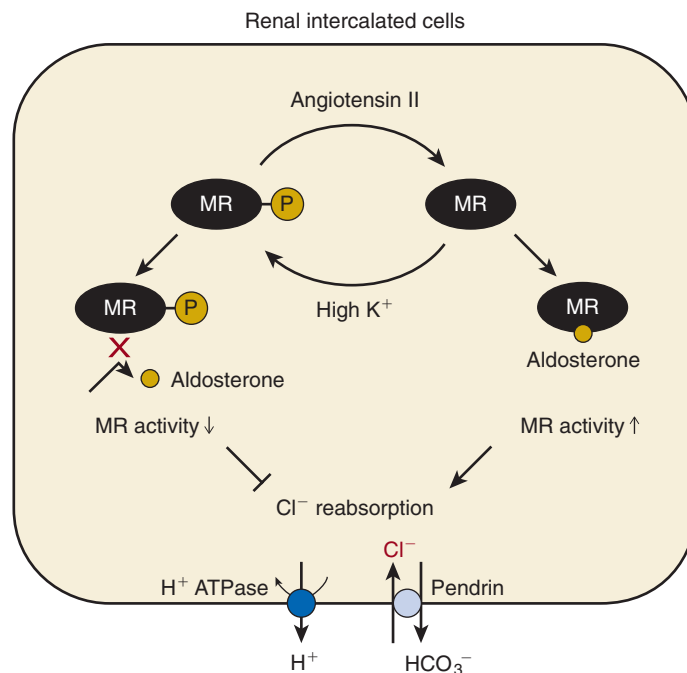


Fig. 12.8 The role of mineralocorticoid receptor (MR) ligand-hormone-binding domain (LBD) phosphorylation in controlling chloride reabsorption by intercalated cells. When phosphorylated at Ser-843 in the LBD, MR cannot bind ligand and hence cannot be activated. This phosphorylated state of MR is found only in intercalated cells, not in neighboring principal cells. In states of volume depletion, an elevated angiotensin II level decreases MR phosphorylation at Ser-843, allowing activation. In intercalated cells, MR mediates stimulation of both the proton pump and Cl^- - HCO_3^- exchangers, thereby increasing Cl^- reabsorption and promoting increased plasma volume while inhibiting K^+ secretion. In contrast, in states of hyperkalemia, phosphorylation of Ser-843 is increased, and hence Cl^- reabsorption by intercalated cells is decreased and the principal cell-dependent K^+ secretion is increased. (From Shibata S, Rinehart J, Zhang J, et al. Regulated mineralocorticoid receptor phosphorylation controls ligand binding and renal response to volume depletion and hyperkalemia. *Cell Metab.* 2013;18:660–671.)

implicated them in electroneutral NaCl transport via the combined actions of the Na^+ -dependent Cl^- - HCO_3^- exchanger (NDCBE)¹⁶⁴ and the apical Cl^- - HCO_3^- exchanger, pendrin.^{165,166} Thus, when MR is active in these cells (S843 dephosphorylated), electroneutral NaCl transport occurs, without enhancing the driving force for K^+ secretion. Because intercalated cells lack 11 β -HSD2 under these conditions, it is cortisol that binds to and activates MR. When intercalated cell MR is inactive (S843 phosphorylated), aldosterone acts in principal cells to stimulate ENaC-dependent electrogenic Na^+ transport, which enhances K^+ secretion.

ALDOSTERONE-INDEPENDENT ENAC-MEDIATED SODIUM REABSORPTION IN THE DISTAL NEPHRON

The term “aldosterone-sensitive distal nephron” emphasizes the primacy of this key steroid in the control of ion transport in this region of the nephron. However, ENaC activity and aldosterone sensitivity exhibit axial heterogeneity from the late distal convoluted tubule (DCT2) through the connecting

tubule to the cortical collecting duct and, finally, to the medullary collecting duct. In mice on a standard sodium diet, total ENaC expression increases with progression from the DCT2 to the connecting tubule,¹⁶⁷ although ENaC apical localization and activity are higher in the DCT2. Only under conditions of a low-sodium diet or aldosterone administration does the primacy of the connecting tubule in particular, and to a lesser extent the cortical collecting duct, emerge. The total luminal surface area in the DCT2 and connecting tubule is several-fold higher than in the cortical collecting duct,¹⁶⁸ and together these two segments appear to be sufficient to maintain sodium balance, even in the absence of detectable ENaC along the collecting duct. Mice lacking ENaC selectively in the collecting duct come into balance, even on a low-sodium diet.¹⁶⁹ Congruent with these findings, deletion of α -ENaC from the DCT2, connecting tubule, and collecting duct results in severe sodium wasting.¹⁷⁰ Notably, it is the connecting tubule that appears to be most important in the response to aldosterone, whereas DCT2 has the highest baseline transport in the absence of MR activation.¹⁷¹ The cortical collecting duct is not as critical as was originally thought for either baseline or aldosterone-stimulated sodium reabsorption, probably due to its smaller surface area compared with the DCT2 and connecting tubule.

As we continue to traverse the nephron, further sodium reabsorption is minimal in the medullary collecting duct, on a normal sodium diet, and is not significantly stimulated by aldosterone.¹³⁹

SITES OF MINERALOCORTICOID RECEPTOR EXPRESSION AND LOCUS OF ACTION ALONG THE NEPHRON

ALDOSTERONE-SENSITIVE DISTAL NEPHRON

In the kidney, MR is expressed at the highest levels in distal nephron cells extending from the last third of the DCT through the medullary collecting duct,¹⁷² which is frequently referred to as the ASDN (Fig. 12.9).¹ This pattern of expression was first demonstrated using labeled hormone-binding studies performed before the cloning of MR¹⁷³ and has been confirmed since by several methods, including the polymerase chain reaction assay,¹⁷⁴ in situ hybridization,¹⁷⁵ and immunohistochemical analysis.¹⁷⁶ Effects of aldosterone on electrogenic Na^+ and K^+ transport in principal cells have been found consistently in these nephron segments,¹⁷² which also express ENaC, and 11β -HSD2, as addressed in detail earlier.

Collecting duct intercalated cells also express MR and respond specifically to aldosterone and alter proton secretion. Aldosterone directly increases the activity of the H^+ -ATPase in the collecting duct, and its absence results in decreased proton secretion.^{177–179} Interestingly, nongenomic stimulation of H^+ -ATPase activity in type A intercalated cells has been demonstrated in isolated murine collecting ducts.¹⁸⁰ Consistent with these effects, aldosterone deficiency results in distal renal tubular acidosis type 4, and excess aldosterone results in metabolic alkalosis.¹⁸¹ It should be noted that aldosterone also stimulates H^+ secretion due to effects on principal cell Na^+ transport, which alter the electrical gradient. These older studies must also now be interpreted in the context of more recent data,¹⁶² which, as noted earlier, demonstrate that the effect of aldosterone on intercalated cells depends on the genesis of the signal.

OTHER SITES OF EXPRESSION

MR has been identified at some level in all parts of the nephron examined, including the glomerulus.^{174,182–187} Its effects, at least at some of these sites, are likely to be physiologically relevant in states of volume depletion and acid–base disturbances; however, the data are not as robust and consistent as those for the ASDN.

Glomerulus

MR (but not 11β -HSD2) is expressed in glomerular mesangial cells, where it is thought to affect proliferation and production of reactive oxygen species^{188,189} and to have profibrotic effects through SGK1.¹⁹⁰ These effects have been suggested to be important in the progression of renal damage, particularly in diabetic nephropathy,¹⁹¹ in which glucocorticoids mimic the activity of aldosteronism in the context of tissue damage. However, the physiologic role of mesangial cell MR is uncertain.

Proximal Convoluted Tubule

Hierholzer and Stolte have shown, through elegant microperfusion studies, that the sodium reabsorptive capacity of the proximal convolution is decreased in adrenalectomized animals and restored by administration of aldosterone.¹⁹² Chronic volume depletion increases sodium reabsorption in the proximal convoluted tubule, which is in part mediated by MR. The mechanisms of action in this nephron segment are controversial. Some studies have indicated an MR-dependent increase in the activity of Na^+ - H^+ -exchanger isoform 3, possibly through an increase in trafficking of the transporter to the membrane.^{193–196} This transporter contributes to sodium and bicarbonate reabsorption. MR activation, in turn, may activate the Na^+ - K^+ -ATPase in the basolateral membrane of the proximal convoluted tubule to maintain a gradient for sodium reabsorption.^{197–200}

Medullary Thick Ascending Limb

In the medullary thick ascending limb, mineralocorticoids but not glucocorticoids increase sodium and chloride reabsorption. In rodents, adrenalectomy impairs the reabsorption of NaCl in the medullary thick ascending limb, and aldosterone restores this process.^{201,202} This reabsorptive defect may contribute to the urinary concentrating and diluting abnormality measured in patients with Addison disease and in mice lacking aldosterone synthase.^{178,192,203} The medullary thick ascending limb also participates in the regulation of acid–base balance by reabsorbing most of the filtered HCO_3^- that is not reabsorbed by the proximal tubule. In this context, aldosterone has been shown to stimulate the Na^+ - H^+ exchanger in the amphibian thick ascending limb, possibly through a rapid nongenomic effect.²⁰⁴ Other studies have also implicated regulation of the Na - K - 2Cl cotransporter type 2 in the thick ascending limb—as well as the NCC in the DCT (see later)—by oxidative stress response kinase 1 (OSR1) and STE20/SPS1-related proline-alanine-rich kinase (SPAK) (OSR1/SPAK).^{205,206}

Distal Convoluted Tubule

The studies of potassium regulation described earlier have also provided insight into the role of direct or indirect aldosterone-induced NCC transport. Aldosterone increases

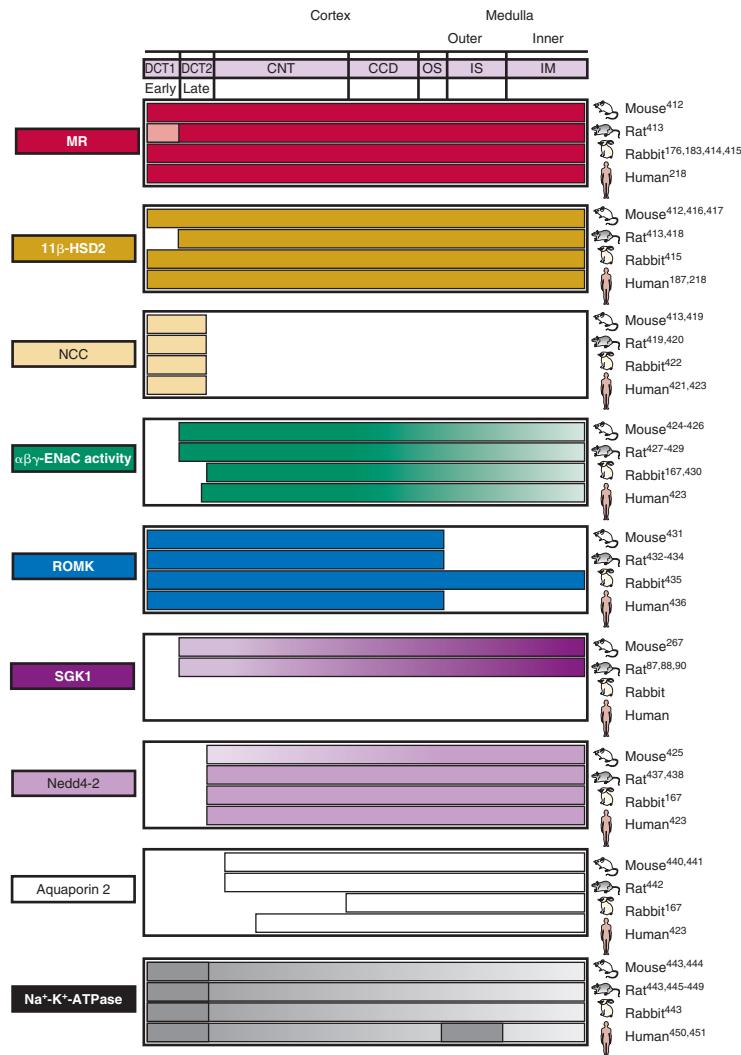


Fig. 12.9 Expression and/or activity of the mineralocorticoid-dependent transport machinery in principal cells along the mature aldosterone-sensitive distal nephron (ASDN). Mineralocorticoid specificity is conferred by the presence of the mineralocorticoid receptor (*MR*) and 11β-hydroxysteroid dehydrogenase type 2 (*11β-HSD2*), beginning primarily from the latter part of the distal convoluted tubule (*DCT*). The thiazide-sensitive sodium chloride cotransporter (*NCC*) is expressed exclusively in the DCT, but after the transition from the DCT to the connecting tubule (CNT), sodium reabsorption is distinctly determined by amiloride-sensitive sodium channel (*ENaC*) activity. *ENaC* activity is strongest in the CNT and decreases down to the inner medulla collecting duct. Variation in gene expression or activity along the nephron is indicated by the intensity of shading. Note that there is some variation in gene expression from mouse to human. However, the machinery for sodium reabsorption in the ASDN is predominantly conserved across species. Each nephron segment is drawn to scale, but expression of channels and transporters in intercalated cells is omitted. Expression and/or activity is based on messenger RNA, protein, and biochemical studies. G, Glomerulus; *PCT*, proximal convoluted tubule; *ROMK channel*, renal outer medullary potassium; *SGK1*, serum- and glucocorticoid-regulated kinase 1. (Modified from Loffing J, Korbacher C: Regulated sodium transport in the renal connecting tubule [CNT] via the epithelial sodium channel [ENaC]. *Pflügers Archiv*. 2009;458[1]:111–135.)

NCC phosphorylation and total protein²⁰⁷ abundance but, until recently, the mechanism of this upregulation was unclear. Elegant work by several groups have shown that aldosterone indirectly stimulates NCC via the activation of ENaC-mediated sodium transport and the resultant potassium secretion^{208,209} and hypokalemia. In turn, a lower plasma potassium level activates NCC.^{152,153} Another potential mechanism is through direct regulation by aldosterone. In rodent studies, aldosterone stimulates serum- and glucocorticoid kinase 1, which inhibits the ubiquitin ligase Nedd4-2, which, in turn, can regulate WNK1 and NCC phosphorylation.²¹⁰ As discussed later, this pathway mirrors a well-known mechanism of aldosterone-mediated disinhibition of Nedd4-2 and subsequent ENaC

degradation in principal cells. The specific physiologic contexts for these different direct and indirect modes of aldosterone-dependent NCC are still unknown.

NONRENAL ALDOSTERONE-RESPONSIVE TIGHT EPITHELIA

The mineralocorticoid effects of aldosterone have predominantly been studied in the distal nephron, but do influence other—mostly ENaC-expressing—tight epithelia. ENaC is present in visceral epithelial cells of the distal colon, distal lung, salivary glands, sweat glands, and taste buds.

COLON

Under physiologic conditions, approximately 1.3 to 1.8 L of electrolyte-rich fluid is reabsorbed per day from the colonic epithelium, which accounts for about 90% of the salt and water that enter the proximal colon from the terminal ileum. In nonmammalian vertebrates, sodium conservation by the colon plays an even more significant role.²¹¹ This transport is regulated by several transporters and channels, including ENaC. Like the nephron, the proximal colon reabsorbs sodium via an electroneutral, ENaC-independent process. In the distal colon, electrogenic Na⁺ absorption via ENaC channels is the predominant mode of sodium transport.^{212–215} In disease states such as inflammatory bowel disease, ENaC-mediated sodium reabsorption can be reduced,²¹⁶ although in diarrheal states, elevated aldosterone levels may attenuate sodium and water loss from the colon.²¹⁷ It should be noted that in the colon, as in the distal nephron, MR signaling is aldosterone selective, reflecting the activity of 11 β -HSD2.²¹⁸ Aldosterone increases electrogenic sodium absorption and potassium secretion and inhibits electroneutral absorption.²¹⁹ This is in contrast to glucocorticoids, which, at higher concentrations, activate GR to stimulate electroneutral absorption in the proximal and distal colon.^{84,220} As in the distal nephron, the aldosterone response can be characterized by an early and late response. The early response gene, *SGK1*, is upregulated by aldosterone via MR.²²¹ However, in contrast with the kidney, aldosterone and a low-salt diet have been shown to stimulate transcription of β -ENaC but not α -ENaC in rat models.^{222,223}

Aldosterone stimulates electrogenic potassium secretion from colonic epithelia. The significance of this secretion is evident in anuric patients. Potassium secretion from the colon is much higher in patients undergoing long-term hemodialysis than in patients not undergoing dialysis.^{224–226} Indeed, administration of fludrocortisone, a mineralocorticoid agonist, to dialysis patients has been shown to reduce hyperkalemia in small clinical trials.²²⁷ Low doses of the common MR antagonist spironolactone do not result in significant hyperkalemia.^{228–230}

LUNG

Vectorial transport of salt and water across the distal airway epithelium (ciliated Clara cells, nonciliated cuboidal cells) and alveoli (types I and II alveoli) primarily determines fluid clearance from the lung. ENaC is the rate-limiting step in sodium transport in the lung and plays a primary role in several physiologic and pathophysiologic conditions determined by fluid clearance.²³¹ At birth, the lung assumes a resorptive phenotype, and lack of functional ENaC channels leads to neonatal respiratory distress syndrome in mouse knockout models.²³² In children, lack of functional ENaC (e.g., autosomal recessive pseudohypoaldosteronism type I)^{233–235} results in increased rates of recurrent infection due to increased airway liquid.²³⁴ In the mature lung, defective ENaC channels can lead to pulmonary edema and pathologic conditions (e.g., acute respiratory distress syndrome,²³⁶ high-altitude pulmonary edema²³⁷). Conversely, hyperabsorption through ENaC is emerging as an important mechanism of decreased mucus clearance in cystic fibrosis.²³⁸

The molecular apparatus for mineralocorticoid-stimulated liquid reabsorption via ENaC (concomitant MR and 11 β -

HSD2) is present in late gestational and mature adult lung in humans^{218,239,240} and rats,²⁴¹ and there is some evidence for a significant physiologic role of aldosterone in ENaC-mediated sodium transport,²⁴¹ although glucocorticoids acting via GR are likely to play the predominant role in lung.^{118,221,242–245} Importantly, glucocorticoids, but not mineralocorticoids, play a critical role in lung maturation in humans, and GR knockout mice, like α -ENaC knockout mice, die of respiratory insufficiency within hours of birth. In contrast, MR knockout mice demonstrate a severe salt-wasting phenotype but no significant lung phenotype.^{177,246}

EXOCRINE GLANDS AND SENSORS

ENaC-mediated sodium reabsorption is also measurable in the salivary and sweat glands.²⁴⁷ The importance of these tissues for sodium and water homeostasis is underscored by rare genetic mutations that result in elevated plasma aldosterone levels and pseudohypoaldosteronism, with normal renal tubular function but significant sodium loss from salivary or sweat glands.^{248,249} ENaC channels also play an important role in transduction of sodium salt taste in the anterior papillae of the tongue.^{250,251} The appropriate molecular machinery for mineralocorticoid-responsive sodium reabsorption is expressed in these organs,^{218,252,253} and these epithelia are model systems for the study of ENaC regulation.^{247,254} As in colonic epithelia, aldosterone stimulates the expression of β - and γ -ENaC and sodium transport in glands and taste buds in animal models.^{251,255} Moreover, in humans, changes in dietary sodium are inversely proportional to sodium transport across salivary epithelia.²⁵⁶ Similar to the aldosterone-responsive distal nephron and distal colon, sodium uptake is coupled with potassium secretion in salivary epithelia. This effect is evident in humans with hyperaldosteronism. Such patients have a salivary [Na⁺]/[K⁺] ratio significantly lower than that of subjects without the disorder,^{257,258} although this has not been accepted as a valid means to screen for hyperaldosteronism.

ROLE OF SERUM- AND GLUCOCORTICOID-REGULATED KINASE IN MEDIATING ALDOSTERONE EFFECTS

INDUCTION OF SGK1 BY ALDOSTERONE

In the early to mid-1960s, primarily from the work of Edelman and colleagues, it became clear that aldosterone, like cortisol, exerted most, if not all, of its key physiologic effects by altering transcription rates of a specific subset of mRNA-encoding genes.²⁵⁹ In particular, hormone-induced changes in gene transcription were shown to be essential for its effects on epithelial Na⁺ transport.⁷³ Transporters involved in Na⁺ reabsorption (Na⁺-K⁺-ATPase and ENaC), are regulated by aldosterone at the transcriptional level. However, these effects are manifest several hours after most of the change in Na⁺ transport has already occurred and hence could not explain the early and greatest proportion of effects of aldosterone.¹ Considerable effort by many groups went into unbiased screening for aldosterone-regulated proteins²⁶⁰ and later aldosterone-regulated mRNAs (reviewed in Verrey¹). In 1999, SGK1 was identified as the first early-onset, aldosterone-induced gene

product, which clearly stimulates ENaC-mediated sodium reabsorption in the distal nephron^{221,261} without pleiotropic effects on other cellular processes. The physiologic relevance of SGK1 has now been firmly established, and investigations by numerous laboratories into its mechanism of action have revealed critical general features of the mechanism underlying hormone-regulated ion transport. It is therefore addressed in some detail here.

SGK1 mRNA levels are increased within 15 minutes, and protein levels within 30 minutes, in cultured cells on stimulation by aldosterone^{262,263} and in the collecting duct by aldosterone or a low-salt diet (a physiologic stimulus for aldosterone secretion).^{87,221,264} Notably, SGK1 is increased more abundantly in the kidney cortex (the connecting tubule and cortical collecting duct) than in the medulla, which is congruent with the potency of aldosterone-induced ENaC activation in these nephron segments, as noted earlier.^{265,266} SGK1 is expressed in other nephron segments, including glomeruli, proximal tubule, and papillae^{87,221,267}; however, its rapid induction in the ASDN appears to provide most of the basis for its role in aldosterone-regulated sodium and potassium transport. SGK1, induced by high sodium in infiltrating T cells, is also important for inflammation in the kidney and hypertension, although the specific nephron segments are unknown.^{268,269}

MOLECULAR MECHANISMS OF SGK1 ACTION IN THE ALDOSTERONE-SENSITIVE DISTAL NEPHRON

SGK1 is a serine-threonine kinase of the AGC protein kinase superfamily,²⁷⁰ and its kinase activity appears to be essential for Na⁺ transport regulation. Although it has effects on proliferation and apoptosis in kidney cells, these effects appear to be minor, and the control of ENaC and other transporters²⁷¹ predominates. SGK1 is interesting as a signaling kinase in that both its expression level and activity are highly regulated. SGK1 transcription is induced by a variety of stimuli in addition to aldosterone. As its name implies, these include serum and glucocorticoids, but also follicle-stimulating hormone, transforming growth factor- β , and osmotic stress.^{272–275}

SGK1 activation is primarily regulated through phosphorylation, which is required for its stimulation of ENaC.^{276–278,279,280} Like that of its close relative, Akt, SGK1 phosphorylation is stimulated by a variety of growth factors, including insulin and insulin-like growth factor-1^{279–282}; these act through PI3K to trigger phosphorylation at two key residues, an activation loop (residue T256) and a hydrophobic motif (S422). Specifically, the α -isoform of the p110 subunit of PI3K stimulates PI3K-dependent kinase 1 (PDK1) to phosphorylate T256, and also stimulates mammalian target of rapamycin [mTOR] in its complex 2 variant (mTORC2, formerly called PDK2) to phosphorylate S422.^{276,280,282–286} mTORC2 also uses a cofactor, SIN1, to specify activation of SGK1 rather than related family members, such as Akt.²⁸⁷ In turn, the upstream kinases, PDK1 and mTORC2, phosphorylate and thereby activate SGK1 kinase. Thus, SGK1 serves as a convergence point for different classes of stimuli, which act on the one hand to control its expression (aldosterone) and on the other to control its activity (insulin and other activators), which results in the coordinate regulation of ENaC.

In the study of the physiologic and pathophysiologic roles of SGK1 in the ASDN, mice lacking SGK1 under different

physiologic stimuli have provided considerable insight. Unlike MR knockout mice,¹⁷⁷ mice lacking SGK1 survive the neonatal period and appear normal when consuming a normal sodium diet, although circulating aldosterone is markedly elevated. When subjected to a low-sodium diet, these mice have a profound sodium-wasting phenotype, akin to pseudohypoaldosteronism type I.^{288,289} Additional mouse models of SGK1 deletion have demonstrated diminished processing of ENaC subunits I.^{290,291} Notably, this is a significantly milder phenotype than with deletion of MR or α -ENaC.²⁹² These comparisons suggest that disruption of SGK1 signaling may be insufficient to eliminate aldosterone-mediated sodium transport due to additional aldosterone-induced and aldosterone-repressed proteins, which could compensate for the lack of SGK1.

SGK1 may also play a significant role in states of aldosterone excess or upregulation of hormonal activators of SGK1 (e.g., insulin). SGK1 knockout mice are protected from the development of salt-sensitive hypertension, which accompanies the hyperinsulinemia of the metabolic syndrome.^{293,294} Taken together, SGK1 is an important component of ENaC regulation to maintain both sodium and potassium homeostasis.

Despite its accepted role as a mediator of aldosterone-stimulated sodium reabsorption, the mechanisms whereby SGK1 stimulates ENaC are not fully characterized. Several mechanistic studies have demonstrated that SGK1 is rapidly induced but also rapidly degraded.^{221,295,296} The N-terminus of the kinase, which distinguishes SGK1 from other kinase family members (e.g., Akt), is important for stimulation of sodium transport but is also the target for rapid degradation of the kinase via the ubiquitin-proteasome system.^{297–303} The pathophysiologic implications of the N-terminus for sodium transport are unclear, but they may involve a negative feedback loop to limit sodium reabsorption in states of hypertension. The molecular mechanisms of ENaC stimulation by SGK1 can be divided into three known categories (Fig. 12.10): (1) posttranslational effects on the E3 ubiquitin ligase Nedd4-2; (2) posttranslational Nedd4-2-independent effects; and (3) transcription of gene products such as α -ENaC.

SGK1 INHIBITS THE UBIQUITIN LIGASE NEDD4-2

Before the discovery of SGK1 as an aldosterone-induced early gene product, the E3 ubiquitin ligase known as “neural developmentally downregulated isoform 4-2” (Nedd4-2) was shown to interact with the C-terminal tails of β -ENaC and γ -ENaC³⁰⁴ and decrease surface expression of the channel via channel ubiquitination, hence inhibiting the sodium current.^{108,305} The genetic defect in Liddle syndrome (ENaC-mediated hypertension, hypokalemia, and metabolic alkalosis) consists of a gain-of-function mutation in the C-terminal tail of these subunits, which results in decreased inhibition by Nedd4-2 and hence increased ENaC activity.³⁰⁶ Lack of Nedd4-2 in vivo results in increased ENaC activity and salt-sensitive hypertension,^{307,308} recapitulating a Liddle syndrome-like phenotype. SGK1 interacts with and phosphorylates Nedd4-2^{263,277} in an ENaC signaling complex¹¹³ and enhances cell surface expression of ENaC,^{262,309} a determinant of ENaC activity (see Fig. 12.10A). This interaction coordinates the phosphorylation-dependent binding of 14-3-3 proteins to inhibit Nedd4-2 and prevent the ubiquitination of ENaC.^{310–313} This disinhibition of ENaC parallels a recurring theme in the regulation of ion transport in the kidney seen with the WNK kinases and NCC, other aldosterone-regulated

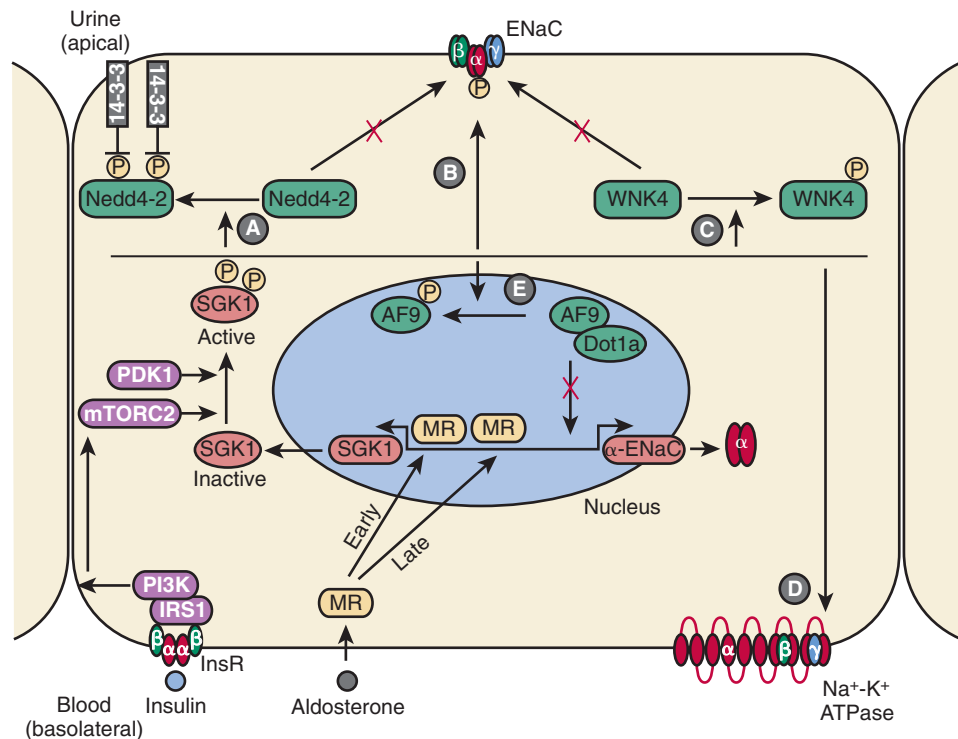


Fig. 12.10 Mechanisms of serum- and glucocorticoid-regulated kinase 1 (SGK1)-mediated stimulation of the amiloride-sensitive sodium channel (ENaC). Within principal cells of the mammalian kidney, SGK1 is transcriptionally upregulated as an early aldosterone-induced gene product. SGK1 is then phosphorylated twice via a phosphatidylinositol-3-kinase (PI3K)-dependent cascade of upstream kinases leading to active SGK1. Active SGK1 has multiple effects: it increases apical plasma membrane ENaC by inhibiting Nedd4-2 and Raf-1, and it induces transcription of the α -ENaC (thereby influencing late effects of aldosterone). (A–E, clockwise) Shown are the individual mechanisms that have been elucidated in principal cells. See text for details. *InsR*, Insulin receptor; *IRS1*, insulin receptor substrate 1; *MR*, mineralocorticoid receptor; *mTORC2*, mammalian target of rapamycin complex 2; *PDK1*, 3-phosphoinositide-dependent protein kinase type 1.

gene products (e.g., GILZ) and ENaC, and NHERF2 and ROMK.^{270,314} Similarly, SGK1 has also been implicated in the stimulation of NCC via inhibition of Nedd4-2 and increased abundance of PY motif-containing WNK1.^{210,315,316}

SGK1 ENHANCES EPITHELIAL SODIUM CHANNEL ACTIVITY INDEPENDENTLY OF NEDD4-2

In cell culture systems, mutation of SGK1 phosphorylation sites on Nedd4-2 does not completely abolish the ability of SGK1 to stimulate ENaC.²⁷⁷ Furthermore, SGK1 has been shown to stimulate ENaC channels with Liddle syndrome mutations, which are unable to bind Nedd4-2.^{221,262} Consequently, other Nedd4-2-independent mechanisms of SGK1 stimulation have been proposed. SGK1 directly phosphorylates a serine residue in the intracellular C-terminal tail of α -ENaC, which directly activates channels at the cell surface (see Fig. 12.10B).^{317,318} SGK1 has been implicated in the stimulation of ENaC via the phosphorylation of WNK4, a kinase mutated in pseudohypoaldosteronism type II (see Fig. 12.10C).³¹⁹ Cell surface-expressed SGK1 may also increase open probability of the channel.^{318,320} In addition to showing effects on ENaC, SGK1 has been found to stimulate the activity of basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$, which separately increases ENaC-mediated sodium transport (see Fig. 12.10D).^{321,322} The time course of these effects and their relative importance compared with Nedd4-2-dependent inhibition have not been explored.³¹⁷ The next generation of molecular studies of SGK1 will elucidate the relative importance of each of these pathways.

SGK1 STIMULATES THE COMPONENTS OF SODIUM TRANSPORT MACHINERY

SGK1 also regulates the expression of late aldosterone-responsive genes, primarily α -ENaC.^{323,324} Active SGK1 is an important mediator of aldosterone-sensitive α -ENaC transcription in vivo via inhibition of a transcriptional repression element, the disruptor of telomeric silencing alternative splice variant a (Dot1a)-ALL1-fused gene from the chromosome 9 (Af9) complex.³²⁴ SGK1 phosphorylates Af9 and reduces the interaction between Dot1a and Af9. This releases suppression of ENaC transcription by this complex (see Fig. 12.10E). Thus, SGK1 not only acts on ENaC channels to enhance sodium channel activity rapidly through the increase of active channels at the apical surface and the increase of $\text{Na}^+\text{-K}^+\text{-ATPase}$ at the basolateral surface, but also stimulates the transcription of elements of the machinery for sodium transport to promote a sustained response to aldosterone. SGK1 is an early-onset gene, but its effects influence both immediate- and long-term aldosterone-stimulated sodium reabsorption.

SGK1 STIMULATES POTASSIUM SECRETION IN THE ALDOSTERONE-SENSITIVE DISTAL NEPHRON

Further evidence of a role for SGK1 in the regulation of sodium transport in the ASDN has been revealed by the study of potassium secretion. If SGK1 enhances ENaC-mediated sodium transport, the potential difference across

the apical to basolateral surface of principal cells should be higher (more negative) and thus should indirectly stimulate potassium secretion. SGK1 knockout mice are unable to secrete potassium adequately in the short and long terms when challenged with a high-potassium diet, and mice with constitutive or inducible deletion of SGK1 are prone to hyperkalemia.^{325–327} Moreover, the potential difference across collecting duct epithelia from these knockout mice indicates that the effect of SGK1 on potassium secretion occurs via ENaC, not through direct regulation of ROMK.³²⁸ SGK1 also directly inhibits Nedd4-2, and deletion of Nedd4-2 predisposes low-potassium-fed mice to hypokalemia via the constitutive stimulation of ENaC-mediated sodium transport.³²⁹ Thus, SGK1 and its effectors have physiologically relevant roles in sodium and potassium transport in the ASDN.

ALTERNATE MODES OF REGULATION OF ENAC-MEDIATED SODIUM TRANSPORT BY ALDOSTERONE

Although SGK1 remains the best characterized aldosterone-regulated gene, other early aldosterone-induced mRNAs, including *K-ras*, *GILZ*, kidney-specific *WNK1*, *Usp45*, melanophilin, and promyelocytic leukemia zinc finger,^{330–335} have also been implicated in the stimulation of ENaC. Their distinct mechanisms of action are beyond the scope of this chapter, but more recent data have suggested that micro-RNAs stimulated by aldosterone may play a prominent role in regulating both SGK1 and ENaC.

Aldosterone can upregulate or downregulate several microRNAs in cultured cells and in vivo. These micro-RNAs can then indirectly increase or decrease protein levels of intermediate regulators of ENaC-mediated transport. The first micro-RNA cluster (*mmu-miR-335-3p*, *mmu-miR-290-5p*, and *mmu-miR-1983*) to be described by Butterworth was downregulated by aldosterone within 24 hours and thereby released the 3' untranslated region of ankyrin 3 to increase apical trafficking of α -ENaC.^{336,337} Aldosterone also increases micro-RNAs that inhibit a negative regulator of ENaC, intersectin 2.³³⁸ Aldosterone has also been shown to promote the rapid induction of SGK1 mRNA by decreasing micro-RNA 466g in cultured cells.³³⁹ Taken together, SGK1 plays a prominent role in transducing the effect of aldosterone to stimulate ENaC for both regulation of blood pressure and potassium homeostasis. Additionally, there are alternate effectors of aldosterone, but the physiologic contexts of these other pathways have not been as well established.

11 β -HYDROXYSTEROID DEHYDROGENASE TYPE 2

ESSENTIAL DETERMINANT OF MINERALOCORTICOID SPECIFICITY

The physiologic glucocorticoid cortisol (corticosterone in rats and mice) has a high affinity for MR, equivalent to that of aldosterone and, as noted earlier, circulates at plasma free concentrations that are 100-fold or more higher than those of aldosterone. Central to the ability of MR to respond to aldosterone selectively in the ASDN is the coexpression of the enzyme 11 β -HSD2.^{5,6} 11 β -HSD2 converts cortisol (corti-

costerone) to receptor-inactive 11-keto steroids (cortisone in humans, 11-dehydrocorticosterone in rats and mice), using nicotinamide adenine dinucleotide (NAD) as a cosubstrate and generating sufficient amounts of the reduced form of NAD (NADH) to alter the redox potential of the cell. This dependence sets it in contrast to 11 β -HSD1, which uses the reduced form of nicotinamide adenine dinucleotide phosphate (NADPH), preferentially catalyzes the conversion of the oxidized to the reduced form, and has received substantial attention as a target for the treatment of metabolic syndrome.³⁴⁰ Aldosterone has a very reactive aldehyde group at carbon 18 (see Fig. 12.1), which forms an 11,18-hemiacetal and is protected from dehydrogenation by 11 β -HSD2.^{5,6}

SITES OF EXPRESSION

In the kidney, 11 β -HSD2 is expressed at high levels throughout the ASDN,^{175,186,341} where it is coexpressed with MR and ENaC (see Fig. 12.9).³⁴² It is also coexpressed in DCT with NCC.^{343,344} Interestingly, expression has also been found in the thick ascending limb,¹⁸⁶ although expression levels appear to be substantially lower, and increase progressively in DCT. Expression is the highest in the connecting tubule and cortical collecting duct.^{341,344} It is also expressed in the aldosterone-sensitive segments of the colon, particularly the distal colon, as is the case for MR, although there is species variability.³⁴⁵ 11 β -HSD2 expression has also been described in several nonepithelial tissues, including placenta,³⁴⁶ the nucleus tractus solitarius in the brain,³⁴⁷ and the vessel wall,⁵⁶ which makes all these tissues potential aldosterone target tissues.

IMPACT ON MINERALOCORTICOID RECEPTOR ACTIVITY

The initial^{5,6} and still widely held interpretation of the role of 11 β -HSD2 was that of excluding active glucocorticoids from epithelial MR, which allowed aldosterone unfettered access. This is only part of the picture, however; to reduce the signal to noise ratio from 100-fold to 10% would require that 999 of every 1000 cortisol molecules entering the cell be metabolized to cortisone, a very tall order in an organ such as the kidney, which commands 20% to 25% of cardiac output. 11 β -HSD2 in epithelia (and in other tissues in which it is expressed) clearly reduces glucocorticoid levels by an order of magnitude³⁴⁸ but still leaves them with intracellular levels well above those of aldosterone. At the same time, although it is clear that when 11 β -HSD2 is operative, glucocorticoid-occupied MR is not transcriptionally active, it is also clear that when enzyme activity is insufficient (as in apparent mineralocorticoid excess) or deficient (as in licorice abuse or by genetic mutation), cortisol can activate MR and ion transport. Although the subcellular mechanisms involved have yet to be established, it appears that glucocorticoid-MR complexes are conformationally distinct from aldosterone-MR complexes. One intriguing possibility is that these hormone-receptor complexes, in contrast to aldosterone-MR complexes, are held inactive by the obligate generation of NADH from the cosubstrate NAD, required for the operation of 11 β -HSD2.³⁴⁹ There is direct evidence to support the idea that redox potential affects the activity of the glucocorticoid receptor through effects on thioredoxin.³⁵⁰

APPARENT MINERALOCORTICOID EXCESS: A DISEASE OF DEFECTIVE 11 β -HYDROXYSTEROID DEHYDROGENASE TYPE 2

Apparent mineralocorticoid excess was first described by New, and the molecular mechanisms responsible were established after an intense but fruitless search for a novel mineralocorticoid.³⁵¹ The condition reflects a partial or complete deficiency of 11 β -HSD2 activity, is more common in consanguinity, and manifests as severe juvenile hypertension (see also Chapters 17 and 44).³⁵² Confectionery licorice (or that added to chewing tobacco) contains glycyrrhizic and glycyrrhetic acids, suicide substrates for 11 β -HSD2, which thus acts as a potent inhibitor of the enzyme. Lack of functional 11 β -HSD2 results in MR activation by cortisol and inappropriate mineralocorticoid-like stimulation of ENaC-mediated Na⁺ reabsorption. This causes severe hypertension, often accompanied by hypokalemia. Plasma renin, angiotensin II, and aldosterone are suppressed. Treatment of apparent mineralocorticoid excess is the use of MR antagonists and additional antihypertensives, as required. Treatment of licorice abuse is moderation.

ROLE IN BLOOD VESSELS

Studies of 11 β -HSD2 in the human vascular wall³⁵³ have defined the activity of aldosterone and cortisol in this physiologic aldosterone target tissue. Aldosterone at nanomolar concentrations causes a rapid rise in the intracellular pH, reflecting nongenomic activation of the Na⁺-H⁺ exchanger. Cortisol alone over a range of doses produced no effect, but when carbenoxolone was added to inhibit 11 β -HSD2, cortisol mimicked aldosterone. Inhibitor studies have revealed that the effects of both aldosterone and cortisol are mediated by classic MR. In other studies involving tissue damage, mineralocorticoid antagonists were protective, whereas aldosterone or cortisol worsened injury. The inference from these results was that cortisol becomes an MR agonist in the context of tissue damage (or when 11 β -HSD2 is pharmacologically inhibited), with alteration of reactive oxygen species generation and redox potential.³⁵⁴ It is further notable that aldosterone has been shown to have both vasodilatory and vasoconstricting effects in animals and humans.³⁵⁵ These contradictory results have not been fully reconciled,³⁵⁶ but may well reflect a combination of direct effects on vascular smooth muscle to stimulate myosin light-chain phosphorylation through ERK activation^{357,358} on the one hand, and stimulatory effects on endothelial cell nitric oxide synthase³⁵⁵ on the other. Finally, it is of considerable interest that vascular smooth muscle cells express ENaC, in addition to MR, and that the channel might play a role in vascular tone.³⁵⁹

SUMMARY OF 11 β -HYDROXYSTEROID DEHYDROGENASE TYPE 2 ROLES

In summary, the enzyme 11 β -HSD2 is crucial for the aldosterone-selective activation of epithelial MR and possibly of MR in other tissues, including blood vessels, nucleus of the solitary tract, and placenta. It does this in part by debulking intracellular glucocorticoids by an order of magnitude, which is not sufficient to account for its blockade of cortisol agonist activity. Current evidence supports the possibility that 11 β -HSD2-mediated generation of NADH renders

glucocorticoid-occupied MR inactive. Partial or complete deficiency of 11 β -HSD2 results in the syndrome of apparent mineralocorticoid excess, which is mimicked by licorice abuse.

NONGENOMIC EFFECTS OF ALDOSTERONE

The classic effects of aldosterone on ion transport are genomic, with MR acting at the nuclear level to regulate DNA-directed, RNA-mediated protein synthesis and thereby sodium transport. Such genomic effects are characterized by a lag period of 45 to 60 minutes before changes in ion transport can be measured, commensurate with a homeostatic role for aldosterone action in regulating sodium and potassium status in response to dietary intake. In other circumstances (e.g., orthostasis, acute blood volume depletion), aldosterone secretion rises rapidly, and acute nongenomic effects are an understandable response. Such rapid effects were first demonstrated over 30 years ago in the laboratory³⁶⁰; in human vascular tissues, they have been amply demonstrated both *in vitro*³⁵³ and *in vivo*.³⁶¹ Although most of these rapid nongenomic effects appear to be mediated via activation of classic MR,^{353,360} there is evidence from atomic force microscopy studies for non-MR membrane sites binding aldosterone with high affinity on cultured endothelial cells.³⁶² Such nongenomic effects are not unique to aldosterone, having been shown for the other recognized classes of steroid hormones³⁶³ and reported for dehydroepiandrosterone (DHEA).³⁵⁵ Genomic effects commonly have a lag period of 20 minutes or longer and are abrogated by inhibitors of transcription, such as actinomycin D. Most nongenomic effects of steroids have time courses from onset to plateau of 5 to 10 minutes and are mediated by a variety of pathways.

MR does not have a myristoylation site (e.g., unlike estrogen receptors³⁶⁴), and there is little evidence for membrane-associated classic MR. Most rapid nongenomic effects of aldosterone appear to be mediated by classic MR in that they are inhibited by the MR antagonist RU 28318. In some cases,³⁵³ spironolactone is ineffective as an inhibitor; exclusive reliance on blockade by spironolactone has led to the assumption of a widely distributed aldosterone receptor distinct from classic MR and a long and unsuccessful search for such a membrane-bound species.³⁶⁵ The physiology of nongenomic aldosterone actions has been slow to be accepted, which in part reflects the major emphasis on the clearly genomic actions of aldosterone in the kidney. The most obvious example is the conjunction of rapid secretion of aldosterone in response to orthostasis and its demonstrated rapid vascular effects.^{356,366} With more interest in the pathophysiologic effects of MR activation, particularly in nonclassic aldosterone target tissues, there has been renewed interest in the rapid nongenomic effects of aldosterone (and the physiologic glucocorticoids) via classic MR. Further details of the nongenomic actions of aldosterone can be found in reports by Funder³⁶⁶ and other sources.^{367,368}

DISEASE STATES

PRIMARY ALDOSTERONISM

Clinically, the most prevalent disorder directly involving aldosterone is Conn syndrome, or primary aldosteronism.³⁶⁹

In this syndrome, aldosterone secretion is elevated and (relatively) autonomous as a result of an adrenal adenoma or, more frequently, bilateral adrenal hyperplasia and, very rarely, adrenal carcinoma or the inherited disorder glucocorticoid remediable aldosteronism (FH-1). Once considered rare (<1% of all cases of hypertension), necessarily characterized by hypokalemia and relatively benign, primary aldosteronism is now thought to account for approximately 8% to 13% of all hypertension, which reflects improved case detection and diagnosis. In contrast with previous teachings, frank hypokalemia is found in only 25% to 30% of cases, and the incidence of cardiovascular pathology (e.g., fibrosis, fibrillation, infarct, stroke) is substantially higher than in age-, gender-, and blood pressure-matched individuals with essential hypertension.^{370,371}

Guidelines for the case detection, diagnosis, and management of primary aldosteronism have been published³⁷² as a first step in addressing what has been increasingly recognized as a major public health issue. It has long been thought and taught that the role of aldosterone in blood pressure regulation reflects its epithelial effects leading to retention of sodium, and with it water, which thus increases circulating volume. This increase, in turn, is reflected in an increased cardiac output, which is reflexively normalized by vasoconstriction and thus elevation of blood pressure (in keeping with the Guyton hypothesis³⁷³). Although the epithelial effects of aldosterone on vascular volume are indisputably homeostatically important, there have been compelling experimental and clinical studies to suggest a role for nonepithelial effects in mineralocorticoid-induced hypertension.^{374,375} In addition to MR-mediated central nervous system and vascular effects in hypertension, roles for macrophages have been demonstrated by two groups using distinct and complementary experimental approaches.^{376,377}

Other studies have suggested a role for a mutated K⁺ channel (KCNJ5) in the pathogenesis of aldosterone-producing adrenal adenomas and in the rare condition of FH type III.^{25,26} See earlier, “Aldosterone Synthesis,” for additional details.

CONGESTIVE HEART FAILURE

Aldosterone has been implicated in the pathophysiology of congestive heart failure since soon after its discovery in the mid-1950s.^{378,379} Until fairly recently, most of the focus has been on the counterproductive effects of aldosterone in epithelia. More recently, the beneficial effects of MR antagonists in congestive heart failure have suggested an additional effect in myocardium itself.³⁸⁰ In the Randomized Aldactone Evaluation Study (RALES),³⁸⁰ addition of low-dose (mean, 26 mg/day) spironolactone to standard of care treatment in patients with progressive heart failure produced a 30% reduction in mortality and 35% fewer hospitalizations. This result is often attributed to spironolactone antagonizing the effect of aldosterone on cardiomyocyte MR, but actually reflects its antagonizing of cortisol acting as an MR agonist under ischemic conditions. Subsequently, the Eplerenone Post-Acute Myocardial Infarction Heart Failure Efficacy and Survival Study (EPHESUS) examined the effect of eplerenone, an MR antagonist with improved specificity relative to spironolactone, on heart failure due to systolic dysfunction complicating acute myocardial infarction. The study showed

that adding eplerenone (25 mg/day) to conventional therapy significantly decreased mortality due to all causes (31%) and cardiovascular mortality (13%).³⁸¹ Potassium concentration was only slightly higher in the eplerenone-treated group than in the placebo-treated group (4.47 mmol/L and 4.54 mmol/L, respectively). Coupled with studies on the direct vascular effects of aldosterone addressed earlier, these data suggest that MR antagonists have a beneficial effect that cannot be accounted for by diuretic actions in the kidney alone.³⁸²

It is also notable that a trial (Eplerenone in Mild Patients Hospitalization And Survival Study in Heart Failure [EMPHASIS-HF]) examining the effect of eplerenone in New York Heart Association (NYHA) class II heart failure (milder than previously examined), was stopped early because a significant benefit was found in the treated group.³⁸³ In summary, the pathophysiologic effects of aldosterone excess on the cardiovascular system in primary aldosteronism have been well documented, and MR also plays an important role in essential hypertension and heart failure. Importantly, MR expressed in cardiac and vascular cells may commonly be activated by cortisol rather than aldosterone, which is present in serum at levels that are higher by 100-fold or more and mimics aldosterone in the context of tissue damage.

CHRONIC KIDNEY DISEASE

The role of MR blockade in slowing the progression of chronic kidney disease has been considered in a recent study and commentary^{384,385} and in Chapter 59. The study, a double-blind, randomized, placebo-controlled trial, examined the antialbuminuric effect of the MR antagonist eplerenone in nondiabetic hypertensive patients with albuminuria.

NONEPITHELIAL ACTIONS OF ALDOSTERONE

In addition to the classic epithelial tissues involved in ion transport—kidney, colon, sweat gland, salivary gland—there are documented effects of aldosterone in the brain, vascular wall, and possibly the placenta, as previously noted. Many other tissues and organs have been postulated as physiologic aldosterone target tissues, largely on the inadequate evidence that they express MR and can be shown *in vitro* to respond by some measure to aldosterone. What underpins these hypotheses is the misconception that aldosterone is the cognate ligand for MR, which is true for epithelia but not for cells not expressing 11 β -HSD2, coupled with disregard for the role of cortisol. Cortisol was not only the ligand for MR in cartilaginous and bony fish, millions of years before the appearance of aldosterone synthase, but is the overwhelming occupant of MR that is not protected by 11 β -HSD2 (primarily nonepithelial MR) throughout the body. It is notable that some nonepithelial MR is also protected by 11 β -HSD2—for example, in the nucleus tractus solitarius.

The fact that aldosterone can activate MR under experimental conditions without 11 β -HSD2 was illustrated by the work of Gómez-Sánchez and colleagues more than 2 decades ago.³⁸⁶ Very low doses of aldosterone that did not affect blood pressure when infused systemically elevated blood pressure when infused into the lateral ventricle of conscious, free-living rats. That this did not reflect a physiologic role for aldosterone,

however, was shown by the co-infusion of one, two, and five times the dose of corticosterone, which progressively blocked the blood pressure effect of the infused aldosterone, evidence for the absence of 11β -HSD2 in the hypothalamic nuclei involved and the overwhelming occupancy of their MR by the physiologic glucocorticoid.

The two established nonepithelial aldosterone target tissues are the vascular wall and nucleus tractus solitarius in the brain. Both these tissues express 11β -HSD2, as noted earlier, allowing aldosterone-selective MR activation; both can be reasonably envisaged as having important ancillary roles supporting the primary epithelial role of aldosterone on fluid and electrolyte homeostasis. Aldosterone vasoconstricts blood vessels, acutely and in the longer term, in response to volume depletion; similarly, it acts on the nucleus tractus solitarius to stimulate salt appetite. Both actions are thus harnessed into the physiologic role of aldosterone in maintaining fluid and electrolyte balance.

It is commonly assumed that in pathophysiologic states of high aldosterone levels, such as primary aldosteronism, the deleterious effects are mediated by aldosterone occupying and inappropriately activating nonprotected MR in cardiomyocytes, for example. It is plausible that instead of the approximately 1% physiologic occupancy (given the ≈ 100 -fold higher levels of plasma free cortisol), aldosterone occupancy of cardiomyocyte MR might rise to 3% to 5%. Relatively minor degrees of MR occupancy have been shown to be effective for spironolactone, acting as a protective inverse agonist³⁸⁷; similarly, therefore, minor degrees of cardiomyocyte MR occupancy by aldosterone could potentially produce the deleterious effects seen.

This explanation, however, is almost certainly incorrect. Plasma aldosterone levels are as high or higher in chronic sodium deficiency (or in the effectively volume-depleted condition of secondary hyperaldosteronism), with no deleterious cardiovascular effects. In primary and secondary aldosteronism, and in chronic sodium deficiency, physiologic target tissues, both renal tubular and coronary vascular, are exposed to (and respond to) maintained high levels of aldosterone. It is thus unlikely that the cardiovascular damage in primary aldosteronism reflects increased MR activation in blood vessels, coronary and peripheral. The key difference between these circumstances is that primary aldosteronism is a state of aldosterone and sodium excess and the others of sodium and volume depletion.

A plausible but untested mechanism of aldosterone-induced damage is that it is secondary to increased renal sodium reabsorption and the action of endogenous ouabain on blood vessels. Endogenous ouabain is incompletely explored, but its levels are elevated in primary aldosteronism.³⁸⁸ Like aldosterone, its secretion is elevated by ACTH and angiotensin (the latter via AT_2R); in stark contrast with aldosterone, it is raised (not lowered) in states of sodium excess.^{389–391} It acts via Na^+K^+ -ATPase in vessel walls as a vasoconstrictor, presumably physiologically to produce a pressure natriuresis as a homeostatic response. Thus, it may be that the cardiovascular damage in primary aldosteronism reflects a combination of the effects of aldosterone plus endogenous ouabain on the vasculature; if this is the case, the source and origin of the nonepithelial effects of aldosterone remain squarely in the renal tubule and the exaggerated sodium retention therein.

RECENT ADVANCES IN NONRENAL MINERALOCORTICOID RECEPTOR-MEDIATED DISEASE PATHOLOGY

Recent developments in the use of MR antagonists for nonrenal disease are not described in detail here. See the report by Jaisser and Farman³⁹² for a review. It is particularly worth noting the surprising recent use of spironolactone for the treatment of retinal diseases. MR is expressed in several retinal cell types, including glial cells, which are essential for retinal water and ion homeostasis, epithelial cells, and choroidal endothelial cells. The activation of MR has been implicated in multiple retinal pathologies, and MR antagonists, particularly spironolactone, have been shown to have therapeutic benefit.^{392,393}

ACKNOWLEDGMENTS

Our coauthor, colleague, and friend John Stokes died in 2012, just before work on the 10th edition of this textbook began. John was senior author on the chapter “Aldosterone Regulation of Ion Transport,” written as a new chapter for the 9th edition of Brenner and Rector, and inspired our writing and focus for subsequent editions, including the present chapter for the 11th edition. John had a profound knowledge of aldosterone action in the renal tubules, a sharp wit, and unsurpassed work ethic. For all these reasons, he was the model coauthor and colleague. John also made enormous primary contributions to aldosterone and renal tubule research, which enriched all of us, as did his humor and enthusiasm for life. For those interested in reading more about John’s inspiring life and contributions to nephrology and renal research, see the eloquent eulogy at <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3715930/>.

 Complete reference list available at ExpertConsult.com.

KEY REFERENCES

- Simpson SA, Tait JF. A quantitative method for the bioassay of the effect of adrenal cortical steroids on mineral metabolism. *Endocrinology*. 1952;50(2):150–161.
- Simpson SA, Tait JF. Physicochemical methods of detection of a previously unidentified adrenal hormone. *Mem Soc Endocrinol*. 1953;2:9–24.
- Simpson SA, et al. Constitution of aldosterone, a new mineralocorticoid. *Experientia*. 1954;10(3):132–133.
- Funder JW, et al. Mineralocorticoid action: target tissue specificity is enzyme, not receptor, mediated. *Science*. 1988;242(4878):583–585.
- Lifton RP, et al. A chimaeric 11β -hydroxylase/aldosterone synthase gene causes glucocorticoid-remediable aldosteronism and human hypertension. *Nature*. 1992;355:262–265.
- Choi M, et al. K^+ channel mutations in adrenal aldosterone-producing adenomas and hereditary hypertension. *Science*. 2011;331:768–772.
- Boukroun S, et al. Prevalence, clinical, and molecular correlates of KCNJ5 mutations in primary aldosteronism. *Hypertension*. 2012;59(3):592–598.
- Azizan EAB, et al. Somatic mutations in *ATP1A1* and *CACNA1D* underlie a common subtype of adrenal hypertension. *Nat Genet*. 2013;45:1055.
- Bhargava A, Pearce D. Mechanisms of mineralocorticoid action: determinants of receptor specificity and actions of regulated gene products. *Trends Endocrinol Metab*. 2004;15(4):147–153.
- Li Y, et al. Structural and biochemical mechanisms for the specificity of hormone binding and coactivator assembly by mineral corticoid receptor. *Mol Cell*. 2005;19:367–380.

54. Geller DS, et al. Autosomal dominant pseudohypoaldosteronism type 1: mechanisms, evidence for neonatal lethality, and phenotypic expression in adults. *J Am Soc Nephrol*. 2006;17(5):1429–1436.
55. Krozowski ZS, Funder JW. Renal mineralocorticoid receptors and hippocampal corticosterone-binding species have identical intrinsic steroid specificity. *Proc Natl Acad Sci U. S. A.* 1983;80(19):6056–6060.
59. Pearce D, Yamamoto KR. Mineralocorticoid and glucocorticoid receptor activities distinguished by nonreceptor factors at a composite response element. *Science*. 1993;259(5098):1161–1165.
61. Pascual-Le Tallec L, et al. The elongation factor ELL (eleven-nineteen lysine-rich leukemia) is a selective coregulator for steroid receptor functions. *Mol Endocrinol*. 2005;19(5):1158–1169.
66. Hong H, et al. GRIP1, a novel mouse protein that serves as a transcriptional coactivator in yeast for the hormone binding domains of steroid receptors. *Proc Natl Acad Sci U. S. A.* 1996;93(10):4948–4952.
67. Onate SA, et al. Sequence and characterization of a coactivator for the steroid hormone receptor superfamily. *Science*. 1995;270(5240):1354–1357.
75. Conn JW, Knopf RF, Nesbit RM. Clinical characteristics of primary aldosteronism from an analysis of 145 cases. *Am J Surg*. 1964;107:159–172.
77. Canessa CM, et al. Amiloride-sensitive epithelial Na⁺ channel is made of three homologous subunits. *Nature*. 1994;367(6462):463–467.
79. Garty H, Palmer LG. Epithelial sodium channels: function, structure, and regulation. *Physiol Rev*. 1997;77:359–396.
88. Chen S, et al. Epithelial sodium channel regulated by aldosterone-induced protein sgk. *Proc Natl Acad Sci U. S. A.* 1999;96:2514–2519.
97. Frindt G, Ergonul Z, Palmer LG. Surface expression of epithelial Na channel protein in rat kidney. *J Gen Physiol*. 2008;131(6):617–627.
130. Wade JB, et al. Morphological and physiological responses to aldosterone: time course and sodium dependence. *Am J Physiol*. 1990;259(1 Pt 2):F88–F94.
138. Knight KK, et al. Liddle's syndrome mutations increase Na⁺ transport through dual effects on epithelial Na⁺ channel surface expression and proteolytic cleavage. *Proc Natl Acad Sci U. S. A.* 2006;103(8):2805–2808.
139. Stokes JB. Potassium secretion by cortical collecting tubule: relation to sodium absorption, luminal sodium concentration, and transepithelial voltage. *Am J Physiol Renal Fluid Electrolyte Physiol*. 1981;241:F395–F402.
141. Estilo G, Liu W, Pastor-Soler N, et al. Effect of aldosterone on BK channel expression in mammalian cortical collecting duct. *Am J Physiol Renal Physiol*. 2008;295:F780–F788.
162. Shibata S, et al. Regulated mineralocorticoid receptor phosphorylation controls ligand binding, allowing distinct physiologic responses to aldosterone. *Cell Metab*. 2013;18(November 5):660–671.
176. Farman N, et al. Immunolocalization of gluco- and mineralocorticoid receptors in rabbit kidney. *Am J Physiol*. 1991;260(2 Pt 1):C226–C233.
177. Berger S, et al. Mineralocorticoid receptor knockout mice: pathophysiology of Na⁺ metabolism. *Proc Natl Acad Sci U. S. A.* 1998;95(16):9424–9429.
184. Vandewalle A, et al. Aldosterone binding along the rabbit nephron: an autoradiographic study on isolated tubules. *Am J Physiol*. 1981;240(3):F172–F179.
206. Ko B, Mistry AC, Hanson L, et al. Aldosterone acutely stimulates NCC activity via a SPAK-mediated pathway. *Am J Physiol Renal Physiol*. 2013;305:F645–F652.
217. Levitan R, Ingelfinger FJ. Effect of *d*-aldosterone on salt and water absorption from the intact human colon. *J Clin Invest*. 1965;44:801–808.
219. Turnamian SG, Binder HJ. Regulation of active sodium and potassium transport in the distal colon of the rat: role of the aldosterone and glucocorticoid receptors. *J Clin Invest*. 1989;84(6):1924–1929.
225. Hayes CP Jr, McLeod ME, Robinson RR. An extrarenal mechanism for the maintenance of potassium balance in severe chronic renal failure. *Trans Assoc Am Physicians*. 1967;80:207–216.
226. Sandle GI, et al. Enhanced rectal potassium secretion in chronic renal insufficiency: evidence for large intestinal potassium adaptation in man. *Clin Sci*. 1986;71(4):393–401.
270. Bhalla V, et al. Disinhibitory pathways for control of sodium transport: regulation of ENaC by SGK1 and GILZ. *Am J Physiol Renal Physiol*. 2006;291(4):F714–F721.
277. Debonneville C, et al. Phosphorylation of Nedd4-2 by Sgk1 regulates epithelial Na(+) channel cell surface expression. *EMBO J*. 2001;20(24):7052–7059.
278. Snyder PM, Olson DR, Thomas BC. Serum- and glucocorticoid-regulated kinase modulates Nedd4-2-mediated inhibition of the epithelial Na⁺ channel. *J Biol Chem*. 2002;277(1):5–8.
288. Wulff P, et al. Impaired renal Na(+) retention in the sgk1-knockout mouse. *J Clin Invest*. 2002;110(9):1263–1268.
335. Soundararajan R, et al. A novel role for glucocorticoid-induced leucine zipper protein in epithelial sodium channel-mediated sodium transport. *J Biol Chem*. 2005;280(48):39970–39981.
304. Staub O, et al. WW domains of Nedd4 bind to the proline-rich PY motifs in the epithelial Na⁺ channel deleted in Liddle's syndrome. *EMBO J*. 1996;15(10):2371–2380.
347. Geerling JC, Loewy AD. Aldosterone in the brain. *Am J Physiol Renal Physiol*. 2009;297(3):F559–F576.
349. Funder JW. Is aldosterone bad for the heart? *Trends Endocrinol Metab*. 2004;15(4):139–142.
356. Oberleithner H. Is the vascular endothelium under the control of aldosterone? Facts and hypothesis. *Pflugers Arch*. 2007;454(2):187–193.
366. Funder JW. The nongenomic actions of aldosterone. *Endocr Rev*. 2005;26(3):313–321.
369. Young WF. Primary aldosteronism: renaissance of a syndrome. *Clin Endocrinol (Oxf)*. 2007;66(5):607–618.
386. Gómez-Sánchez EP, et al. ICV infusion of corticosterone antagonizes ICV-aldosterone hypertension. *Am J Physiol*. 1990;258(4 Pt 1):E649–E653.
387. Milhailidou AS, et al. Glucocorticoids activate cardiac mineralocorticoid receptors during experimental myocardial infarction. *Hypertension*. 2009;54:1306–1312.
389. Dostanic-Larson I, et al. The highly conserved cardiac glycoside binding site of Na,K-ATPase plays a role in blood pressure regulation. *PNAS*. 2005;102(44):15845–15850.
395. McCormick JA, Yang CL, Ellison DH. WNK kinases and renal sodium transport in health and disease: an integrated view. *Hypertension*. 2008;51(3):588–596.
403. Kim GH, et al. The thiazide-sensitive Na-Cl cotransporter is an aldosterone-induced protein. *Proc Natl Acad Sci U. S. A.* 1998;95(24):14552–14557.
404. Rozansky DJ, et al. Aldosterone mediates activation of the thiazide-sensitive Na-Cl cotransporter through an SGK1 and WNK4 signaling pathway. *J Clin Invest*. 2009;119(9):2601–2612.
415. Bonvalet JP, et al. Distribution of 11 beta-hydroxysteroid dehydrogenase along the rabbit nephron. *J Clin Invest*. 1990;86:832–837.

REFERENCES

- Verrey F. Early aldosterone action: toward filling the gap between transcription and transport. *Am J Physiol*. 1999;277(3 Pt 2):F319–F327.
- Simpson SA, Tait JF. A quantitative method for the bioassay of the effect of adrenal cortical steroids on mineral metabolism. *Endocrinology*. 1952;50(2):150–161.
- Simpson SA, Tait JF. Physicochemical methods of detection of a previously unidentified adrenal hormone. *Mem Soc Endocrinol*. 1953;2:9–24.
- Simpson SA, et al. Constitution of aldosterone, a new mineralocorticoid. *Experientia*. 1954;10(3):132–133.
- Funder JW, et al. Mineralocorticoid action: target tissue specificity is enzyme, not receptor, mediated. *Science*. 1988;242(4878):583–585.
- Edwards CR, et al. Localisation of 11 beta-hydroxysteroid dehydrogenase—tissue specific protector of the mineralocorticoid receptor. *Lancet*. 1988;2(8618):986–989.
- Colombe L, et al. A mineralocorticoid-like receptor in the rainbow trout, *Oncorhynchus mykiss*: cloning and characterization of its steroid binding domain. *Steroids*. 2006;65(6):319–328.
- Nuclear Receptors Nomenclature Committee. A unified nomenclature system for the nuclear receptor superfamily. *Cell*. 1999;97(2):161–163.
- Hu X, Funder JW. The evolution of mineralocorticoid receptors. *Mol Endocrinol*. 2006;20(7):1471–1478.
- Makhanova N, et al. Disturbed homeostasis in sodium-restricted mice heterozygous and homozygous for aldosterone synthase gene disruption. *Hypertension*. 2006;48(6):1151–1159.
- Shibata S, et al. Regulated mineralocorticoid receptor phosphorylation controls ligand binding, allowing distinct physiologic responses to aldosterone. *Cell Metab*. 2013;18(5):660–671.
- Funder JW. Angiotensin retains sodium by dephosphorylating mineralocorticoid receptors in renal intercalated cells. *Cell Metab*. 2013;18(5):609–610.
- Funder JW. Aldosterone and mineralocorticoid receptors: lessons from gene deletion studies. *Hypertension*. 2006;48(6):1018–1019.
- Todkar A, Picard N, Loffing-Cueni D, et al. Mechanisms of renal control of potassium homeostasis in complete aldosterone deficiency. *J Am Soc Nephrol*. 2015;26:425–438.
- Mornet E, et al. Characterization of two genes encoding human steroid 11 β -hydroxylase (P-450 11 β). *J Biol Chem*. 1989;264:20961–20967.
- Curnow KM, et al. The product of the CYP11B2 gene is required for aldosterone biosynthesis in the human adrenal cortex. *Mol Endocrinol*. 1991;5:1513–1522.
- Lifton RP, et al. A chimaeric 11 β -hydroxylase/aldosterone synthase gene causes glucocorticoid-remediable aldosteronism and human hypertension. *Nature*. 1992;355:262–265.
- Young DB, et al. Effects of sodium intake on steady-state potassium excretion. *AJP-Renal*. 1984;246(6 Pt 2):F772–F778.
- Ehrhart-Bornstein M, et al. Human adipocytes secrete mineralocorticoid-releasing factors. *Proc Natl Acad Sci U. S. A.* 2003;100(24):14211–14216.
- Nogueira EF, Bollag WB, Rainey WE. Angiotensin II regulation of adrenocortical gene transcription. *Mol Cell Endocrinol*. 2009;302(2):230–236.
- Miller WL. StAR search—what we know about how the steroidogenic acute regulatory protein mediates mitochondrial cholesterol import. *Mol Endocrinol*. 2007;21(3):589–601.
- Spat A, Hunyady L, Szanda G. Signaling interactions in the adrenal cortex. *Front Endocrinol (Lausanne)*. 2016;7:17. PMID: 26973596. PMC: PMC4770035.
- Ganguly A, Davis JS. Role of calcium and other mediators in aldosterone secretion from the adrenal glomerulosa cells. *Pharmacol Rev*. 1994;46(4):417–447.
- Foster RH, MacFarlane CH, Bustamante MO. Recent progress in understanding aldosterone secretion. *Gen Pharmacol*. 1997;28(5):647–651.
- Schrier RW. Aldosterone “escape” vs. “breakthrough.” *Nat Rev Nephrol*. 2010;6(2):61.
- Choi M, et al. K⁺ channel mutations in adrenal aldosterone-producing adenomas and hereditary hypertension. *Science*. 2011;331:768–772.
- Geller DS, et al. A novel form of human mendelian hypertension featuring nonglucocorticoid-remediable aldosteronism. *J Clin Endocrinol Metab*. 2008;93:3117–3123.
- Boukroun S, et al. Prevalence, clinical, and molecular correlates of KCNJ5 mutations in primary aldosteronism. *Hypertension*. 2012;59(3):592–598.
- Scholl UI, et al. Somatic and germline CACNA1D calcium channel mutations in aldosterone-producing adenomas and primary aldosteronism. *Nat Genet*. 2013;45:1050–1054.
- Azizan EAB, et al. Somatic mutations in ATP1A1 and CACNA1D underlie a common subtype of adrenal hypertension. *Nat Genet*. 2013;45:1055–1060.
- DeFranco DB. Navigating steroid hormone receptors through the nuclear compartment. *Mol Endocrinol*. 2002;16(7):1449–1455.
- Fejes-Toth G, Pearce D, Naray-Fejes-Toth A. Subcellular localization of mineralocorticoid receptors in living cells: effects of receptor agonists and antagonists. *Proc Natl Acad Sci U. S. A.* 1998;95(6):2973–2978.
- Pratt WB, et al. Role of hsp90 and the hsp90-binding immunophilins in signalling protein movement. *Cell Signal*. 2004;16(8):857–872.
- Gallo LI, et al. Differential recruitment of tetratricopeptide repeat domain immunophilins to the mineralocorticoid receptor influences both heat-shock protein 90-dependent retrotransport and hormone-dependent transcriptional activity. *Biochemistry*. 2007;46(49):14044–14057.
- Vienghcareun S, et al. The mineralocorticoid receptor: insights into its molecular and (patho)physiological biology. *Nucl Recept Signal*. 2007;5:e012.
- Arriza JL, et al. The neuronal mineralocorticoid receptor as a mediator of glucocorticoid response. *Neuron*. 1988;1(9):887–900.
- Bhargava A, Pearce D. Mechanisms of mineralocorticoid action: determinants of receptor specificity and actions of regulated gene products. *Trends Endocrinol Metab*. 2004;15(4):147–153.
- Liu W, et al. Steroid receptor heterodimerization demonstrated in vitro and in vivo. *Proc Natl Acad Sci U. S. A.* 1995;92(26):12480–12484.
- Trapp T, et al. Heterodimerization between mineralocorticoid and glucocorticoid receptor: a new principle of glucocorticoid action in the CNS. *Neuron*. 1994;13(6):1457–1462.
- Reichardt HM, et al. DNA binding of the glucocorticoid receptor is not essential for survival. *Cell*. 1998;93(4):531–541.
- Liu W, et al. Steroid receptor transcriptional synergy is potentiated by disruption of the DNA-binding domain dimer interface. *Mol Endocrinol*. 1996;10(11):1399–1406.
- Kaspar F, et al. A mutant androgen receptor from patients with Reifenstein syndrome: identification of the function of a conserved alanine residue in the D box of steroid receptors. *Mol Cell Biol*. 1993;13(12):7850–7858.
- Sartorato P, et al. Different inactivating mutations of the mineralocorticoid receptor in fourteen families affected by type I pseudohypoaldosteronism. *J Clin Endocrinol Metab*. 2003;88(6):2508–2517.
- Picard D, Yamamoto KR. Two signals mediate hormone-dependent nuclear localization of the glucocorticoid receptor. *EMBO J*. 1987;6(3333):3333–3340.
- Walther RF, et al. A serine/threonine-rich motif is one of three nuclear localization signals that determine unidirectional transport of the mineralocorticoid receptor to the nucleus. *J Biol Chem*. 2005;280(17):17549–17561.
- Starr DB, et al. Intracellular receptors use a common mechanism to interpret signaling information at response elements. *Genes Dev*. 1996;10(10):1271–1283.
- Prefontaine GG, et al. Recruitment of octamer transcription factors to DNA by glucocorticoid receptor. *Mol Cell Biol*. 1998;18(6):3416–3430.
- Iniguez-Lluhi JA, Pearce D. A common motif within the negative regulatory regions of multiple factors inhibits their transcriptional synergy. *Mol Cell Biol*. 2000;20(16):6040–6050.
- Fuse H, Kitagawa H, Kato S. Characterization of transactivational property and coactivator mediation of rat mineralocorticoid receptor activation function-1 (AF-1). *Mol Endocrinol*. 2000;14(6):889–899.
- Savory JG, et al. Glucocorticoid receptor homodimers and glucocorticoid-mineralocorticoid receptor heterodimers form in the cytoplasm through alternative dimerization interfaces. *Mol Cell Biol*. 2001;21(3):781–793.
- Rogerson FM, Fuller PJ. Interdomain interactions in the mineralocorticoid receptor. *Mol Cell Endocrinol*. 2003;200(1–2):45–55.
- Li Y, et al. Structural and biochemical mechanisms for the specificity of hormone binding and coactivator assembly by mineralocorticoid receptor. *Mol Cell*. 2005;19(3):367–380.
- Fagart J, et al. Crystal structure of a mutant mineralocorticoid receptor responsible for hypertension. *Nat Struct Mol Biol*. 2005;12(6):554–555.

53. Geller DS, et al. Activating mineralocorticoid receptor mutation in hypertension exacerbated by pregnancy. *Science*. 2000; 289(5476):119–123.
54. Geller DS, et al. Autosomal dominant pseudohypoaldosteronism type I: mechanisms, evidence for neonatal lethality, and phenotypic expression in adults. *J Am Soc Nephrol*. 2006;17(5):1429–1436.
55. Krozowski ZS, Funder JW. Renal mineralocorticoid receptors and hippocampal corticosterone-binding species have identical intrinsic steroid specificity. *Proc Natl Acad Sci U. S. A.* 1983;80(19):6056–6060.
56. Funder JW, et al. Vascular type I aldosterone binding sites are physiological mineralocorticoid receptors. *Endocrinology*. 1989; 125(4):2224–2226.
57. De Kloet ER. Hormones and the stressed brain. *Ann N Y Acad Sci*. 2004;1018:1–15.
58. Kitagawa H, et al. Ligand-selective potentiation of rat mineralocorticoid receptor activation function 1 by a CBP-containing histone acetyltransferase complex. *Mol Cell Biol*. 2002;22(11):3698–3706.
59. Pearce D, Yamamoto KR. Mineralocorticoid and glucocorticoid receptor activities distinguished by nonreceptor factors at a composite response element. *Science*. 1993;259(5098):1161–1165.
60. Tallec LP, et al. Protein inhibitor of activated signal transducer and activator of transcription 1 interacts with the N-terminal domain of mineralocorticoid receptor and represses its transcriptional activity: implication of small ubiquitin-related modifier 1 modification. *Mol Endocrinol*. 2003;17(12):2529–2542.
61. Pascual-Le Tallec L, et al. The elongation factor ELL (eleven-nineteen lysine-rich leukemia) is a selective coregulator for steroid receptor functions. *Mol Endocrinol*. 2005;19(5):1158–1169.
62. Choudhry MA, Ball A, McEwan IJ. The role of the general transcription factor IIF in androgen receptor-dependent transcription. *Mol Endocrinol*. 2006;20(9):2052–2061.
63. Roeder RG. Transcriptional regulation and the role of diverse coactivators in animal cells. *FEBS Lett*. 2005;579(4):909–915.
64. Lee DY, et al. Role of protein methylation in regulation of transcription. *Endocr Rev*. 2005;26(2):147–170.
65. Watson JD. *Molecular Biology of the Gene*. 6th ed. San Francisco: Benjamin Cummings; 2007.
66. Hong H, et al. GRIP1, a novel mouse protein that serves as a transcriptional coactivator in yeast for the hormone binding domains of steroid receptors. *Proc Natl Acad Sci U. S. A.* 1996;93(10): 4948–4952.
67. Onate SA, et al. Sequence and characterization of a coactivator for the steroid hormone receptor superfamily. *Science*. 1995;270(5240):1354–1357.
68. Metivier R, et al. Estrogen receptor- α directs ordered, cyclical, and combinatorial recruitment of cofactors on a natural target promoter. *Cell*. 2003;115(6):751–763.
69. Yokota K, et al. Coactivation of the N-terminal transactivation of mineralocorticoid receptor by Ubc9. *J Biol Chem*. 2007;282(3):1998–2010.
70. Endoh H, et al. Purification and identification of p68 RNA helicase acting as a transcriptional coactivator specific for the activation function 1 of human estrogen receptor α . *Mol Cell Biol*. 1999;19(8):5363–5372.
71. Faresse N. Post-translational modifications of the mineralocorticoid receptor: how to dress the receptor according to the circumstances? *J Steroid Biochem Mol Biol*. 2014;143:334–342.
72. Ganong WF, Mulrow PJ. Rate of change in sodium and potassium excretion after injection of aldosterone into the aorta and renal artery of the dog. *Am J Physiol*. 1958;195(2):337–342.
73. Edelman IS, Fimognari GM. On the biochemical mechanism of action of aldosterone. *Recent Prog Horm Res*. 1968;24(1):1–44.
74. Ganguly A. Primary aldosteronism. *N Engl J Med*. 1998;339:1828–1834.
75. Conn JW, Knopf RF, Nesbit RM. Clinical characteristics of primary aldosteronism from an analysis of 145 cases. *Am J Surg*. 1964;107:159–172.
76. Palmer LG, Antonian L, Frindt G. Regulation of apical K and Na channels and Na/K pumps in rat cortical collecting tubule by dietary K. *J Gen Physiol*. 1994;104:693–710.
77. Canessa CM, et al. Amiloride-sensitive epithelial Na⁺ channel is made of three homologous subunits. *Nature*. 1994;367(6462):463–467.
78. Canessa CM, Horisberger JD, Rossier BC. Epithelial sodium channel related to proteins involved in neurodegeneration. *Nature*. 1993;361(6411):467–470.
79. Garty H, Palmer LG. Epithelial sodium channels: function, structure, and regulation. *Physiol Rev*. 1997;77:359–396.
80. Hummler E, et al. Early death due to defective neonatal lung liquid clearance in α ENaC-deficient mice. *Nat Genet*. 1996;12:325–328.
81. Barker PM, et al. Role of γ ENaC subunit in lung liquid clearance and electrolyte balance in newborn mice. Insights into perinatal adaptation and pseudohypoaldosteronism. *J Clin Invest*. 1998;102:1634–1640.
82. McDonald FJ, et al. Disruption of the β subunit of the epithelial Na⁺ channel in mice: hyperkalemia and neonatal death associated with a pseudohypoaldosteronism phenotype. *Proc Natl Acad Sci U. S. A.* 1999;96:1727–1731.
83. Lifton RP, Gharavi AG, Geller DS. Molecular mechanisms of human hypertension. *Cell*. 2001;104(4):545–556.
84. Kunzelmann K, Mall M. Electrolyte transport in the mammalian colon: mechanisms and implications for disease. *Physiol Rev*. 2002; 82(1):245–289.
85. Giebisch G. Renal potassium transport: mechanisms and regulation. *Am J Physiol*. 1998;274(5 Pt 2):F817–F833.
86. Palmer LG. Potassium secretion and the regulation of distal nephron K channels. *Am J Physiol*. 1999;277(6 Pt 2):F821–F825.
87. Loffing J, et al. Aldosterone induces rapid apical translocation of ENaC in early portion of renal collecting system: possible role of SGK. *Am J Physiol Renal Physiol*. 2001;280(4):F675–F682.
88. Chen S, et al. Epithelial sodium channel regulated by aldosterone-induced protein sgk. *Proc Natl Acad Sci U. S. A.* 1999;96:2514–2519.
89. Naray-Fejes-Toth A, et al. *sgk* is an aldosterone-induced kinase in the renal collecting duct. Effects on epithelial Na⁺ channels. *J Biol Chem*. 1999;274:16973–16978.
90. Shigaev A, et al. Regulation of *sgk* by aldosterone and its effects on the epithelial Na⁺ channel. *Am J Physiol*. 2000;278(4): F613–F619.
91. McCormick JA, et al. SGK1: a rapid aldosterone-induced regulator of renal sodium reabsorption. *Physiology*. 2005;20(2):134–139.
92. Vallon V, et al. Role of Sgk1 in salt and potassium homeostasis. *Am J Physiol Regul Integr Comp Physiol*. 2005;288(1):R4–R10.
93. Schild L, et al. A mutation in the epithelial sodium channel causing Liddle disease increases channel activity in the *Xenopus laevis* oocyte expression system. *Proc Natl Acad Sci U. S. A.* 1995;92:5699–5703.
94. Snyder PM, et al. Mechanism by which Liddle's syndrome mutations increase activity of a human epithelial Na⁺ channel. *Cell*. 1995;83:969–978.
95. Loffing J, et al. Aldosterone induces rapid apical translocation of ENaC in early portion of renal collecting system: possible role of SGK. *Am J Physiol Renal Physiol*. 2001;280(4):F675–F682.
96. Masilamani S, et al. Aldosterone-mediated regulation of ENaC α , β , and γ subunit proteins in rat kidney. *J Clin Invest*. 1999;104:R19–R23.
97. Frindt G, Ergonul Z, Palmer LG. Surface expression of epithelial Na channel protein in rat kidney. *J Gen Physiol*. 2008;131(6): 617–627.
98. Butterworth MB, et al. Acute ENaC stimulation by cAMP in a kidney cell line is mediated by exocytic insertion from a recycling channel pool. *J Gen Physiol*. 2004;125(1):81–101.
99. Goldfarb SB, et al. Differential effects of Hsc70 and Hsp70 on the intracellular trafficking and functional expression of epithelial sodium channels. *Proc Natl Acad Sci U. S. A.* 2006;103(15):5817–5822.
100. Butterworth MB, et al. PKA-dependent ENaC trafficking requires the SNARE-binding protein complexin. *Am J Physiol Renal Physiol*. 2005;289(5):F969–F977.
101. Martel JA, et al. Melanophilin, a novel aldosterone-induced gene in mouse cortical collecting duct cells. *Am J Physiol Renal Physiol*. 2007;293(3):F904–F913.
102. Soundararajan R, et al. A novel role for glucocorticoid-induced leucine zipper protein in epithelial sodium channel-mediated sodium transport. *J Biol Chem*. 2005;280(48):39970–39981.
103. Goulet CC, et al. Inhibition of the epithelial Na⁺ channel by interaction of Nedd4 with a PY motif deleted in Liddle's syndrome. *J Biol Chem*. 1998;273:30012–30017.
104. Schild L, et al. Identification of PY motif in the epithelial Na channel subunits as a target sequence for mutations causing channel activation found in Liddle syndrome. *EMBO J*. 1996;15:2381–2387.
105. Wang H, et al. Clathrin-mediated endocytosis of the epithelial sodium channel: role of epsin. *J Biol Chem*. 2006;281(20):14129–14135.
106. Shimkets RA, Lifton RP, Canessa CM. The activity of the epithelial sodium channel is regulated by clathrin-mediated endocytosis. *J Biol Chem*. 1997;272:25537–25541.

107. Malik B, et al. Regulation of epithelial sodium channels by the ubiquitin-proteasome proteolytic pathway. *Am J Physiol Renal Physiol*. 2006;290(6):F1285–F1294.
108. Staub O, et al. Regulation of stability and function of the epithelial Na⁺ channel (ENaC) by ubiquitination. *EMBO J*. 1997;16(21):6325–6336.
109. Butterworth MB, et al. Regulation of the epithelial sodium channel by membrane trafficking. *Am J Physiol Renal Physiol*. 2009;296(1):F10–F24.
110. Pochynyuk O, et al. Regulation of the epithelial Na⁺ channel (ENaC) by phosphatidylinositides. *Am J Physiol Renal Physiol*. 2006;290(5):F949–F957.
111. Tong Q, et al. Direct activation of the epithelial Na⁺ channel by phosphatidylinositol 3,4,5-trisphosphate and phosphatidylinositol 3,4-bisphosphate produced by phosphoinositide 3-OH kinase. *J Biol Chem*. 2004;279(21):22654–22663.
112. Stockand JD, et al. Regulation of Na⁺ reabsorption by the aldosterone-induced small G protein K-Ras2A. *J Biol Chem*. 1999;274(50):35449–35454.
113. Soundararajan R, et al. Epithelial sodium channel regulated by differential composition of a signaling complex. *Proc Natl Acad Sci U. S. A*. 2009;106(19):7804–7809.
114. Staruschenko A, et al. Ras activates the epithelial Na⁺ channel through phosphoinositide 3-OH kinase signaling. *J Biol Chem*. 2004;279(36):37771–37778.
115. Falin RA, Cotton CU. Acute downregulation of ENaC by EGF involves the PY motif and putative ERK phosphorylation site. *J Gen Physiol*. 2007;130(3):313–328.
116. Shi H, et al. Interactions of beta and gamma ENaC with Nedd4 can be facilitated by an ERK-mediated phosphorylation. *J Biol Chem*. 2002;277(16):13539–13547.
117. Staruschenko A, Pochynyuk O, Stockand JD. Regulation of epithelial Na⁺ channel activity by conserved serine/threonine switches within sorting signals. *J Biol Chem*. 2005;280(47):39161–39167.
118. Stokes JB, Sigmund RD. Regulation of rENaC mRNA by dietary NaCl and steroids: organ, tissue, and steroid heterogeneity. *Am J Physiol Cell Physiol*. 1998;274:C1699–C1707.
119. Masilamani S, et al. Time course of renal Na-K-ATPase, NHE3, NKCC2, NCC, and ENaC abundance changes with dietary NaCl restriction. *Am J Physiol Renal Physiol*. 2002;283(4):F648–F657.
120. Johnson DW, et al. TGF- β 1 dissociates human proximal tubule cell growth and Na⁺-H⁺ exchange activity. *Kidney Int*. 1998;53:1601–1607.
121. Ergonul Z, Frindt G, Palmer LG. Regulation of maturation and processing of ENaC subunits in the rat kidney. *Am J Physiol Renal Physiol*. 2006;291(3):F683–F693.
122. Husted RF, et al. Discordant effects of corticosteroids and expression of subunits on ENaC activity. *Am J Physiol Renal Physiol*. 2007;293(3):F813–F820.
123. Crabbe J. Site of action of aldosterone on the toad bladder. *Nature*. 1963;200(4908):787–788.
124. Sharp GW, et al. Evidence for a mucosal effect of aldosterone on sodium transport in the toad bladder. *J Clin Invest*. 1966;45(10):1640–1647.
125. Pellanda AM, et al. Sodium-independent effect of aldosterone on initial rate of ouabain binding in A6 cells. *Am J Physiol*. 1992;C899–C906.
126. Verrey F, et al. Regulation by aldosterone of Na⁺, K⁺-ATPase mRNAs, protein synthesis, and sodium transport in cultured kidney cells. *J Cell Biol*. 1987;104(5):1231–1237.
127. Park CS, Edelman IS. Dual action of aldosterone on toad bladder: Na⁺ permeability and Na⁺ pump modulation. *Am J Physiol*. 1984;F517–F525.
128. Palmer LG, Antonian L, Frindt G. Regulation of the Na-K pump of the rat cortical collecting tubule by aldosterone. *J Gen Physiol*. 1993;102(1):43–57.
129. Coutry N, et al. Time course of sodium-induced Na⁺-K⁺-ATPase recruitment in rabbit cortical collecting tubule. *Am J Physiol Cell Physiol*. 1992;263:C61–C68.
130. Wade JB, et al. Morphological and physiological responses to aldosterone: time course and sodium dependence. *Am J Physiol*. 1990;259(1 Pt 2):F88–F94.
131. Verrey F, Krachenbuhl JP, Rossier BC. Aldosterone induces a rapid increase in the rate of Na, K-ATPase gene transcription in cultured kidney cells. *Mol Endocrinol*. 1989;3(9):1369–1376.
132. Kolla V, Robertson NM, Litwack G. Identification of a mineralocorticoid/glucocorticoid response element in the human Na/K ATPase alpha gene promoter. *Biochem Biophys Res Commun*. 1999;266(1):5–14.
133. Nagel W. Rheogenic sodium transport in a tight epithelium, the amphibian skin. *J Physiol*. 1980;302:281–295.
134. Blot-Chabaud M, et al. Cell sodium-induced recruitment of Na⁺-K⁺-ATPase pumps in rabbit cortical collecting tubules is aldosterone-dependent. *J Biol Chem*. 1990;265:11676–11681.
135. Kleyman TR, Carattino MD, Hughey RP. ENaC at the cutting edge: regulation of epithelial sodium channels by proteases. *J Biol Chem*. 2009;284:20447–20451.
136. Nariyo T, et al. Regulation of prostasin by aldosterone in the kidney. *J Clin Invest*. 2002;109(3):401–408.
137. Wakida N, et al. Inhibition of prostasin-induced ENaC activities by PN-1 and regulation of PN-1 expression by TGF- β 1 and aldosterone. *Kidney Int*. 2006;70(8):1432–1438.
138. Knight KK, et al. Liddle's syndrome mutations increase Na⁺ transport through dual effects on epithelial Na⁺ channel surface expression and proteolytic cleavage. *Proc Natl Acad Sci U. S. A*. 2006;103(8):2805–2808.
139. Stokes JB. Potassium secretion by cortical collecting tubule: relation to sodium absorption, luminal sodium concentration, and transepithelial voltage. *Am J Physiol Renal Fluid Electrolyte Physiol*. 1981;241:F395–F402.
140. Palmer LG, Antonian L, Frindt G. Regulation of apical K and Na channels and Na/K pumps in rat cortical collecting tubule by dietary K. *J Gen Physiol*. 1994;104:693–710.
141. Estilo G, Liu W, Pastor-Soler N, et al. Effect of aldosterone on BK channel expression in mammalian cortical collecting duct. *Am J Physiol Renal Physiol*. 2008;295:F780–F788.
142. Diezi J, et al. Micropuncture study of electrolyte transport across papillary collecting duct of the rat. *Am J Physiol*. 1973;224:623–634.
143. Stokes JB. Ion transport by the cortical and outer medullary collecting tubule. *Kidney Int*. 1982;22:473–484.
144. Koeppen BM. Conductive properties of the rabbit outer medullary collecting duct: inner stripe. *Am J Physiol Renal Physiol*. 1985;248(4):F500–F506.
145. Sorensen MV, Grossmann S, Roesinger M, et al. Rapid dephosphorylation of the renal sodium chloride cotransporter in response to oral potassium intake in mice. *Kidney Int*. 2013;83:811–824.
146. Rengarajan S, Lee DH, Oh YT, et al. Increasing plasma [K⁺] by intravenous potassium infusion reduces NCC phosphorylation and drives kaliuresis and natriuresis. *Am J Physiol Renal Physiol*. 2014;306:F1059–F1068.
147. Young DB. Relationship between plasma potassium concentration and renal potassium excretion. *Am J Physiol*. 1982;242:F599–F603.
148. Young DB, Jackson TE, Tipayamongtri U, et al. Effects of sodium intake on steady-state potassium excretion. *Am J Physiol*. 1984;246:F772–F778.
149. Palla AT, Moon TM, Akella R, et al. Chloride sensing by WNK1 involves inhibition of autophosphorylation. *Sci Signal*. 2014;7:ra41. PMID: 24803536. PMC: PMC4123527.
150. Lazrak A, Liu Z, Huang CL. Antagonistic regulation of ROMK by long and kidney-specific WNK1 isoforms. *Proc Natl Acad Sci U. S. A*. 2006;103(5):1615–1620.
151. Kahle KT, Ring AM, Lifton RP. Molecular physiology of the WNK kinases. *Annu Rev Physiol*. 2008;70(1):329–355.
152. Terker AS, Zhang C, McCormick JA, et al. Potassium modulates electrolyte balance and blood pressure through effects on distal cell voltage and chloride. *Cell Metab*. 2015;21:39–50. PMID: 25565204.
153. Cuevas CA, Su XT, Wang MX, et al. Potassium sensing by renal distal tubules requires Kir4.1. *J Am Soc Nephrol*. 2017;28(6):1814–1825.
154. Grimm PR, Sansom SC. BK channels in the kidney. *Curr Opin Nephrol Hypertens*. 2007;16:430–436.
155. Le Moellie C, et al. Aldosterone and tight junctions: modulation of claudin-4 phosphorylation in renal collecting duct cells. *Am J Physiol Cell Physiol*. 2005;289(6):C1513–C1521.
156. Wall SM, et al. NaCl restriction upregulates renal Slc26a4 through subcellular redistribution: role in Cl[−] conservation. *Hypertension*. 2004;44(6):982–987.
157. Hirohama D, Ayuzawa N, Ueda K, et al. Aldosterone is essential for angiotensin II-induced upregulation of pendrin. *J Am Soc Nephrol*. 2017;PMID: 29021385.

158. Pech V, et al. Angiotensin II increases chloride absorption in the cortical collecting duct in mice through a pendrin-dependent mechanism. *Am J Physiol Renal Physiol*. 2007;292(3):F914–F920.
159. Vallet M, et al. Pendrin regulation in mouse kidney primarily is chloride-dependent. *J Am Soc Nephrol*. 2006;17(8):2153–2163.
160. Stokes JB. Physiologic resistance to the action of aldosterone. *Kidney Int*. 2000;57:1319–1323.
161. Harris RC, Breyer MD. Physiological regulation of cyclooxygenase-2 in the kidney. *Am J Physiol Renal Physiol*. 2001;281(1):F1–F11.
162. Shibata S, et al. Regulated mineralocorticoid receptor phosphorylation controls ligand binding, allowing distinct physiologic responses to aldosterone. *Cell Metab*. 2013;18(5):660–671.
163. Funder JW. Angiotensin retains sodium by dephosphorylating mineralocorticoid receptors in renal intercalated cells. *Cell Metab*. 2013;18(5):609–610.
164. Levief F, Hubner CA, Houillier P, et al. The Na⁺-dependent chloride-bicarbonate exchanger SLC4A8 mediates an electroneutral Na⁺ reabsorption process in the renal cortical collecting ducts of mice. *J Clin Invest*. 2010;120:1627–1635.
165. Frische S, Kwon TH, Frokier J, et al. Regulated expression of pendrin in rat kidney in response to chronic NH₄Cl or NaHCO₃ loading. *Am J Physiol Renal Physiol*. 2003;284:F584–F593.
166. Kim YH, Pech V, Spencer KB, et al. Reduced ENaC protein abundance contributes to the lower blood pressure observed in pendrin-null mice. *Am J Physiol Renal Physiol*. 2007;293:F1314–F1324.
167. Loffing J, et al. Localization of epithelial sodium channel and aquaporin-2 in rabbit kidney cortex. *Am J Physiol Renal Physiol*. 2000;278:F530–F539.
168. Zhai XY, Thomsen JS, Birn H, et al. Three-dimensional reconstruction of the mouse nephron. *J Am Soc Nephrol*. 2006;17:77–88.
169. Rubera I, et al. Collecting duct-specific gene inactivation of α ENaC in the mouse kidney does not impair sodium and potassium balance. *J Clin Invest*. 2003;112(4):554–565.
170. Christensen BM, Perrier R, Wang Q, et al. Sodium and potassium balance depends on α ENaC expression in connecting tubule. *J Am Soc Nephrol*. 2010;21:1942–1951.
171. Nesterov V, Dahlmann A, Krueger B, et al. Aldosterone-dependent and -independent regulation of the epithelial sodium channel (ENaC) in mouse distal nephron. *Am J Physiol Renal Physiol*. 2012;303:F1289–F1299.
172. Marver D, Kokko JP. Renal target sites and the mechanism of action of aldosterone. *Miner Electrolyte Metab*. 1983;9(1):1–18.
173. Bonvalet JP. Binding and action of aldosterone, dexamethasone, 1-25(OH)₂D₃, and estradiol along the nephron. *J Steroid Biochem*. 1987;27(4–6):953–961.
174. Todd-Turla KM, et al. Distribution of mineralocorticoid and glucocorticoid receptor mRNA along the nephron. *Am J Physiol*. 1993;F781–F791.
175. Roland BL, Krozowski ZS, Funder JW. Glucocorticoid receptor, mineralocorticoid receptors, 11 beta-hydroxysteroid dehydrogenase-1 and -2 expression in rat brain and kidney: in situ studies. *Mol Cell Endocrinol*. 1995;111(1):R1–R7.
176. Farman N, et al. Immunolocalization of gluco- and mineralocorticoid receptors in rabbit kidney. *Am J Physiol*. 1991;260(2 Pt 1):C226–C233.
177. Berger S, et al. Mineralocorticoid receptor knockout mice: pathophysiology of Na⁺ metabolism. *Proc Natl Acad Sci U. S. A*. 1998;95(16):9424–9429.
178. Makhanova N, et al. Kidney function in mice lacking aldosterone. *Am J Physiol Renal Physiol*. 2006;290(1):F61–F69.
179. Stone DK, et al. Mineralocorticoid modulation of rabbit medullary collecting duct acidification. A sodium-independent effect. *J Clin Invest*. 1983;72(1):77–83.
180. Winter C, et al. Nongenomic stimulation of vacuolar H⁺-ATPases in intercalated renal tubule cells by aldosterone. *Proc Natl Acad Sci U. S. A*. 2004;101(8):2636–2641.
181. Wagner CA, et al. Regulated acid-base transport in the collecting duct. *Pflügers Arch*. 2009;458(1):137–156.
182. Marver D, Schwartz MJ. Identification of mineralocorticoid target sites in the isolated rabbit cortical nephron. *Proc Natl Acad Sci U. S. A*. 1980;77(6):3672–3676.
183. Doucet A, Katz AI. Mineralocorticoid receptors along the nephron: [3H]aldosterone binding in rabbit tubules. *Am J Physiol*. 1981;241(6):F605–F611.
184. Vandewalle A, et al. Aldosterone binding along the rabbit nephron: an autoradiographic study on isolated tubules. *Am J Physiol*. 1981;240(3):F172–F179.
185. Gnionsahe A, et al. Aldosterone binding sites along nephron of *Xenopus* and rabbit. *Am J Physiol*. 1989;257(1 Pt 2):R87–R95.
186. Krozowski Z, et al. Immunohistochemical localization of the 11 beta-hydroxysteroid dehydrogenase type II enzyme in human kidney and placenta. *J Clin Endocrinol Metab*. 1995;80(7):2203–2209.
187. Kyosseff Z, Walker PD, Reeves WB. Immunolocalization of NAD-dependent 11 beta-hydroxysteroid dehydrogenase in human kidney and colon. *Kidney Int*. 1996;49(1):271–281.
188. Miyata K, et al. Aldosterone stimulates reactive oxygen species production through activation of NADPH oxidase in rat mesangial cells. *J Am Soc Nephrol*. 2005;16(10):2906–2912.
189. Nishiyama A, et al. Involvement of aldosterone and mineralocorticoid receptors in rat mesangial cell proliferation and deformability. *Hypertension*. 2005;45(4):710–716.
190. Terada Y, et al. Aldosterone-stimulated SGK1 activity mediates profibrotic signaling in the mesangium. *J Am Soc Nephrol*. 2008;19(2):298–309.
191. Cha DR, et al. Role of aldosterone in diabetic nephropathy. *Nephrology (Carlton)*. 2005;10(suppl):S37–S39.
192. Hierholzer K, Stolte H. The proximal and distal tubular action of adrenal steroids on Na reabsorption. *Nephron*. 1969;6(3):188–204.
193. Wiederholt M, et al. Sodium conductance changes by aldosterone in the rat kidney. *Pflügers Arch*. 1974;348(2):155–165.
194. Pergher PS, Leite-Dellova D, de Mello-Aires M. Direct action of aldosterone on bicarbonate reabsorption in vivo cortical proximal tubule. *Am J Physiol Renal Physiol*. 2009;296(5):F1185–F1193.
195. Leite-Dellova DC, et al. Genomic and nongenomic dose-dependent biphasic effect of aldosterone on Na⁺/H⁺ exchanger in proximal S3 segment: role of cytosolic calcium. *Am J Physiol Renal Physiol*. 2008;295(5):F1342–F1352.
196. Krug AW, et al. Aldosterone stimulates surface expression of NHE3 in renal proximal brush borders. *Pflügers Arch*. 2003;446(4):492–496.
197. ElMernissi G, Doucet A. Short-term effects of aldosterone and dexamethasone on Na-K-ATPase along the rabbit nephron. *Pflügers Arch*. 1983;399:147–151.
198. Garg LC, Knepper MA, Burg MB. Mineralocorticoid effects on Na-K-ATPase in individual nephron segments. *Am J Physiol*. 1981;240(6):F536–F544.
199. Schmidt U, et al. Sodium- and potassium-activated ATPase: a possible target of aldosterone. *J Clin Invest*. 1975;55(3):655–660.
200. Schmid H, et al. Hormonal effects on Na-K-ATPase of various parts of the rat nephron. *Curr Probl Clin Biochem*. 1975;4:214–217.
201. Stanton BA. Regulation by adrenal corticosteroids of sodium and potassium transport in loop of Henle and distal tubule of rat kidney. *J Clin Invest*. 1986;78(6):1612–1620.
202. Work J, Jamison RL. Effect of adrenalectomy on transport in the rat medullary thick ascending limb. *J Clin Invest*. 1987;80(4):1160–1164.
203. Crabbe J. The role of aldosterone in the renal concentration mechanism in man. *Clin Sci*. 1962;23:39–46.
204. Oberleithner H, et al. Aldosterone activates Na⁺/H⁺ exchange and raises cytoplasmic pH in target cells of the amphibian kidney. *Proc Natl Acad Sci U. S. A*. 1987;84:1464–1468.
205. Ponce-Coria J, et al. Regulation of NKCC2 by a chloride-sensing mechanism involving the WNK3 and SPAK kinases. *Proc Natl Acad Sci U. S. A*. 2008;105(24):8458–8463.
206. Ko B, Mistry AC, Hanson L, et al. Aldosterone acutely stimulates NCC activity via a SPAK-mediated pathway. *Am J Physiol Renal Physiol*. 2013;305:F645–F652.
207. Kim GH, Masilamani S, Turner R, et al. The thiazide-sensitive Na-Cl cotransporter is an aldosterone-induced protein. *Proc Natl Acad Sci U. S. A*. 1998;95(24):14552–14557.
208. Terker AS, Yarbrough B, Ferdaus MZ, et al. Direct and indirect mineralocorticoid effects determine distal salt transport. *J Am Soc Nephrol*. 2016;27(8):2436–2445.
209. Czogalla J, Vohra T, Penton D, et al. The mineralocorticoid receptor (MR) regulates ENaC but not NCC in mice with random MR deletion. *Pflügers Arch*. 2016;468(5):849–858.
210. Roy A, Al-Qusairi L, Donnelly BF, et al. Alternatively spliced proline-rich cassettes link WNK1 to aldosterone action. *J Clin Invest*. 2015;125(9):3433–3448.

211. Braun EJ. Regulation of renal and lower gastrointestinal function: role in fluid and electrolyte balance. *Comp Biochem Physiol A Mol Integr Physiol*. 2003;136(3):499–505.
212. Levitan R, et al. Water and salt absorption in the human colon. *J Clin Invest*. 1962;41:1754–1759.
213. Clauss W, et al. Ion transport and electrophysiology of the early proximal colon of rabbit. *Pflugers Arch*. 1987;408(6):592–599.
214. Sandle GI, et al. Electrophysiology of the human colon: evidence of segmental heterogeneity. *Gut*. 1986;27(9):999–1005.
215. Yau WM, Makhoul GM. Comparison of transport mechanisms in isolated ascending and descending rat colon. *Am J Physiol*. 1975;228(1):191–195.
216. Greig ER, et al. Decreased expression of apical Na⁺ channels and basolateral Na⁺, K⁺-ATPase in ulcerative colitis. *J Pathol*. 2004;204(1):84–92.
217. Levitan R, Ingelfinger FJ. Effect of *d*-aldosterone on salt and water absorption from the intact human colon. *J Clin Invest*. 1965;44:801–808.
218. Hirasawa G, et al. Colocalization of 11 beta-hydroxysteroid dehydrogenase type II and mineralocorticoid receptor in human epithelia. *J Clin Endocrinol Metab*. 1997;82(11):3859–3863.
219. Turnamian SC, Binder HJ. Regulation of active sodium and potassium transport in the distal colon of the rat: role of the aldosterone and glucocorticoid receptors. *J Clin Invest*. 1989;84(6):1924–1929.
220. Fromm M, Schulzke JD, Hegel U. Control of electrogenic Na⁺ absorption in rat late distal colon by nanomolar aldosterone added in vitro. *Am J Physiol*. 1993;264(1 Pt 1):E68–E73.
221. Shigaev A, et al. Regulation of sgk by aldosterone and its effects on the epithelial Na(+) channel. *Am J Physiol Renal Physiol*. 2000;278(4):F613–F619.
222. Asher C, et al. Aldosterone-induced increase in the abundance of Na⁺ channel subunits. *Am J Physiol*. 1996;271(2 Pt 1):C605–C611.
223. Eppler HJ, et al. Early aldosterone effect in distal colon by transcriptional regulation of ENaC subunits. *Am J Physiol Gastrointest Liver Physiol*. 2000;278(5):G718–G724.
224. Gifford JD, et al. Control of serum potassium during fasting in patients with end-stage renal disease. *Kidney Int*. 1989;35(1):90–94.
225. Hayes CP Jr, McLeod ME, Robinson RR. An extrarenal mechanism for the maintenance of potassium balance in severe chronic renal failure. *Trans Assoc Am Physicians*. 1967;80:207–216.
226. Sandle GI, et al. Enhanced rectal potassium secretion in chronic renal insufficiency: evidence for large intestinal potassium adaptation in man. *Clin Sci*. 1986;71(4):393–401.
227. Imbriano LJ, Durham JH, Maesaka JK. Treating interdialytic hyperkalemia with fludrocortisone. *Semin Dial*. 2003;16(1):5–7.
228. Hussain S, et al. Is spironolactone safe for dialysis patients? *Nephrol Dial Transplant*. 2003;18(11):2364–2368.
229. Saudan P, et al. Safety of low-dose spironolactone administration in chronic haemodialysis patients. *Nephrol Dial Transplant*. 2003;18(11):2359–2363.
230. Gross E, et al. Effect of spironolactone on blood pressure and the renin-angiotensin-aldosterone system in oligo-anuric hemodialysis patients. *Am J Kidney Dis*. 2005;46(1):94–101.
231. Eaton DC, et al. The contribution of epithelial sodium channels to alveolar function in health and disease. *Annu Rev Physiol*. 2009;71:403–423.
232. Hummler E, et al. Early death due to defective neonatal lung liquid clearance in alpha-ENaC-deficient mice. *Nat Genet*. 1996;12(3):325–328.
233. Malagon-Rogers M. A patient with pseudohypoaldosteronism type 1 and respiratory distress syndrome. *Pediatr Nephrol*. 1999;13(6):484–486.
234. Kerem E, et al. Pulmonary epithelial sodium-channel dysfunction and excess airway liquid in pseudohypoaldosteronism. *N Engl J Med*. 1999;341:156–162.
235. Sheridan MB, et al. Mutations in the beta-subunit of the epithelial Na⁺ channel in patients with a cystic fibrosis-like syndrome. *Hum Mol Genet*. 2005;14(22):3493–3498.
236. Matthay MA, Robriquet L, Fang X. Alveolar epithelium: role in lung fluid balance and acute lung injury. *Proc Am Thorac Soc*. 2005;2(3):206–213.
237. Scherrer U, et al. High-altitude pulmonary edema: from exaggerated pulmonary hypertension to a defect in transepithelial sodium transport. *Adv Exp Med Biol*. 1999;474:93–107.
238. Myerburg MM, et al. Airway surface liquid volume regulates ENaC by altering the serine protease-protease inhibitor balance: a mechanism for sodium hyperabsorption in cystic fibrosis. *J Biol Chem*. 2006;281(38):27942–27949.
239. Hirasawa G, et al. 11Beta-hydroxysteroid dehydrogenase type 2 and mineralocorticoid receptor in human fetal development. *J Clin Endocrinol Metab*. 1999;84(4):1453–1458.
240. Suzuki T, et al. 11Beta-hydroxysteroid dehydrogenase type 2 in human lung: possible regulator of mineralocorticoid action. *J Clin Endocrinol Metab*. 1998;83(11):4022–4025.
241. Suzuki S, et al. Modulation of transalveolar fluid absorption by endogenous aldosterone in adult rats. *Exp Lung Res*. 2001;27(2):143–155.
242. Champigny G, et al. Regulation of expression of the lung amiloride-sensitive Na⁺ channel by steroid hormones. *EMBO J*. 1994;13(9):2177–2181.
243. Renard S, et al. Localization and regulation by steroids of the alpha, beta and gamma subunits of the amiloride-sensitive Na⁺ channel in colon, lung and kidney. *Pflugers Arch*. 1995;430(3):299–307.
244. Illek B, Fischer H, Clauss W. Aldosterone regulation of basolateral potassium channels in alveolar epithelium. *Am J Physiol*. 1990;259(4 Pt 1):L230–L237.
245. Keller-Wood M, von Reitzenstein M, McCartney J. Is the fetal lung a mineralocorticoid receptor target organ? Induction of cortisol-regulated genes in the ovine fetal lung, kidney and small intestine. *Neonatology*. 2009;95(1):47–60.
246. Berger SA, et al. Molecular genetic analysis of glucocorticoid and mineralocorticoid signaling in development and physiological processes. *Steroids*. 1996;61(4):236–239.
247. Cook DI, et al. Patch-clamp studies on epithelial sodium channels in salivary duct cells. *Cell Biochem Biophys*. 2002;36(2–3):105–113.
248. Anand SK, et al. Pseudohypoaldosteronism due to sweat gland dysfunction. *Pediatr Res*. 1976;10(7):677–682.
249. Sanderson IR, et al. Familial salivary gland insensitivity to aldosterone: a variant of pseudohypoaldosteronism. *Horm Res*. 1989;32(4):145–147.
250. Kretz O, et al. Differential expression of RNA and protein of the three pore-forming subunits of the amiloride-sensitive epithelial sodium channel in taste buds of the rat. *J Histochem Cytochem*. 1999;47(1):51–64.
251. Lin W, et al. Epithelial Na⁺ channel subunits in rat taste cells: localization and regulation by aldosterone. *J Comp Neurol*. 1999;405(3):406–420.
252. Sasano H, et al. Immunolocalization of mineralocorticoid receptor in human kidney, pancreas, salivary, mammary and sweat glands: a light and electron microscopic immunohistochemical study. *J Endocrinol*. 1992;132(2):305–310.
253. Duc C, et al. Cell-specific expression of epithelial sodium channel alpha, beta, and gamma subunits in aldosterone-responsive epithelia from the rat: localization by in situ hybridization and immunocytochemistry. *J Cell Biol*. 1994;127(6):1907–1921.
254. Rauh R, et al. Stimulation of the epithelial sodium channel (ENaC) by the serum- and glucocorticoid-inducible kinase (Sgk) involves the PY motifs of the channel but is independent of sodium feedback inhibition. *Pflugers Arch*. 2006;452(3):290–299.
255. Riad F, et al. Aldosterone regulates salivary sodium secretion in cattle. *J Endocrinol*. 1986;108(3):405–411.
256. Wotman S, et al. Salivary electrolytes, renin, and aldosterone during sodium loading and depletion. *J Appl Physiol*. 1973;35(3):322–324.
257. McVie R, Levine LS, New MI. The biologic significance of the aldosterone concentration in saliva. *Pediatr Res*. 1979;13(6):755–759.
258. Adlin EV, Marks AD, Channick BJ. Racial difference in salivary sodium-potassium ratio in low renin essential hypertension. *Arch Intern Med*. 1982;142(4):703–706.
259. Porter GA, Bogoroch R, Edelman IS. On the mechanism of action of aldosterone on sodium transport: the role of RNA synthesis. *Proc Natl Acad Sci U. S. A*. 1964;52:1326–1333.
260. Blazer-Yost B, Cox M. Aldosterone-induced proteins: characterization using lectin-affinity chromatography. *Am J Physiol*. 1985;C215–C225.
261. Naray-Fejes-Toth A, et al. *sgk* is an aldosterone-induced kinase in the renal collecting duct: effects on epithelial Na⁺ channels. *J Biol Chem*. 1999;274(24):16973–16978.
262. Alvarez de la Rosa D, et al. The serum and glucocorticoid kinase *sgk* increases the abundance of epithelial sodium channels in the plasma membrane of *Xenopus* oocytes. *J Biol Chem*. 1999;274(53):37834–37839.

263. Flores SY, et al. Aldosterone-induced serum and glucocorticoid-induced kinase 1 expression is accompanied by Nedd4-2 phosphorylation and increased Na⁺ transport in cortical collecting duct cells. *J Am Soc Nephrol*. 2005;16(8):2279–2287.
264. Bhargava A, et al. The serum- and glucocorticoid-induced kinase is a physiological mediator of aldosterone action. *Endocrinology*. 2001;142(4):1587–1594.
265. Loffing J, Korbmayer C. Regulated sodium transport in the renal connecting tubule (CNT) via the epithelial sodium channel (ENaC). *Pflugers Archiv*. 2009;458(1):111–135.
266. Nesterov V, Dahlmann A, Krueger B, et al. Aldosterone-dependent and -independent regulation of the epithelial sodium channel (ENaC) in mouse distal nephron. *Am J Physiol Renal Physiol*. 2012;303(9):F1289–F1299.
267. Hou J, et al. Sgk1 gene expression in kidney and its regulation by aldosterone: spatio-temporal heterogeneity and quantitative analysis. *J Am Soc Nephrol*. 2002;13(5):1190–1198.
268. Wu C, Yosef N, Thallhammer T, et al. Induction of pathogenic TH17 cells by inducible salt-sensing kinase SGK1. *Nature*. 2013;496(7446):513–517.
269. Norlander AE, Saleh MA, Pandey AK, et al. A salt-sensing kinase in T lymphocytes, SGK1, drives hypertension and hypertensive end-organ damage. *JCI Insight*. 2017;2(13):pii: 92801.
270. Bhalla V, et al. Disinhibitory pathways for control of sodium transport: regulation of ENaC by SGK1 and GILZ. *Am J Physiol Renal Physiol*. 2006;291(4):F714–F721.
271. Lang F, et al. (Patho)physiological significance of the serum- and glucocorticoid-inducible kinase isoforms. *Physiol Rev*. 2006;86(4):1151–1178.
272. Waldegger S, et al. h-sgk serine-threonine protein kinase gene as transcriptional target of transforming growth factor beta in human intestine. *Gastroenterology*. 1999;116:1081–1088.
273. Gonzalez-Robayna JJ, et al. Follicle-stimulating hormone (FSH) stimulates phosphorylation and activation of protein kinase B (PKB/Akt) and serum and glucocorticoid-induced kinase (Sgk): evidence for a kinase-independent signaling by FSH in granulosa cells. *Mol Endocrinol*. 2000;14(8):1283–1300.
274. Webster MK, et al. Characterization of sgk, a novel member of the serine/threonine protein kinase gene family which is transcriptionally induced by glucocorticoids and serum. *Mol Cell Biol*. 1993;13(4):2031–2040.
275. Rozansky DJ, et al. Hypotonic induction of SGK1 and Na⁺ transport in A6 cells. *Am J Physiol Renal Physiol*. 2002;283(1):F105–F113.
276. Wang J, et al. Activity of the p110- α subunit of phosphatidylinositol-3-kinase is required for activation of epithelial sodium transport. *Am J Physiol Renal Physiol*. 2008;295(3):F843–F850.
277. Debonneville C, et al. Phosphorylation of Nedd4-2 by Sgk1 regulates epithelial Na⁺ channel cell surface expression. *EMBO J*. 2001;20(24):7052–7059.
278. Snyder PM, Olson DR, Thomas BC. Serum- and glucocorticoid-regulated kinase modulates Nedd4-2-mediated inhibition of the epithelial Na⁺ channel. *J Biol Chem*. 2002;277(1):5–8.
279. Wang J, et al. SGK integrates insulin and mineralocorticoid regulation of epithelial sodium transport. *Am J Physiol Renal Physiol*. 2001;280(2):F303–F313.
280. Park J, et al. Serum- and glucocorticoid-inducible kinase (SGK) is a target of the PI 3-kinase-stimulated signaling pathway. *EMBO J*. 1999;18(11):3024–3033.
281. Kobayashi T, et al. Characterization of the structure and regulation of two novel isoforms of serum- and glucocorticoid-induced protein kinase. *Biochem J*. 1999;344(Pt 1):189–197.
282. Kobayashi T, Cohen P. Activation of serum- and glucocorticoid-regulated protein kinase by agonists that activate phosphatidylinositol 3-kinase is mediated by 3-phosphoinositide-dependent protein kinase-1 (PDK1) and PDK2. *Biochem J*. 1999;339(Pt 2):319–328.
283. Lu M, et al. mTOR complex-2 activates ENaC by phosphorylating SGK1. *J Am Soc Nephrol*. 2010;21:811–818.
284. Garcia-Martinez JM, Alessi DR. mTOR complex 2 (mTORC2) controls hydrophobic motif phosphorylation and activation of serum- and glucocorticoid-induced protein kinase 1 (SGK1). *Biochem J*. 2008;416:375–385.
285. Garcia-Martinez JM, Alessi DR. mTOR complex 2 (mTORC2) controls hydrophobic motif phosphorylation and activation of serum- and glucocorticoid-induced protein kinase 1 (SGK1). *Biochem J*. 2008;416(3):375–385.
286. Jones KT, et al. Rictor/TORC2 regulates *Caenorhabditis elegans* fat storage, body size, and development through sgk-1. *PLoS Biol*. 2009;7(3):e60.
287. Lu M, Wang J, Jones KT, et al. mTOR complex-2 activates ENaC by phosphorylating SGK1. *J Am Soc Nephrol*. 2010;21:811–818.
288. Wulff P, et al. Impaired renal Na⁺ retention in the sgk1-knockout mouse. *J Clin Invest*. 2002;110(9):1263–1268.
289. Fejes-Toth G, et al. Epithelial Na⁺ channel activation and processing in mice lacking SGK1. *Am J Physiol Renal Physiol*. 2008;294(6):F1298–F1305.
290. Fejes-Tóth G, Frindt G, Náray-Fejes-Tóth A, et al. Epithelial Na⁺ channel activation and processing in mice lacking SGK1. *Am J Physiol Renal Physiol*. 2008;294(6):F1298–F1305.
291. Yang L, Frindt G, Lang F, et al. SGK1-dependent ENaC processing and trafficking in mice with high dietary K intake and elevated aldosterone. *Am J Physiol Renal Physiol*. 2017;312(1):F65–F76.
292. Hummler E, Barker P, Talbot C, et al. A mouse model for the renal salt-wasting syndrome pseudohypoaldosteronism. *Proc Natl Acad Sci U. S. A.* 1997;94(21):11710–11715.
293. Huang DY, et al. Blunted hypertensive effect of combined fructose and high-salt diet in gene-targeted mice lacking functional serum- and glucocorticoid-inducible kinase SGK1. *Am J Physiol Regul Integr Comp Physiol*. 2006;290(4):R935–R944.
294. Huang DY, et al. Resistance of mice lacking the serum- and glucocorticoid-inducible kinase SGK1 against salt-sensitive hypertension induced by a high-fat diet. *Am J Physiol Renal Physiol*. 2006;291(6):F1264–F1273.
295. Bhargava A, Wang J, Pearce D. Regulation of epithelial ion transport by aldosterone through changes in gene expression. *Mol Cell Endocrinol*. 2004;217(1–2):189–196.
296. Webster MK, Goya L, Firestone GL. Immediate-early transcriptional regulation and rapid mRNA turnover of a putative serine/threonine protein kinase. *J Biol Chem*. 1993;268(16):11482–11485.
297. Brickley DR, et al. Ubiquitin modification of serum and glucocorticoid-induced protein kinase-1 (SGK-1). *J Biol Chem*. 2002;277(45):43064–43070.
298. Naray-Fejes-Toth A, et al. Regulation of sodium transport in mammalian collecting duct cells by aldosterone-induced kinase, SGK1: structure/function studies. *Mol Cell Endocrinol*. 2004;217(1–2):197–202.
299. Pao AC, et al. NH2 terminus of serum and glucocorticoid-regulated kinase 1 binds to phosphoinositides and is essential for isoform-specific physiological functions. *Am J Physiol Renal Physiol*. 2007;292(6):F1741–F1750.
300. Zhou R, Snyder PM. Nedd4-2 phosphorylation induces serum- and glucocorticoid-regulated kinase (SGK) ubiquitination and degradation. *J Biol Chem*. 2005;280(6):4518–4523.
301. Arteaga MF, et al. Multiple translational isoforms give functional specificity to serum- and glucocorticoid-induced kinase 1. *Mol Biol Cell*. 2007;18(6):2072–2080.
302. Arteaga MF, et al. A brain-specific SGK1 splice isoform regulates expression of ASIC1 in neurons. *Proc Natl Acad Sci U. S. A.* 2008;105(11):4459–4464.
303. Raikwar NS, Snyder PM, Thomas CP. An evolutionarily conserved N-terminal Sgk1 variant with enhanced stability and improved function. *Am J Physiol Renal Physiol*. 2008;295(5):F1440–F1448.
304. Staub O, et al. WW domains of Nedd4 bind to the proline-rich PY motifs in the epithelial Na⁺ channel deleted in Liddle's syndrome. *EMBO J*. 1996;15(10):2371–2380.
305. Abriel H, et al. Defective regulation of the epithelial Na⁺ channel by Nedd4 in Liddle's syndrome. *J Clin Invest*. 1999;103(5):667–673.
306. Kamynina E, et al. A novel mouse Nedd4 protein suppresses the activity of the epithelial Na⁺ channel. *FASEB J*. 2001;15(1):204–214.
307. Shi PP, et al. Salt-sensitive hypertension and cardiac hypertrophy in mice deficient in the ubiquitin ligase Nedd4-2. *Am J Physiol Renal Physiol*. 2008;295:F462–F470.
308. Snyder PM, Steines JC, Olson DR. Relative contribution of Nedd4 and Nedd4-2 to ENaC regulation in epithelia determined by RNA interference. *J Biol Chem*. 2004;279(6):5042–5046.
309. Alvarez de la Rosa D, Canessa CM. Role of SGK in hormonal regulation of epithelial sodium channel in A6 cells. *Am J Physiol Cell Physiol*. 2003;284(2):C404–C414.
310. Bhalla V, et al. Serum- and glucocorticoid-regulated kinase 1 regulates ubiquitin ligase neural precursor cell-expressed, developmentally down-regulated protein 4-2 by inducing interaction with 14-3-3. *Mol Endocrinol*. 2005;19(12):3073–3084.

311. Liang X, et al. 14-3-3 isoforms are induced by aldosterone and participate in its regulation of epithelial sodium channels. *J Biol Chem*. 2006;281(24):16323–16332.
312. Ichimura T, et al. 14-3-3 proteins modulate the expression of epithelial Na⁺ channels by phosphorylation-dependent interaction with Nedd4-2 ubiquitin ligase. *J Biol Chem*. 2005;280(13):13187–13194.
313. Chandran S, Li H, Dong W, et al. Neural precursor cell-expressed developmentally down-regulated protein 4-2 (Nedd4-2) regulation by 14-3-3 protein binding at canonical serum and glucocorticoid kinase 1 (SGK1) phosphorylation sites. *J Biol Chem*. 2012;286:37830–37840.
314. Rossier BC. Negative regulators of sodium transport in the kidney: key factors in understanding salt-sensitive hypertension? *J Clin Invest*. 2003;111(7):947–950.
315. Arroyo JP, Lagnaz D, Ronzaud C, et al. Nedd4-2 modulates renal Na⁺-Cl⁻ cotransporter via the aldosterone-SGK1-Nedd4-2 pathway. *J Am Soc Nephrol*. 2011;22:1707–1719.
316. Faresse N, Lagnaz D, Debonneville A, et al. Inducible kidney-specific Sgk1 knockout mice show a salt-losing phenotype. *Am J Physiol Renal Physiol*. 2012;302:F977–F985.
317. Diakov A, Korbacher C. A novel pathway of epithelial sodium channel activation involves a serum- and glucocorticoid-inducible kinase consensus motif in the C terminus of the channel's alpha-subunit. *J Biol Chem*. 2004;279(37):38134–38142.
318. Thomas SV, Kathalia PP, Rajagopal M, et al. Epithelial sodium channel regulation by cell surface-associated serum- and glucocorticoid-regulated kinase 1. *J Biol Chem*. 2011;286:32074–32085.
319. Ring AM, et al. An SGK1 site in WNK4 regulates Na⁺ channel and K⁺ channel activity and has implications for aldosterone signaling and K⁺ homeostasis. *Proc Natl Acad Sci U. S. A*. 2007;104(10):4025–4029.
320. Vuagniaux G, et al. Synergistic activation of ENaC by three membrane-bound channel-activating serine proteases (mCAP1, mCAP2, and mCAP3) and serum- and glucocorticoid-regulated kinase (Sgk1) in *Xenopus* oocytes. *J Gen Physiol*. 2002;120(2):191–201.
321. Alvarez de la Rosa D, et al. SGK1 activates Na⁺-K⁺-ATPase in amphibian renal epithelial cells. *Am J Physiol Cell Physiol*. 2006;290(2):C492–C498.
322. Zecevic M, et al. SGK1 increases Na, K-ATP cell-surface expression and function in *Xenopus laevis* oocytes. *Pflugers Arch*. 2004;448(1):29–35.
323. Boyd C, Naray-Fejes-Toth A. Gene regulation of ENaC subunits by serum- and glucocorticoid-inducible kinase-1. *Am J Physiol Renal Physiol*. 2005;288(3):F505–F512.
324. Zhang W, et al. Aldosterone-induced Sgk1 relieves Dot1a-Af9-mediated transcriptional repression of epithelial Na⁺ channel alpha. *J Clin Invest*. 2007;117(3):773–783.
325. Huang DY, Wulff P, Völkl H, et al. Impaired regulation of renal K⁺ elimination in the sgk1-knockout mouse. *J Am Soc Nephrol*. 2004;15(4):885–891.
326. Al-Qusairi L, Basquin D, Roy A, et al. Renal tubular SGK1 deficiency causes impaired K⁺ excretion via loss of regulation of NEDD4-2/WNK1 and ENaC. *Am J Physiol Renal Physiol*. 2016;311(2):F330–F342.
327. Yang L, Frindt G, Lang F, et al. SGK1-dependent ENaC processing and trafficking in mice with high dietary K intake and elevated aldosterone. *Am J Physiol Renal Physiol*. 2017;312(1):F65–F76.
328. Huang DY, et al. Impaired regulation of renal K⁺ elimination in the sgk1-knockout mouse. *J Am Soc Nephrol*. 2004;15(4):885–891.
329. Al-Qusairi L, Basquin D, Roy A, et al. Renal tubular ubiquitin-protein ligase NEDD4-2 is required for renal adaptation during long-term potassium depletion. *J Am Soc Nephrol*. 2017;28(8):2431–2442.
330. Fakitsas P, et al. Early aldosterone-induced gene product regulates the epithelial sodium channel by deubiquitylation. *J Am Soc Nephrol*. 2007;18(4):1084–1092.
331. Martel JA, et al. Melanophilin, a novel aldosterone-induced gene in mouse cortical collecting duct cells. *Am J Physiol Renal Physiol*. 2007;293(3):F904–F913.
332. Mastroberardino L, et al. Ras pathway activates epithelial Na⁺ channel and decreases its surface expression in *Xenopus* oocytes. *Mol Biol Cell*. 1998;9(12):3417–3427.
333. Naray-Fejes-Toth A, Boyd C, Fejes-Toth G. Regulation of epithelial sodium transport by promyelocytic leukemia zinc finger protein. *Am J Physiol Renal Physiol*. 2008;295(1):F18–F26.
334. Naray-Fejes-Toth A, Snyder PM, Fejes-Toth G. The kidney-specific WNK1 isoform is induced by aldosterone and stimulates epithelial sodium channel-mediated Na⁺ transport. *Proc Natl Acad Sci U. S. A*. 2004;101(50):17434–17439.
335. Soundararajan R, et al. A novel role for glucocorticoid-induced leucine zipper protein in epithelial sodium channel-mediated sodium transport. *J Biol Chem*. 2005;280(48):39970–39981.
336. Edinger RS, Coronello C, Bodnar AJ, et al. Aldosterone regulates microRNAs in the cortical collecting duct to alter sodium transport. *J Am Soc Nephrol*. 2014;25(11):2445–2457.
337. Klemens CA, Edinger RS, Kightlinger L, et al. Ankyrin G expression regulates apical delivery of the epithelial sodium channel (ENaC). *J Biol Chem*. 2017;292(1):375–385.
338. Liu X, Edinger RS, Klemens CA, et al. A microRNA cluster miR-23-24-27 is upregulated by aldosterone in the distal kidney nephron where it alters sodium transport. *J Cell Physiol*. 2017;232(6):1306–1317.
339. Jacobs ME, Kathalia PP, Chen Y, et al. SGK1 regulation by miR-466g in cortical collecting duct cells. *Am J Physiol Renal Physiol*. 2016;310(11):F1251–F1257.
340. Tomlinson JW, Stewart PM. Mechanisms of disease: selective inhibition of 11beta-hydroxysteroid dehydrogenase type 1 as a novel treatment for the metabolic syndrome. *Nat Clin Pract Endocrinol Metab*. 2005;1(2):92–99.
341. Naray-Fejes-Toth A, Fejes-Toth G. Extranuclear localization of endogenous 11beta-hydroxysteroid dehydrogenase-2 in aldosterone target cells. *Endocrinology*. 1998;139(6):2955–2959.
342. Bachmann S, et al. Sodium transport-related proteins in the mammalian distal nephron—distribution, ontogeny and functional aspects. *Anat Embryol (Berl)*. 1999;200(5):447–468.
343. Velazquez H, et al. Rabbit distal convoluted tubule coexpresses NaCl cotransporter and 11 beta-hydroxysteroid dehydrogenase II mRNA. *Kidney Int*. 1998;54(2):464–472.
344. Bostanjoglo M, et al. 11Beta-hydroxysteroid dehydrogenase, mineralocorticoid receptor, and thiazide-sensitive Na-Cl cotransporter expression by distal tubules. *J Am Soc Nephrol*. 1998;9(8):1347–1358.
345. Whorwood CB, et al. 11 Beta-hydroxysteroid dehydrogenase and corticosteroid hormone receptors in the rat colon. *Am J Physiol*. 1993;264(6 Pt 1):E951–E957.
346. Challis JR, Connor K. Glucocorticoids, 11beta-hydroxysteroid dehydrogenase: mother, fetus, or both? *Endocrinology*. 2009;150(3):1073–1074.
347. Geerling JC, Loewy AD. Aldosterone in the brain. *Am J Physiol Renal Physiol*. 2009;297(3):F559–F576.
348. Funder J, Myles K. Exclusion of corticosterone from epithelial mineralocorticoid receptors is insufficient for selectivity of aldosterone action: in vivo binding studies. *Endocrinology*. 1996;137(12):5264–5268.
349. Funder JW. Is aldosterone bad for the heart? *Trends Endocrinol Metab*. 2004;15(4):139–142.
350. Makino Y, et al. Direct association with thioredoxin allows redox regulation of glucocorticoid receptor function. *J Biol Chem*. 1999;274(5):3182–3188.
351. Wilson RC, et al. Steroid 21-hydroxylase deficiency: genotype may not predict phenotype. *J Clin Endocrinol Metab*. 1995;80(8):2322–2329.
352. Wilson RC, Nimkarn S, New MI. Apparent mineralocorticoid excess. *Trends Endocrinol Metab*. 2001;12(3):104–111.
353. Alzamora R, Michea L, Marusic ET. Role of 11beta-hydroxysteroid dehydrogenase in nongenomic aldosterone effects in human arteries. *Hypertension*. 2000;35(5):1099–1104.
354. Mihailidou KS, et al. Glucocorticoids activate cardiac mineralocorticoid receptors during experimental myocardial infarction. *Hypertension*. 2009;54(6):1306–1312.
355. Liu SL, et al. Aldosterone regulates vascular reactivity: short-term effects mediated by phosphatidylinositol 3-kinase-dependent nitric oxide synthase activation. *Circulation*. 2003;108(19):2400–2406.
356. Oberleithner H. Is the vascular endothelium under the control of aldosterone? Facts and hypothesis. *Pflugers Arch*. 2007;454(2):187–193.
357. Molnar GA, et al. Glucocorticoid-related signaling effects in vascular smooth muscle cells. *Hypertension*. 2008;51(5):1372–1378.
358. Gros R, et al. Rapid effects of aldosterone on clonal human vascular smooth muscle cells. *Am J Physiol Cell Physiol*. 2007;292(2):C788–C794.
359. Kusche-Vihrog K, et al. The epithelial sodium channel (ENaC): mediator of the aldosterone response in the vascular endothelium? *Steroids*. 2009;75(8–9):544–549.
360. Moura AM, Worcel M. Direct action of aldosterone on transmembrane 22Na efflux from arterial smooth muscle: rapid and delayed effects. *Hypertension*. 1984;6(3):425–430.
361. Romagnì P, et al. Aldosterone induces contraction of the resistance arteries in man. *Atherosclerosis*. 2003;166(2):345–349.

362. Wildling L, et al. Aldosterone receptor sites on plasma membrane of human vascular endothelium detected by a mechanical nanosensor. *Pflugers Arch.* 2009;458(2):223–230.
363. Kelly MJ, Levin ER. Rapid actions of plasma membrane estrogen receptors. *Trends Endocrinol Metab.* 2001;12(4):152–156.
364. Wang Z, et al. A variant of estrogen receptor- α , hER- α 36: transduction of estrogen- and antiestrogen-dependent membrane-initiated mitogenic signaling. *Proc Natl Acad Sci U. S. A.* 2006;103(24):9063–9068.
365. Boldyreff B, Wehling M. Rapid aldosterone actions: from the membrane to signaling cascades to gene transcription and physiological effects. *J Steroid Biochem Mol Biol.* 2003;85(2–5):375–381.
366. Funder JW. The nongenomic actions of aldosterone. *Endocr Rev.* 2005;26(3):313–321.
367. Chun TY, Pratt JH. Nongenomic renal effects of aldosterone: dependency on NO and genomic actions. *Hypertension.* 2006;47(4):636–637.
368. Good DW. Nongenomic actions of aldosterone on the renal tubule. *Hypertension.* 2007;49(4):728–739.
369. Young WF. Primary aldosteronism: renaissance of a syndrome. *Clin Endocrinol (Oxf).* 2007;66(5):607–618.
370. Stowasser M, et al. Evidence for abnormal left ventricular structure and function in normotensive individuals with familial hyperaldosteronism type I. *J Clin Endocrinol Metab.* 2005;90(9):5070–5076.
371. Milliez P, et al. Evidence for an increased rate of cardiovascular events in patients with primary aldosteronism. *J Am Coll Cardiol.* 2005;45(8):1243–1248.
372. Funder JW, et al. Case detection, diagnosis, and treatment of patients with primary aldosteronism: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab.* 2008;93(9):3266–3281.
373. Guyton AC, et al. Salt balance and long-term blood pressure control. *Annu Rev Med.* 1980;31:15–27.
374. Gomez-Sanchez EP, Fort CM, Gomez-Sanchez CE. Intracerebroventricular infusion of RU28318 blocks aldosterone-salt hypertension. *Am J Physiol.* 1990;258(3 Pt 1):E482–E484.
375. Levy DG, Rocha R, Funder JW. Distinguishing the antihypertensive and electrolyte effects of eplerenone. *J Clin Endocrinol Metab.* 2004;89(6):2736–2740.
376. Machnik A, et al. Macrophages regulate salt-dependent volume and blood pressure by a vascular endothelial growth factor-C-dependent buffering mechanism. *Nat Med.* 2009;15(5):545–552.
377. Rickard AJ, et al. Deletion of mineralocorticoid receptors from macrophages protects against deoxycorticosterone/salt-induced cardiac fibrosis and increased blood pressure. *Hypertension.* 2009;54(3):537–543.
378. Ball WC Jr, et al. Increased excretion of aldosterone in urine from dogs with right-sided congestive heart failure and from dogs with thoracic inferior vena cava constriction. *Am J Physiol.* 1956;187(1):45–50.
379. Luetscher JA Jr, Neher R, Wettstein A. Isolation of crystalline aldosterone from the urine of patients with congestive heart failure. *Experientia.* 1956;12(1):22–23.
380. Pitt B, et al. The effect of spironolactone on morbidity and mortality in patients with severe heart failure. Randomized Aldactone Evaluation Study Investigators. *N Engl J Med.* 1999;341(10):709–717.
381. Pitt B, et al. Eplerenone reduces mortality 30 days after randomization following acute myocardial infarction in patients with left ventricular systolic dysfunction and heart failure. *J Am Coll Cardiol.* 2005;46(3):425–431.
382. Rossi G, et al. Aldosterone as a cardiovascular risk factor. *Trends Endocrinol Metab.* 2005;16(3):104–107.
383. Zannad F, et al. Rationale and design of the Eplerenone in Mild Patients Hospitalization and Survival Study in Heart Failure (EMPHASIS-HF). *Eur J Heart Fail.* 2010;12(6):617–622.
384. Ando K, Ohtsu H, Uchida S, et al. Anti-albuminuric effect of the aldosterone blocker eplerenone in non-diabetic hypertensive patients with albuminuria: a double-blind, randomised, placebo-controlled trial. *Lancet Diabetes Endocrinol.* 2014;2(12):944–953.
385. Epstein M. Mineralocorticoid receptor antagonists: part of an emerging treatment paradigm for chronic kidney disease. *Lancet Diabetes Endocrinol.* 2014;2(12):925–927.
386. Gómez-Sánchez EP, et al. ICV infusion of corticosterone antagonizes ICV-aldosterone hypertension. *Am J Physiol.* 1990;258:E649–E653.
387. Milhailidou AS, et al. Glucocorticoids activate cardiac mineralocorticoid receptors during experimental myocardial infarction. *Hypertension.* 2009;54:1306–1312.
388. Rossi G, et al. Immunoreactive endogenous ouabain in primary aldosteronism and essential hypertension: relationship with plasma renin, aldosterone and blood pressure levels. *J Hypertens.* 1995;13:1181–1191.
389. Dostanic-Larson I, et al. The highly conserved cardiac glycoside binding site of Na,K-ATPase plays a role in blood pressure regulation. *PNAS.* 2005;102(44):15845–15850.
390. Laredo J, et al. Angiotensin II stimulates secretion of endogenous ouabain from bovine adrenocortical cells via angiotensin type 2 receptors. *Hypertension.* 1997;29:401–407.
391. Hasegawa T, et al. Increase in plasma ouabain-like inhibitor of Na⁺ K⁺-ATPase with high sodium intake in patients with essential hypertension. *J Clin Hypertens.* 1987;3:419–429.
392. Jaisser F, Farman N. Emerging roles of the mineralocorticoid receptor in pathology: toward new paradigms in clinical pharmacology. *Pharmacol Rev.* 2016;68:49–75.
393. Bousquet E, Beydoun T, Rothschild PR, et al. Spironolactone for nonresolving central serous chorioretinopathy: a randomized controlled crossover study. *Retina.* 2015;35:2505–2515.
394. Li Y, Suino K, Daugherty J, et al. Structural and biochemical mechanisms for the specificity of hormone binding and coactivator assembly by mineral corticoid receptor. *Mol Cell.* 2005;19:367–380.
395. McCormick JA, Yang CL, Ellison DH. WNK kinases and renal sodium transport in health and disease: an integrated view. *Hypertension.* 2008;51(3):588–596.
396. Rundle SE, et al. Immunocytochemical demonstration of mineralocorticoid receptors in rat and human kidney. *J Steroid Biochem.* 1989;33(6):1235–1242.
397. Gonzalez-Nunez D, et al. In vitro characterization of aldosterone and cAMP effects in mouse distal convoluted tubule cells. *Am J Physiol Renal Physiol.* 2004;286(5):F936–F944.
398. Nielsen J, et al. Sodium transporter abundance profiling in kidney: effect of spironolactone. *Am J Physiol Renal Physiol.* 2002;283(5):F923–F933.
399. Abdallah JG, et al. Loop diuretic infusion increases thiazide-sensitive Na(+)/Cl(-)-cotransporter abundance: role of aldosterone. *J Am Soc Nephrol.* 2001;12(7):1335–1341.
400. Masilamani S, et al. Time course of renal Na-K-ATPase, NHE3, NKCC2, NCC, and ENaC abundance changes with dietary NaCl restriction. *Am J Physiol Renal Physiol.* 2002;283(4):F648–F657.
401. Wang XY, et al. The renal thiazide-sensitive Na-Cl cotransporter as mediator of the aldosterone-escape phenomenon. *J Clin Invest.* 2001;108(2):215–222.
402. Yang SS, et al. Molecular pathogenesis of pseudohypoaldosteronism type II: generation and analysis of a Wnk4(D561A/+) knockin mouse model. *Cell Metab.* 2007;5(5):331–344.
403. Kim GH, et al. The thiazide-sensitive Na-Cl cotransporter is an aldosterone-induced protein. *Proc Natl Acad Sci U. S. A.* 1998;95(24):14552–14557.
404. Rozansky DJ, et al. Aldosterone mediates activation of the thiazide-sensitive Na-Cl cotransporter through an SGK1 and WNK4 signaling pathway. *J Clin Invest.* 2009;119(9):2601–2612.
405. Chiga M, et al. Dietary salt regulates the phosphorylation of OSR1/SPAK kinases and the sodium chloride cotransporter through aldosterone. *Kidney Int.* 2008;74(11):1403–1409.
406. Richardson C, et al. Activation of the thiazide-sensitive Na⁺-Cl⁻-cotransporter by the WNK-regulated kinases SPAK and OSR1. *J Cell Sci.* 2008;121(Pt 5):675–684.
407. Xu BE, et al. WNK1 activates SGK1 by a phosphatidylinositol 3-kinase-dependent and non-catalytic mechanism. *J Biol Chem.* 2005;280(40):34218–34223.
408. Kirsten E, et al. Increased activity of enzymes of the tricarboxylic acid cycle in response to aldosterone in the toad bladder. *Pflugers Arch Gesamte Physiol Menschen Tiere.* 1968;300(4):213–225.
409. Law PY, Edelman IS. Induction of citrate synthase by aldosterone in the rat kidney. *J Membr Biol.* 1978;41:41–64.
410. Chen WS, et al. Leptin deficiency and beta-cell dysfunction underlie type 2 diabetes in compound Akt knockout mice. *Mol Cell Biol.* 2009;29(11):3151–3162.
411. Pearce D. The role of SGK1 in hormone-regulated sodium transport. *Trends Endocrinol Metab.* 2001;12(8):341–347.
412. Kenouch S, et al. Multiple patterns of 11 beta-hydroxysteroid dehydrogenase catalytic activity along the mammalian nephron. *Kidney Int.* 1992;42:56–60.

413. Bostanjoglo M, et al. 11Beta-hydroxysteroid dehydrogenase, mineralocorticoid receptor, and thiazide-sensitive Na-Cl cotransporter expression by distal tubules. *J Am Soc Nephrol*. 1998;9:1347–1358.
414. Lombes M, et al. Immunohistochemical localization of renal mineralocorticoid receptor by using an anti-idiotypic antibody that is an internal image of aldosterone. *Proc Natl Acad Sci U. S. A.* 1990;87:1086–1088.
415. Bonvalet JP, et al. Distribution of 11 beta-hydroxysteroid dehydrogenase along the rabbit nephron. *J Clin Invest*. 1990;86:832–837.
416. Naray-Fejes-Toth A, Fejes-Toth G. Novel mouse strain with Cre recombinase in 11beta-hydroxysteroid dehydrogenase-2-expressing cells. *Am J Physiol Renal Physiol*. 2007;292:F486–F494.
417. Cole TJ. Cloning of the mouse 11 beta-hydroxysteroid dehydrogenase type 2 gene: tissue specific expression and localization in distal convoluted tubules and collecting ducts of the kidney. *Endocrinology*. 1995;136:4693–4696.
418. Smith RE, et al. Immunohistochemical and molecular characterization of the rat 11 beta-hydroxysteroid dehydrogenase type II enzyme. *Endocrinology*. 1997;138:540–547.
419. Campean V, et al. Localization of thiazide-sensitive Na(+)-Cl(-) cotransport and associated gene products in mouse DCT. *Am J Physiol Renal Physiol*. 2001;281:F1028–F1035.
420. Plotkin MD, et al. Localization of the thiazide sensitive Na-Cl cotransporter, rTSC1 in the rat kidney. *Kidney Int*. 1996;50:174–183.
421. Obermuller N, Bernstein P, Velazquez H, et al. Expression of the thiazide-sensitive Na-Cl cotransporter in rat and human kidney. *Am J Physiol*. 1995;269:F900–F910.
422. Bachmann S, et al. Expression of the thiazide-sensitive Na-Cl cotransporter by rabbit distal convoluted tubule cells. *J Clin Invest*. 1995;96:2510–2514.
423. Biner HL, et al. Human cortical distal nephron: distribution of electrolyte and water transport pathways. *J Am Soc Nephrol*. 2002;13:836–847.
424. Ciampolillo F, et al. Cell-specific expression of amiloride-sensitive, Na(+)-conducting ion channels in the kidney. *Am J Physiol*. 1996;271:C1303–C1315.
425. Loffing-Cueni D, et al. Dietary sodium intake regulates the ubiquitin-protein ligase nedd4-2 in the renal collecting system. *J Am Soc Nephrol*. 2006;17:1264–1274.
426. Loffing J, et al. Distribution of transcellular calcium and sodium transport pathways along mouse distal nephron. *Am J Physiol Renal Physiol*. 2001;281:F1021–F1027.
427. Duc C, et al. Cell-specific expression of epithelial sodium channel alpha, beta, and gamma subunits in aldosterone-responsive epithelia from the rat: localization by in situ hybridization and immunocytochemistry. *J Cell Biol*. 1994;127:1907–1921.
428. Palmer LG, Frindt G. Amiloride-sensitive Na channels from the apical membrane of the rat cortical collecting tubule. *Proc Natl Acad Sci U. S. A.* 1986;83:2767–2770.
429. Schmitt R, et al. Developmental expression of sodium entry pathways in rat nephron. *Am J Physiol*. 1999;276:F367–F381.
430. Dijkink L, et al. Time-dependent regulation by aldosterone of the amiloride-sensitive Na⁺ channel in rabbit kidney. *Pflugers Arch*. 1999;438:354–360.
431. Lu M, et al. Absence of small conductance K⁺ channel (SK) activity in apical membranes of thick ascending limb and cortical collecting duct in ROMK (Bartter's) knockout mice. *J Biol Chem*. 2002;277:37881–37887.
432. Xu JZ, et al. Localization of the ROMK protein on apical membranes of rat kidney nephron segments. *Am J Physiol*. 1997;273:F739–F748.
433. Mennitt PA, et al. Localization of ROMK channels in the rat kidney. *J Am Soc Nephrol*. 1997;8:1823–1830.
434. Kohda Y, et al. Localization of the ROMK potassium channel to the apical membrane of distal nephron in rat kidney. *Kidney Int*. 1998;54:1214–1223.
435. Benchimol C, Zavilowitz B, Satlin LM. Developmental expression of ROMK mRNA in rabbit cortical collecting duct. *Pediatr Res*. 2000;47:46–52.
436. Nusing RM, et al. Expression of the potassium channel ROMK in adult and fetal human kidney. *Histochem Cell Biol*. 2005;123:553–559.
437. Staub O, et al. Immunolocalization of the ubiquitin-protein ligase Nedd4 in tissues expressing the epithelial Na⁺ channel (ENaC). *Am J Physiol*. 1997;272:C1871–C1880.
438. Umemura M, et al. Transcriptional diversity and expression of NEDD4L gene in distal nephron. *Biochem Biophys Res Commun*. 2006;339:1129–1137.
439. Velazquez H, et al. The distal convoluted tubule of rabbit kidney does not express a functional sodium channel. *Am J Physiol Renal Physiol*. 2001;280:F530–F539.
440. Breton S, et al. Depletion of intercalated cells from collecting ducts of carbonic anhydrase II-deficient (CAR2 null) mice. *Am J Physiol*. 1995;269:F761–F774.
441. Nelson RD, et al. Expression of an AQP2 Cre recombinase transgene in kidney and male reproductive system of transgenic mice. *Am J Physiol*. 1998;275:C216–C226.
442. Coleman RA, et al. Expression of aquaporins in the renal connecting tubule. *Am J Physiol Renal Physiol*. 2000;279:F874–F883.
443. Katz AI, Doucet A, Morel F. Na-K-ATPase activity along the rabbit, rat, and mouse nephron. *Am J Physiol*. 1979;237:F114–F120.
444. Piepenhagen PA, et al. Differential expression of Na(+)-K(+)-ATPase, ankyrin, fodrin, and E-cadherin along the kidney nephron. *Am J Physiol*. 1995;269:C1417–C1432.
445. Baskin DG, Stahl WL. Immunocytochemical localization of Na⁺, K⁺-ATPase in the rat kidney. *Histochemistry*. 1982;73:535–548.
446. Charles PG, Dowling JP, Fuller PJ. Characterization of renal Na-K-ATPase gene expression by in situ hybridization. *Ren Physiol Biochem*. 1992;15:10–15.
447. McDonough AA, Magyar CE, Komatsu Y. Expression of Na(+)-K(+)-ATPase alpha- and beta-subunits along rat nephron: isoform specificity and response to hypokalemia. *Am J Physiol*. 1994;267:C901–C908.
448. Wetzel RK, Swadner KJ. Immunocytochemical localization of Na-K-ATPase alpha- and gamma-subunits in rat kidney. *Am J Physiol Renal Physiol*. 2001;281:F531–F545.
449. Kashgarian M, et al. Monoclonal antibody to Na, K-ATPase: immunocytochemical localization along nephron segments. *Kidney Int*. 1985;28:899–913.
450. Beeuwkes R 3rd, Rosen S. Renal sodium-potassium adenosine triphosphatase. Optical localization and x-ray microanalysis. *J Histochem Cytochem*. 1975;23:828–839.
451. Kwon O, et al. Distribution of cell membrane-associated proteins along the human nephron. *J Histochem Cytochem*. 1998;46:1423–1434.

BOARD REVIEW QUESTIONS

1. The most appropriate initial treatment for a patient with primary aldosteronism due to bilateral adrenal hyperplasia is:
 - a. Triamterene
 - b. Bilateral adrenalectomy
 - c. Spironolactone
 - d. Beta blockers
 - e. Thiazide diuretics**Answer: c**
2. The enzyme conferring aldosterone-specificity on mineralocorticoid receptors in epithelial tissues is:
 - a. CYP11B1
 - b. CYP11B2
 - c. 11 β OHSD1
 - d. 11 β OHSD2
 - e. Aldosterone synthase**Answer: d**
3. Mineralocorticoid receptors are aldosterone regulated:
 - a. Transcription factors.
 - b. Receptor tyrosine kinases.
 - c. G-protein coupled receptors.
 - d. Mitochondrial enzymes.
 - e. Epithelial sodium channels.**Answer: a**
4. Serum- and glucocorticoid-regulated kinase 1 (SGK1) acts in principal cells to:
 - a. Inhibit K⁺ channels
 - b. Stimulate the epithelial sodium channel (ENaC)
 - c. Stimulate aldosterone uptake
 - d. Stimulate cortisol break down
 - e. Inhibit the Krebs cycle**Answer: b**
5. Which of the following is true?
 - a. Acute aldosterone regulated potassium secretion is primarily via direct stimulation of potassium channels
 - b. Mineralocorticoid receptors are exclusively expressed in the distal nephron
 - c. Hyperkalemia-induced aldosterone secretion leads to high blood pressure
 - d. Sodium reabsorption increases the driving force for potassium and proton efflux in the distal nephron.**Answer: d**